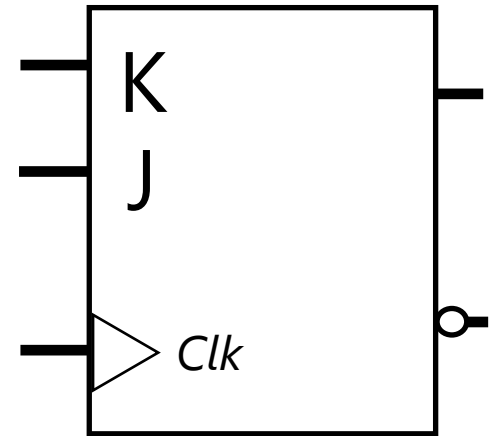
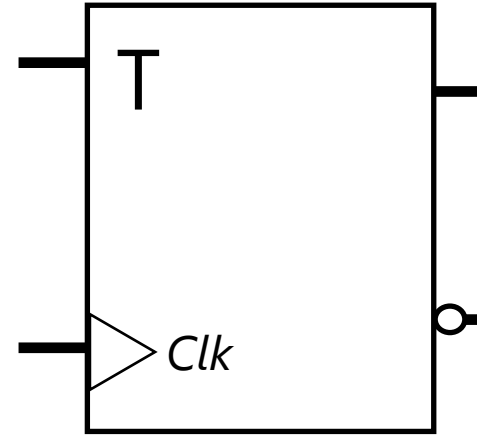
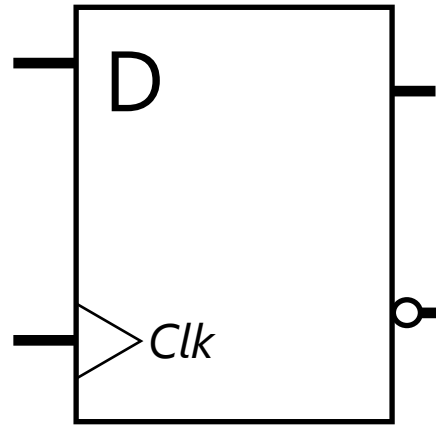
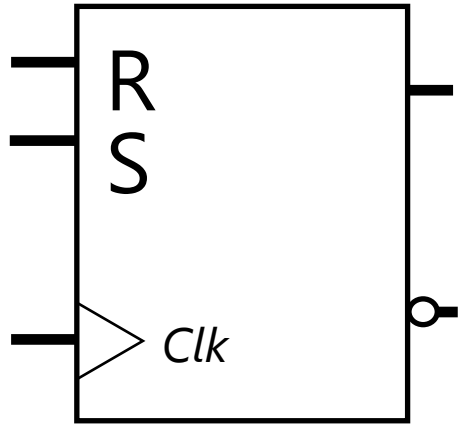

Flip-Flop

A single edge triggered latch



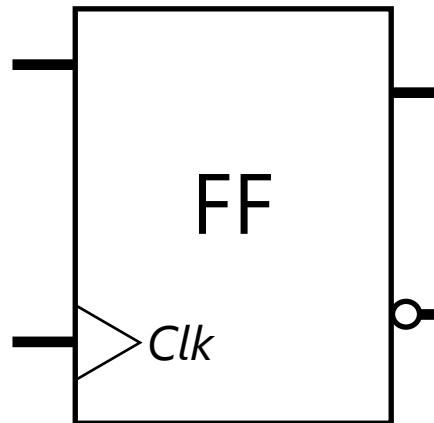
S	R	Q
0	0	Q_t
0	1	0
1	0	1
1	1	$\times$

D	Q
0	0
1	1

T	Q
0	Q_t
1	Q'_t

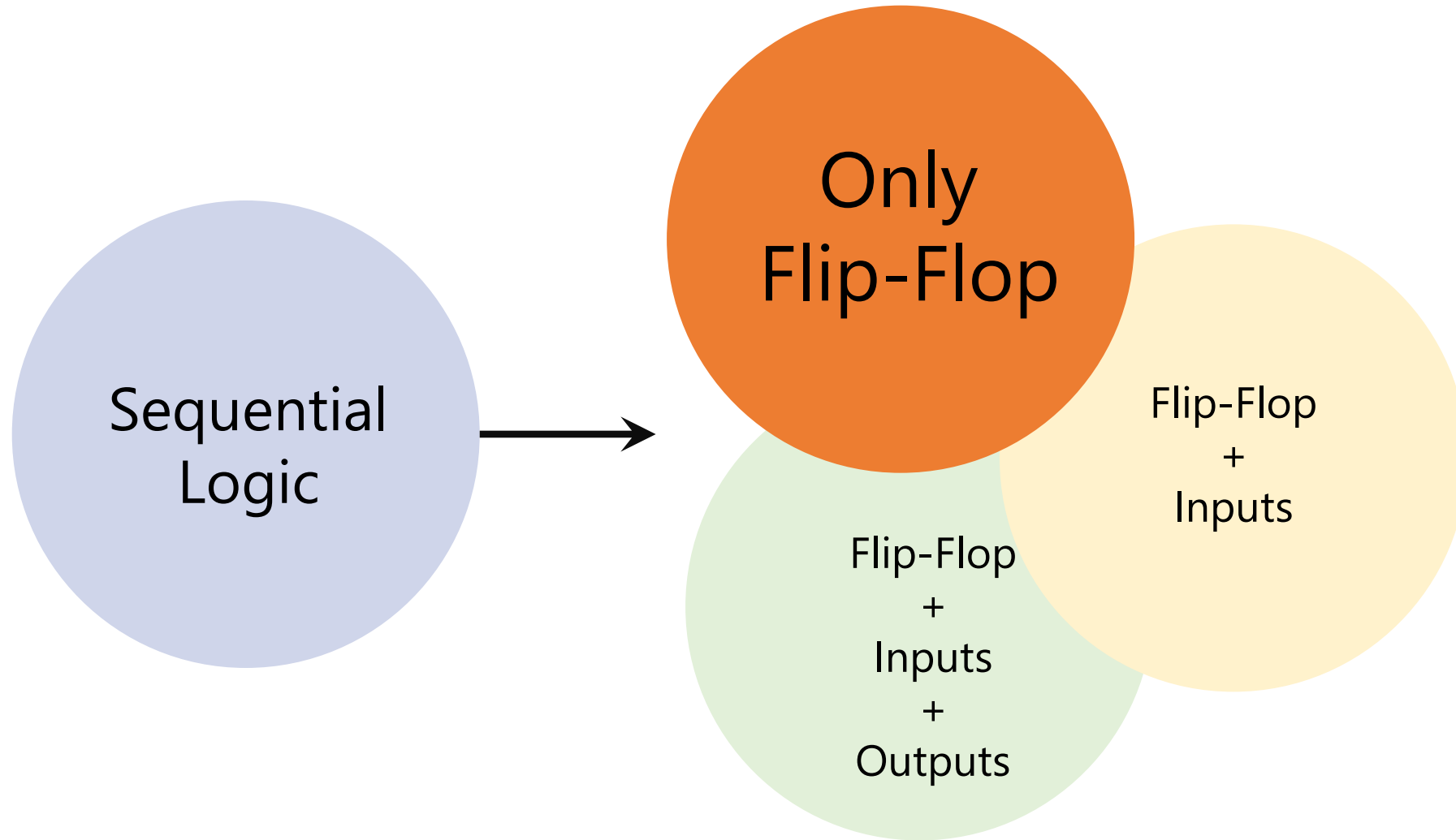
J	K	Q
0	0	Q_t
0	1	0
1	0	1
1	1	Q'_t

Single Edge *Positive*



Analysis: Given a sequential circuit, show the behavior
vs.

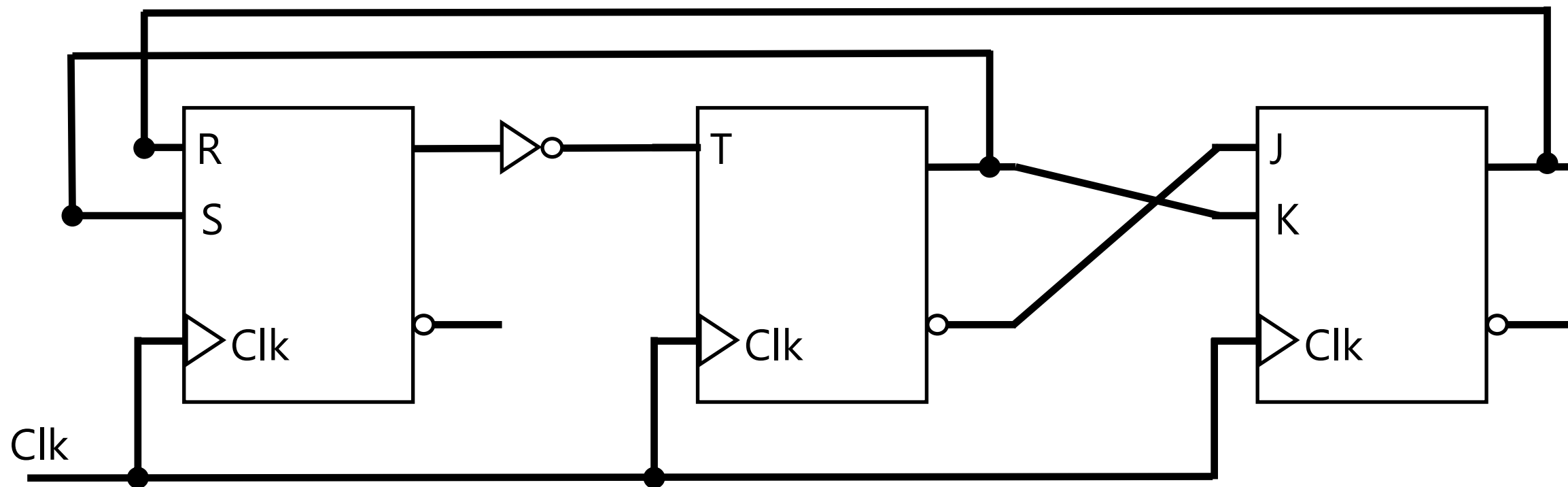
Design: Given a behavior, build the sequential circuit



Analysis (Recap)

0. Is the circuit sequential or combinational? Any FF or feedback → Sequential
 1. What are the flip-flops? RS, D, T, JK, or mixed (e.g., 2 JK, 1 RS, ...)
 2. What are the state combinations? $2^{\#FF}$
 3. Form "State" table:
 - a) Columns: for each FF, two columns:
 - one for current state,
 - one for next state
 - b) Rows: for each state combination
 - In total: $2^{\#FF}$
 4. Fill the state table for next state columns based on:
 - a) the current state
 - b) the inputs to the FFs
 5. Form State Transition Diagram
 6. (Optional) Analyze paths and states in state transition diagram
-

Analysis by an example



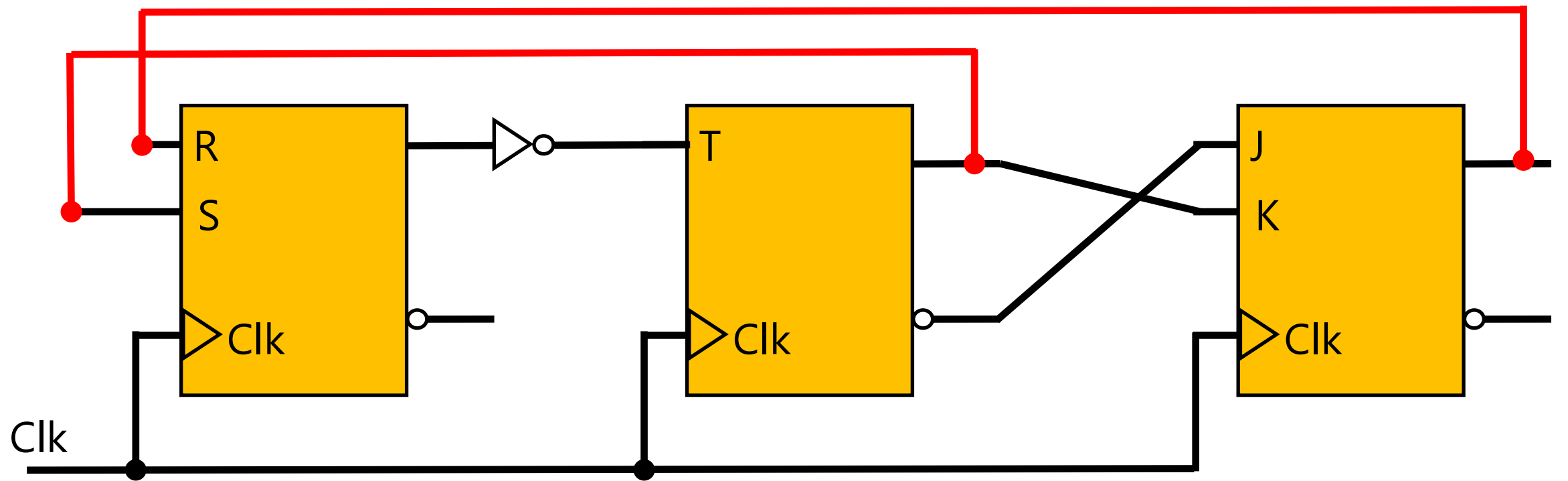
Analysis

0) Is it sequential circuit?

At least one FF $\rightarrow$ Yes

At least one feedback $\rightarrow$ Yes

Otherwise $\rightarrow$ No

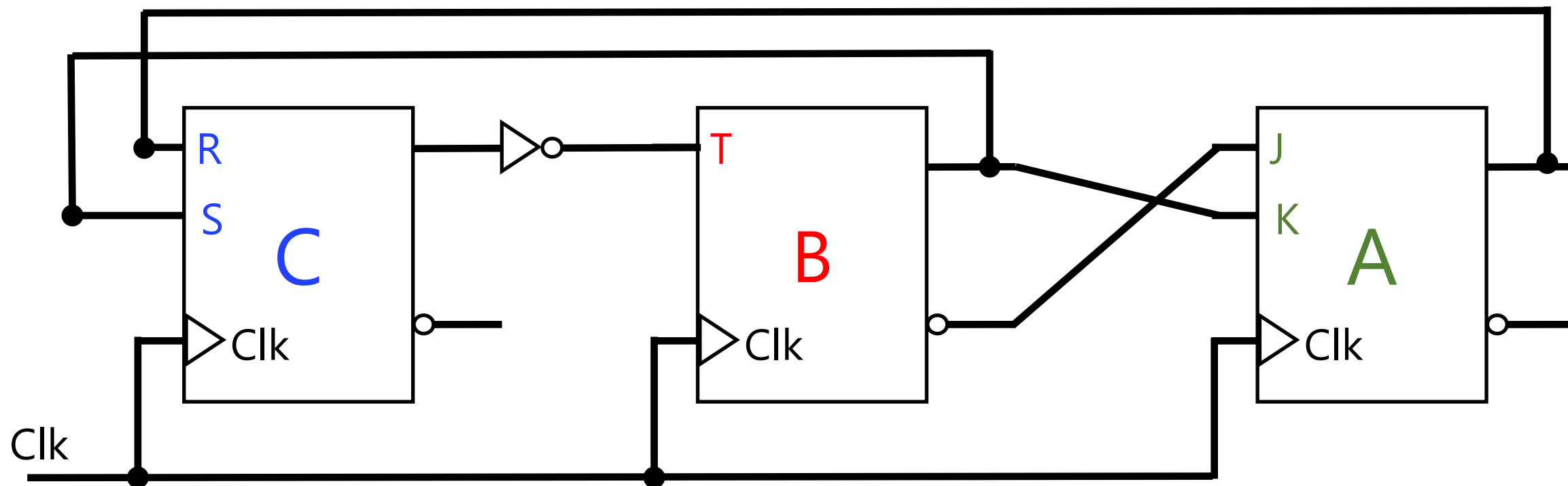


Analysis

1) What are the FFs?

1.1. We pick a name for each FF

1.2. We note the type of FF

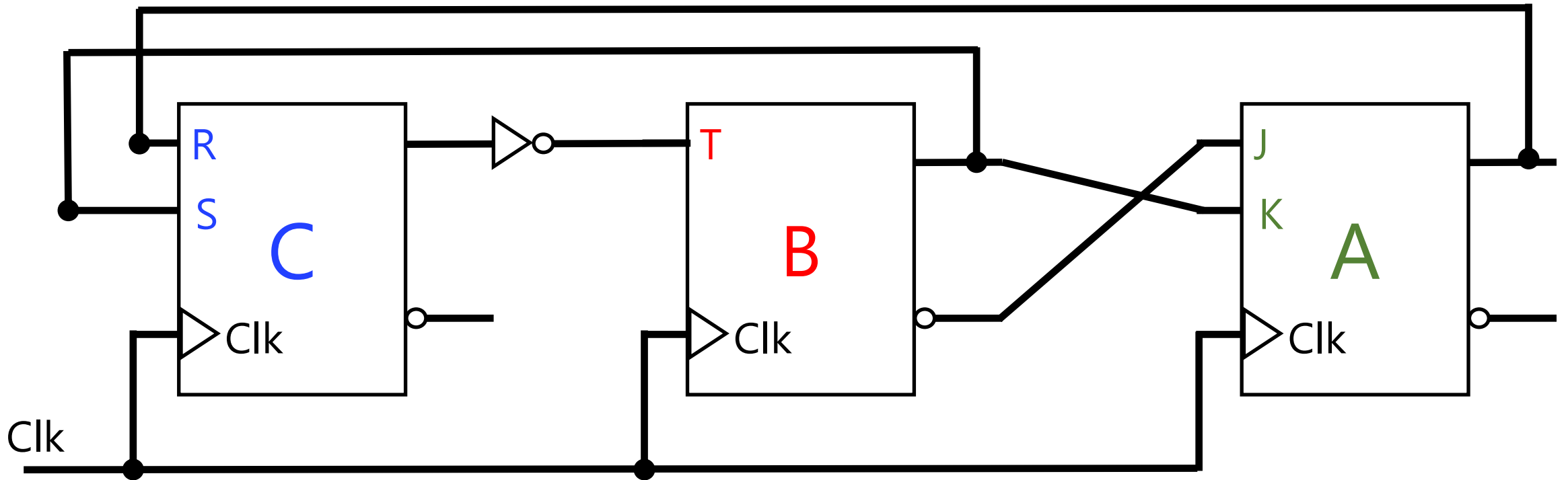


Analysis

2) What are the state combinations
(possibilities)?

Each FF can have $\{0,1\}$ states

In total, $2^{\#FFs}$

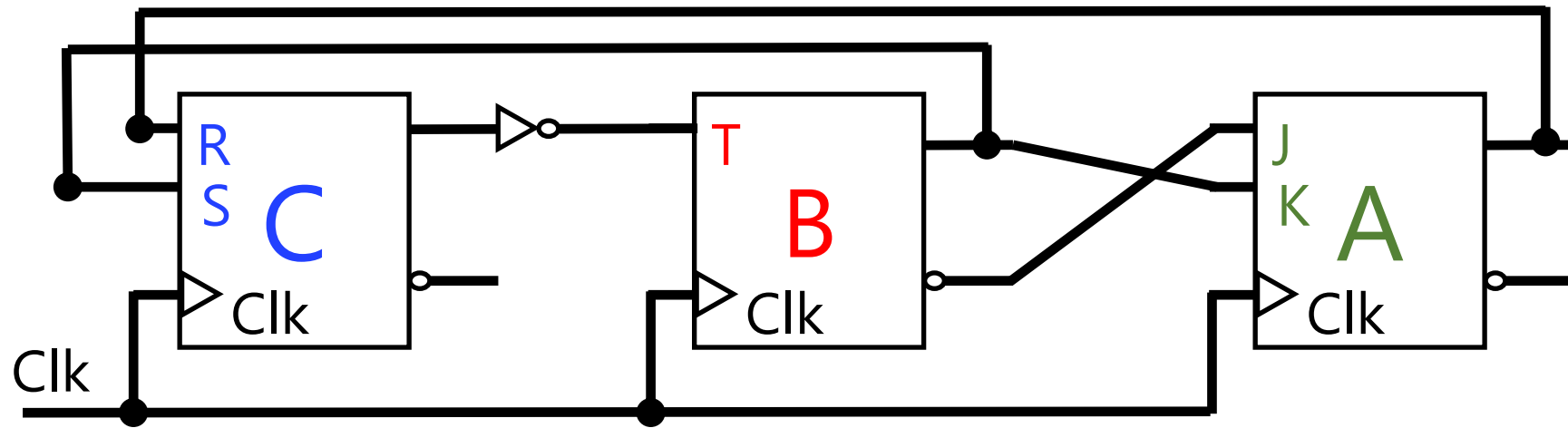


#FFs = 3 $\rightarrow 2^3 = 8$ combinations

Analysis

3) Form a 'State' Table

- 3.1. For each FF, one column for **current** state
- 3.2. For each FF, one column for **next** state
- 3.3. For each combination of current state one row

[illegible]

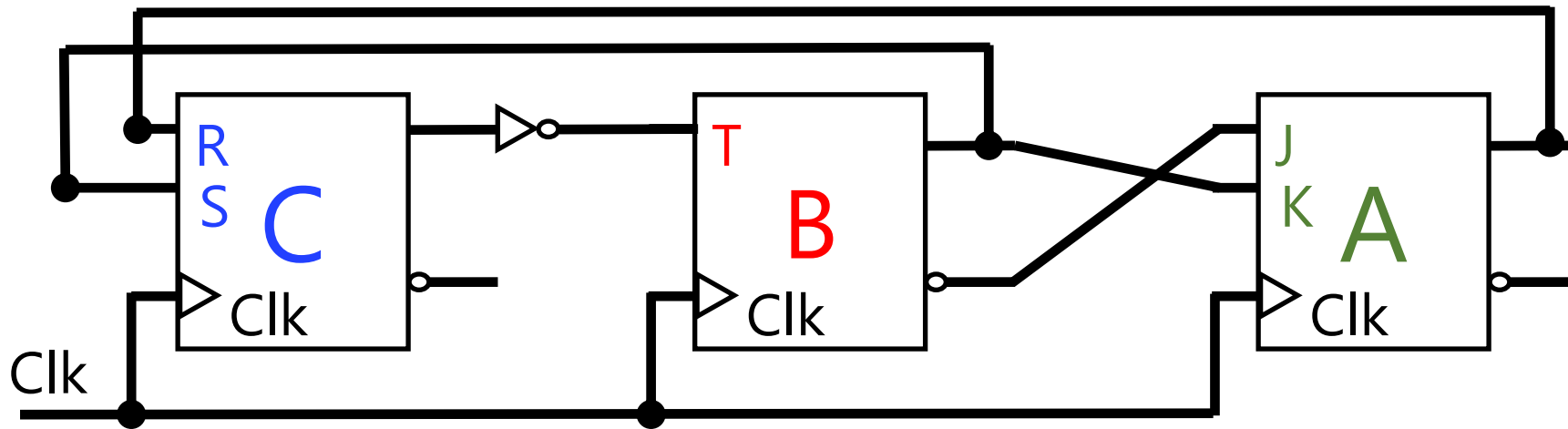
Analysis

4) Fill the 'State' table

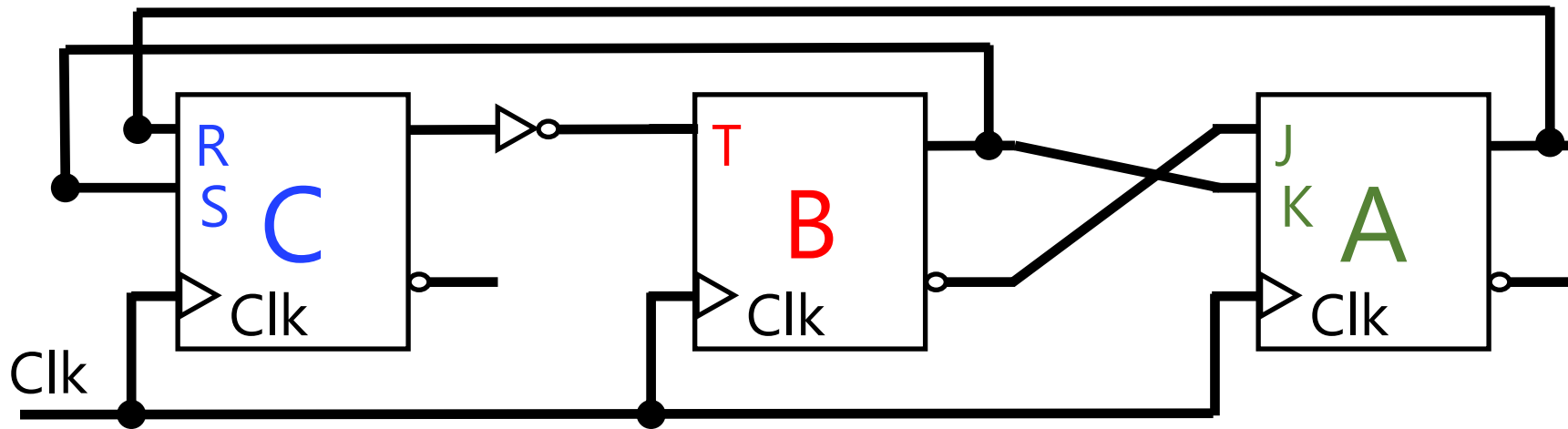
For each FF, we determine the **next** state based on

I) current state

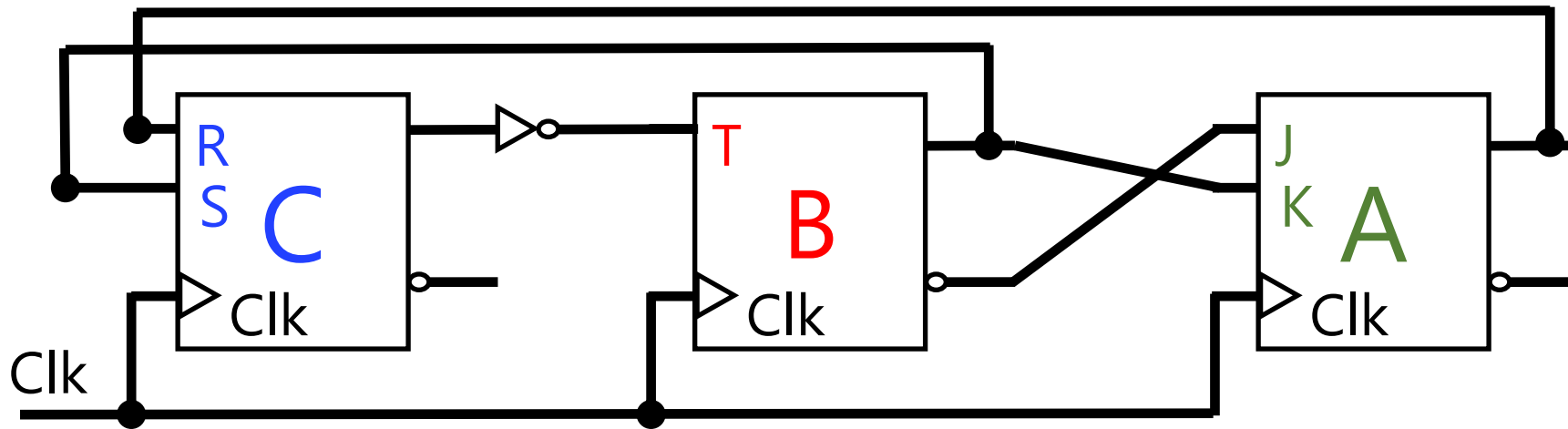
II) the current value of inputs to the FF



Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0			$Q_A(T)=0$ $J_A=Q'_B(T)=1$ $K_A=Q_B(T)=0$ Set Action: 1
0	0	1			
0	1	0			
0	1	1			
1	0	0			



Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0		$Q_B(T)=0$ $T_B=Q'_C(T)=1$ ----- $\text{Comp. } (Q_B(T)) = 1$	1
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			

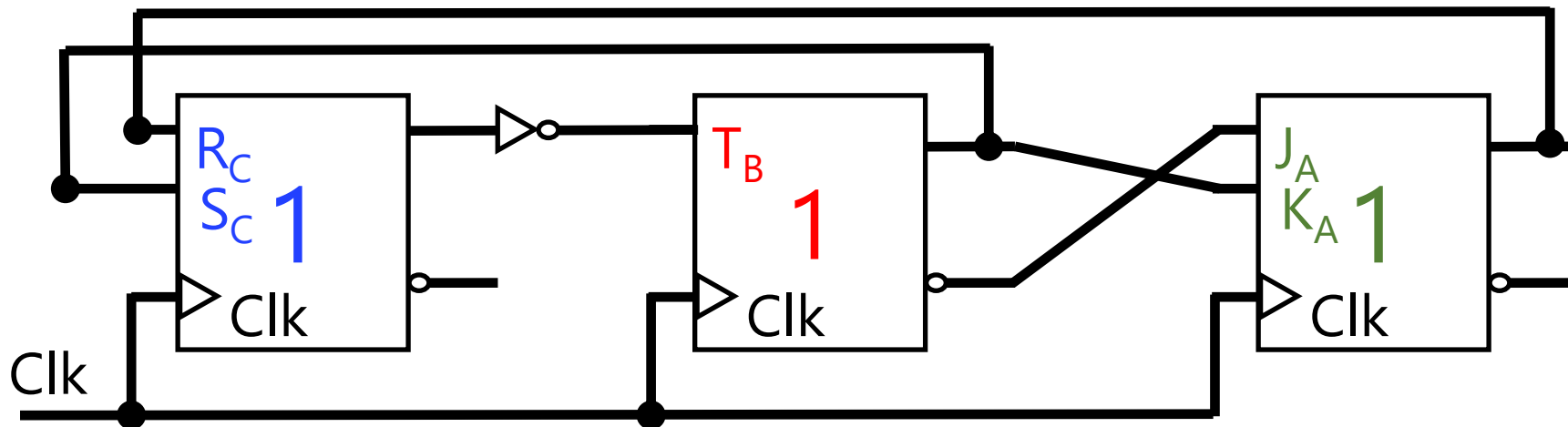


Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	$Q_C(T)=0$ $R_C=Q_A(T)=0$ $S_C=Q_B(T)=0$ ----- Store $Q_C(T) = 0$	1	1
0	0	1			
0	1	0			
0	1	1			
1	0	0			

Analysis

$$Q_A(T) = A, Q'_A(T) = A'$$

For simplicity, the **current status** of a FF can be assume to be as a **binary variable**

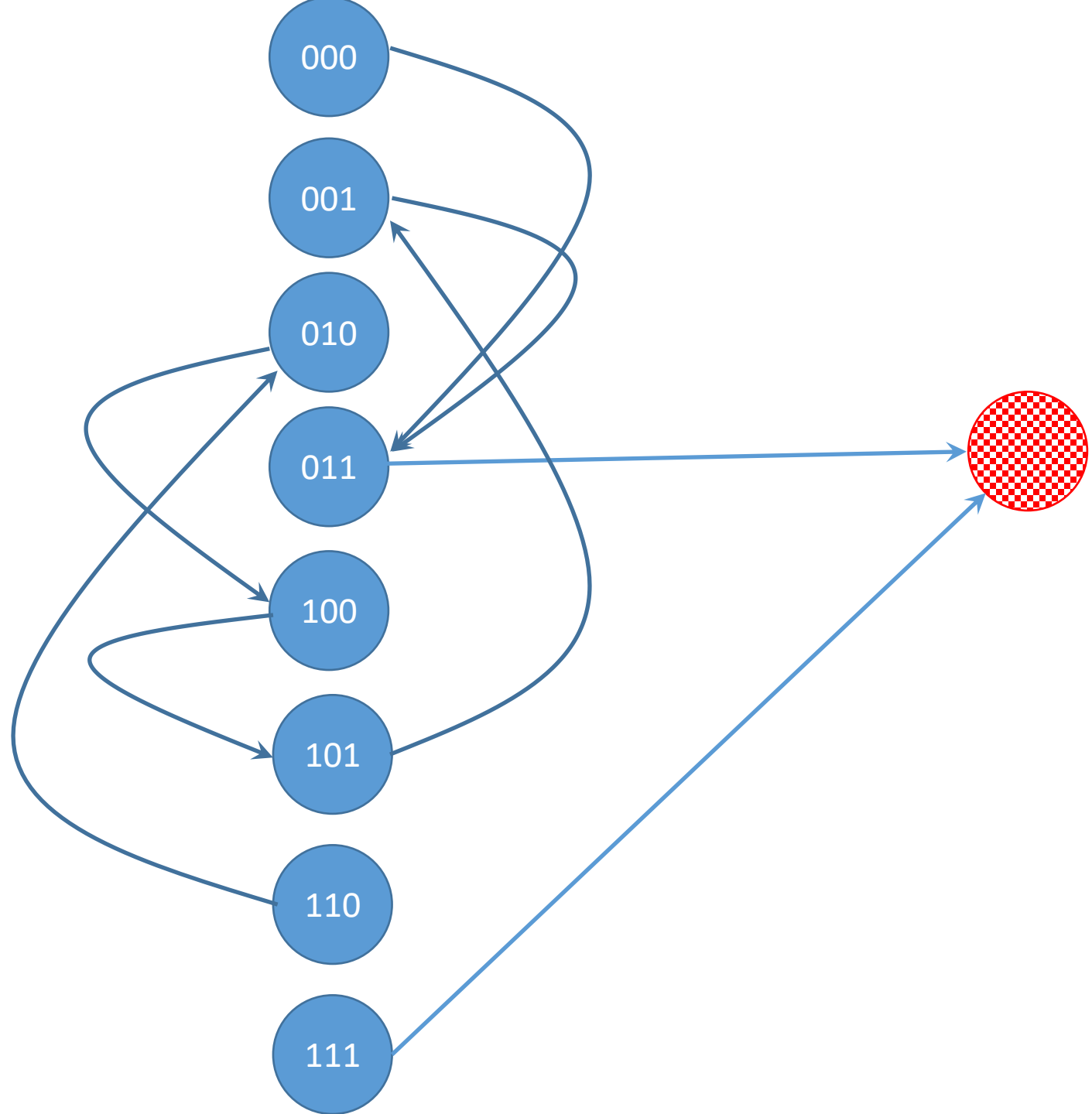


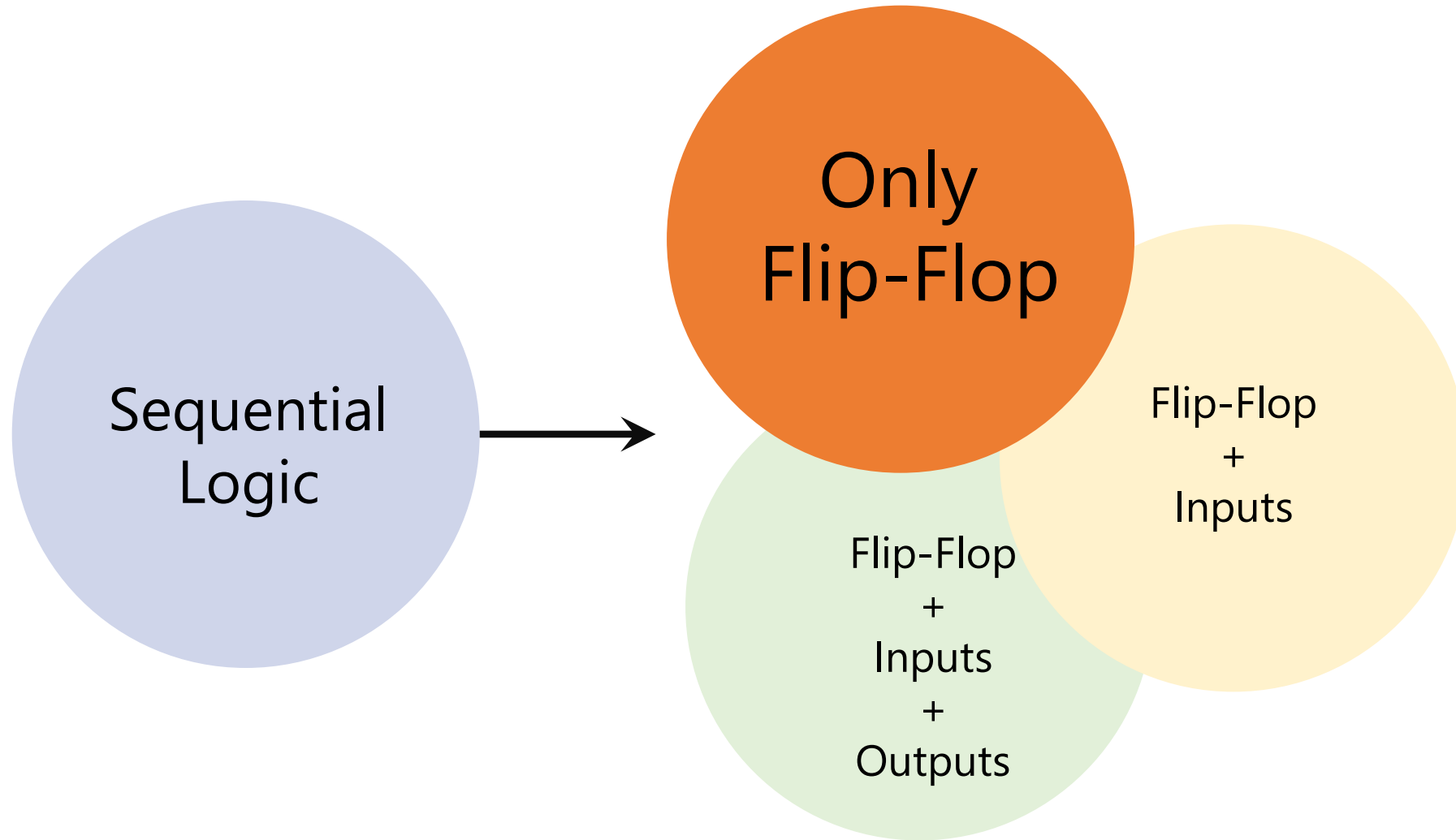
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	1	1
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	×	0	0
1	0	0	1	0	1
1	0	1	0	0	1
1	1	0	0	1	0
1	1	1	×	1	0

Analysis

5) State Transition Diagram

- 5.1. for each state combination (each row), a node
- 5.2. from one state (node) to another state, a directed edge





Design by an example

Counter

Count from 0 to N

Design

0. Do we need combinational logic or sequential logic?

Do we need memory?

Counter

Count from 0 to N

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow \dots \rightarrow N-1 \rightarrow N$

Counter

Count from 0 to N

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow \dots \rightarrow N-1 \rightarrow N$

At each step, we have to see at number we are and then move to next number: $i \rightarrow i+1$

Counter

Count from 0 to N

We need a storage to store current number.

We need a sequential circuit!

Design

1. How many storage (flip-flops)?

Depends on the **storage** you need to store
the **current state** in **binary system**!

Counter

Count from 0 to N

$N = 7$

$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7$

$000 \rightarrow 001 \rightarrow 010 \rightarrow 011 \rightarrow 100 \rightarrow 101 \rightarrow 110 \rightarrow 111$

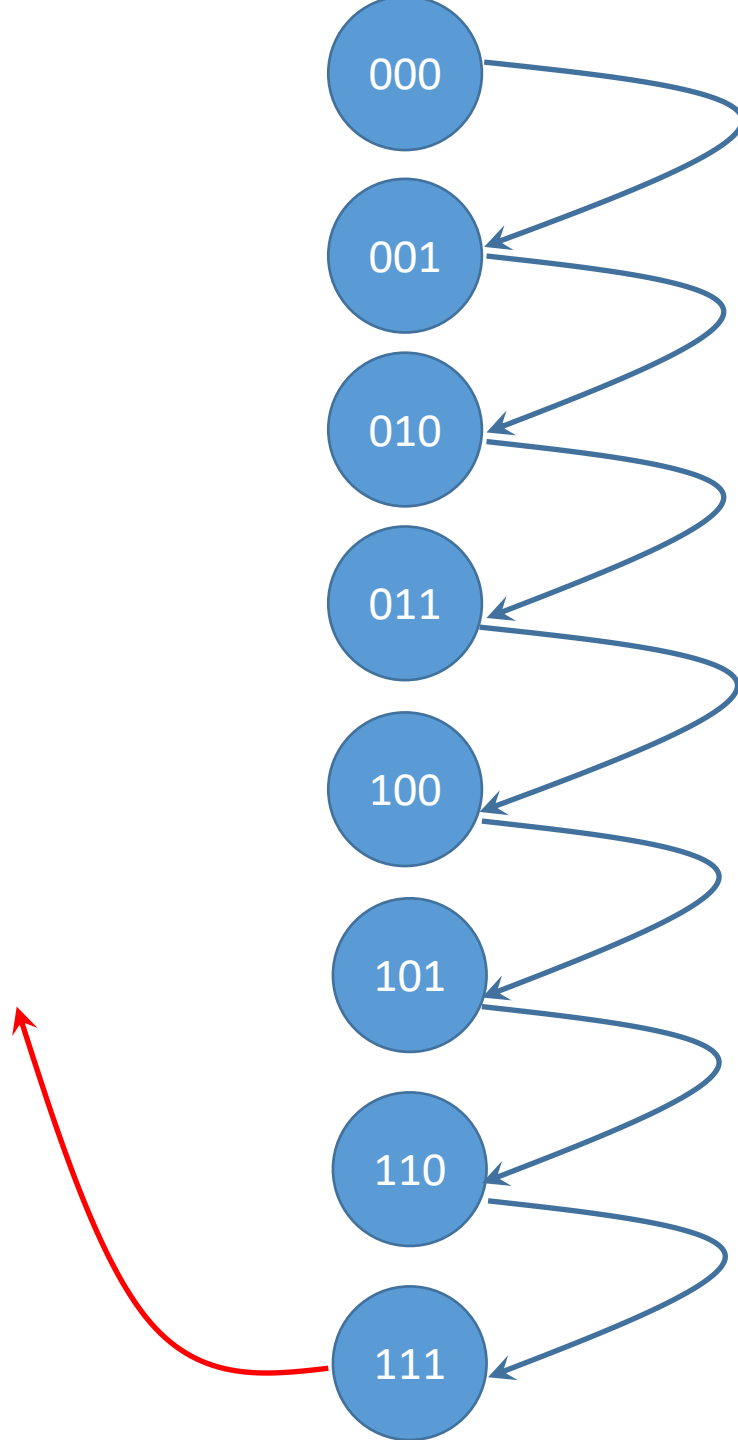
For each intermediate state, we need 3 bits $\rightarrow$ 3 flip-flops

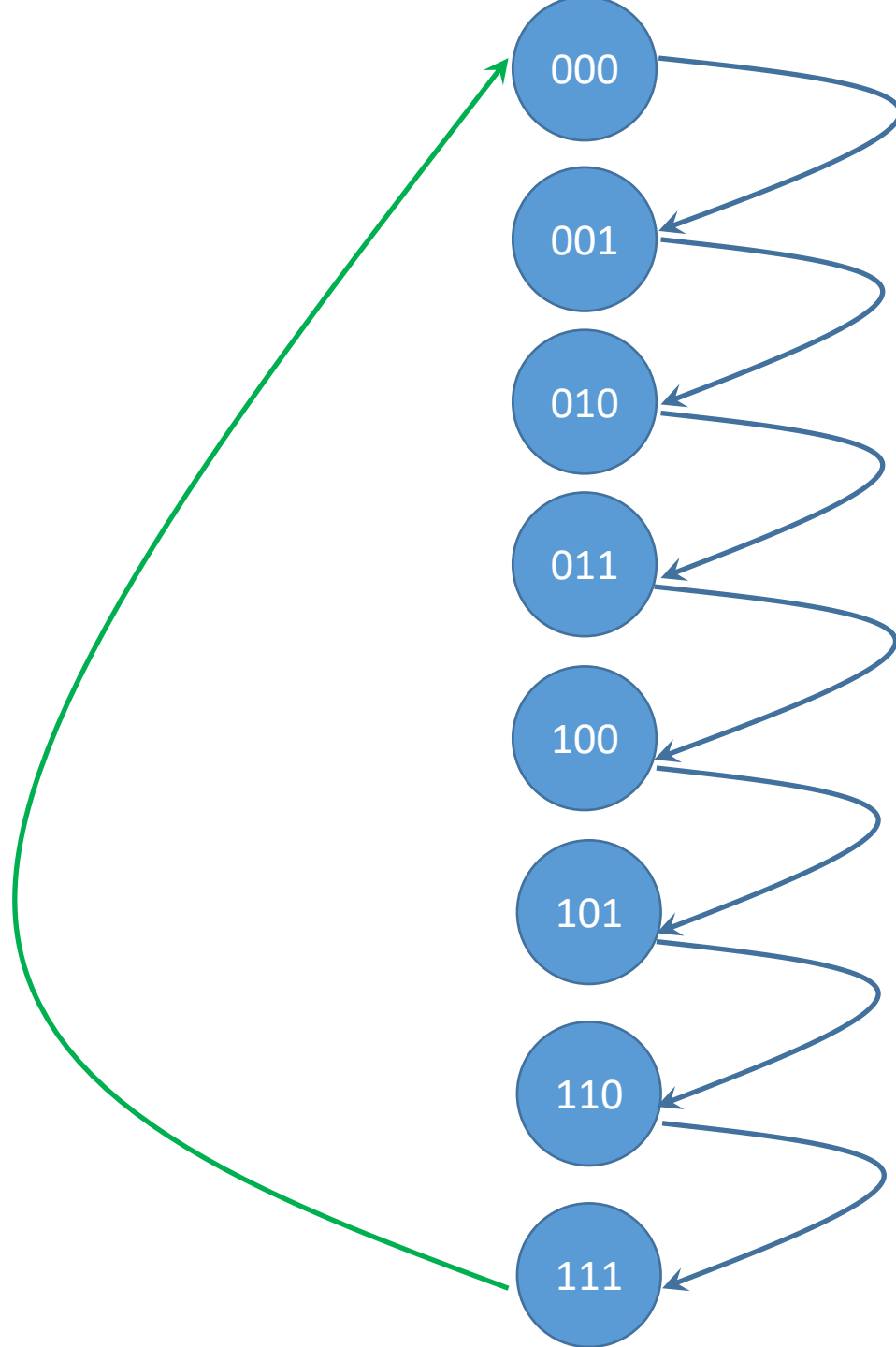
Design

2. Form the state (transition) diagram

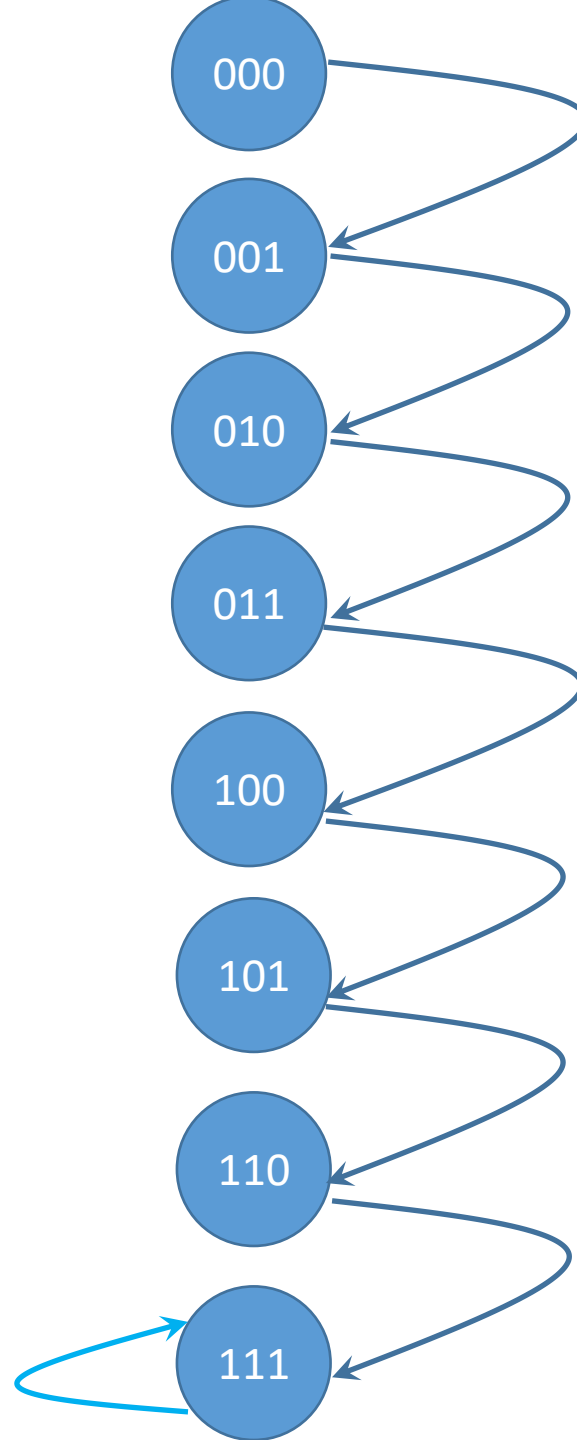
Same as analysis,

- For each state $\rightarrow$ one node
- For each state transition to next state $\rightarrow$ a directed edge





Loop to the
beginning!



Stuck in 7
Just one time counter!

Design

3. Form the state table

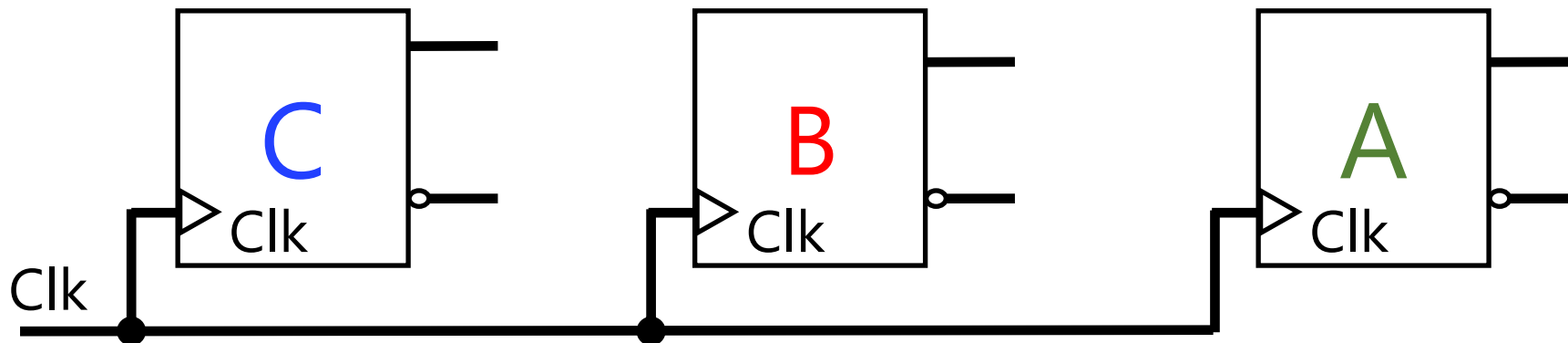
Same as analysis, **two columns** for **each flip-flop** (storage unit)

- a) One for current state $Q(T)$
- b) One for next state $Q(T+1)$

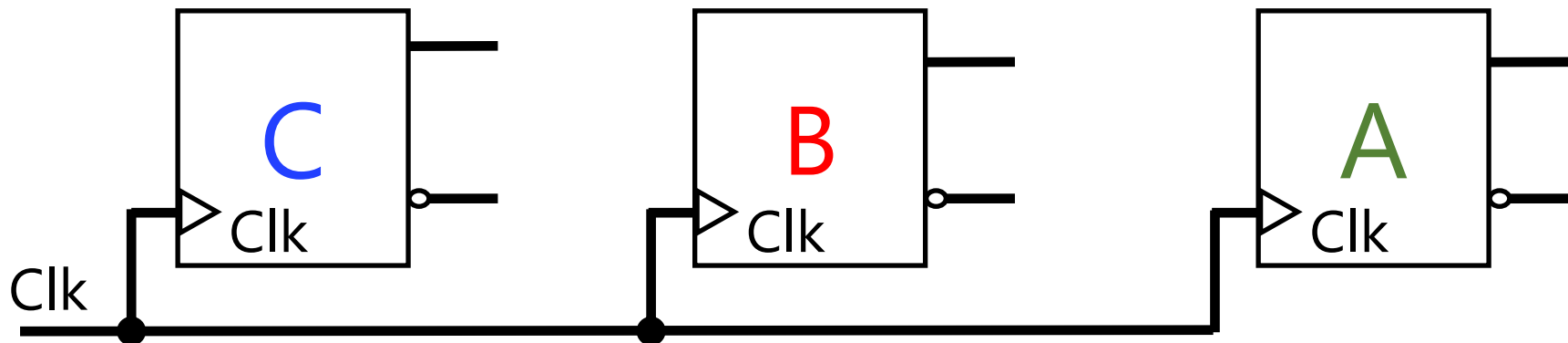
Design

4. Fill the state table

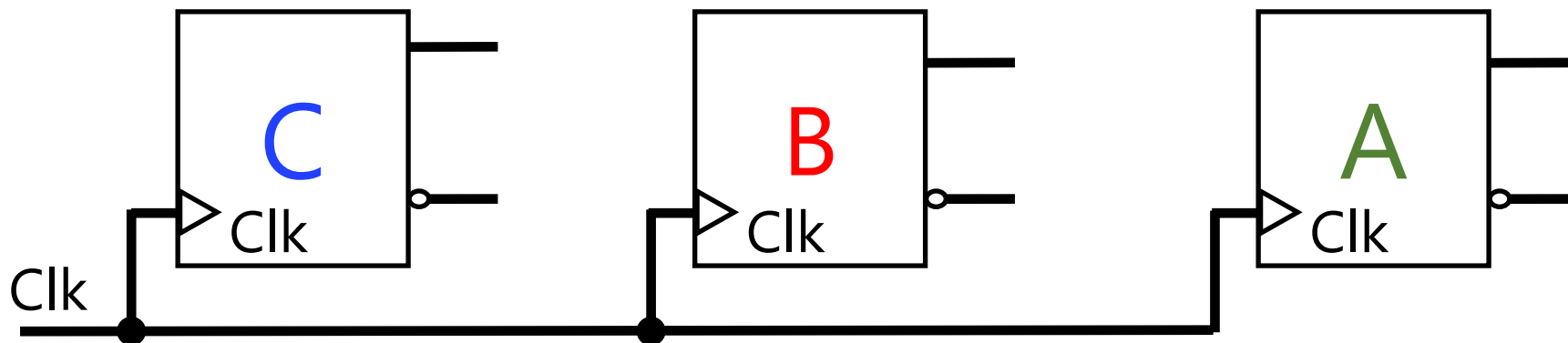
Unlike analysis, here we already know what is going to be the next state $Q(T+1)$ based on current state $Q(T)$



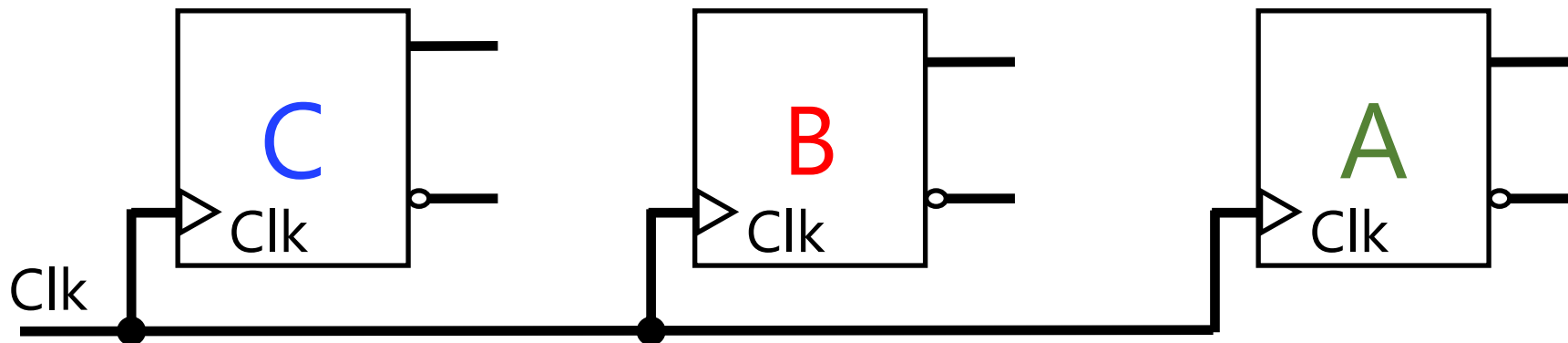
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			



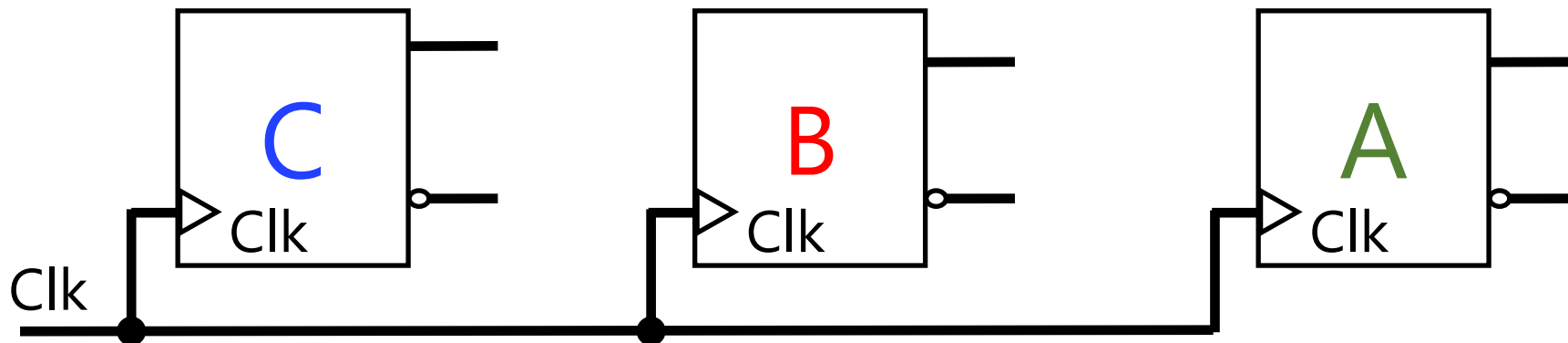
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			



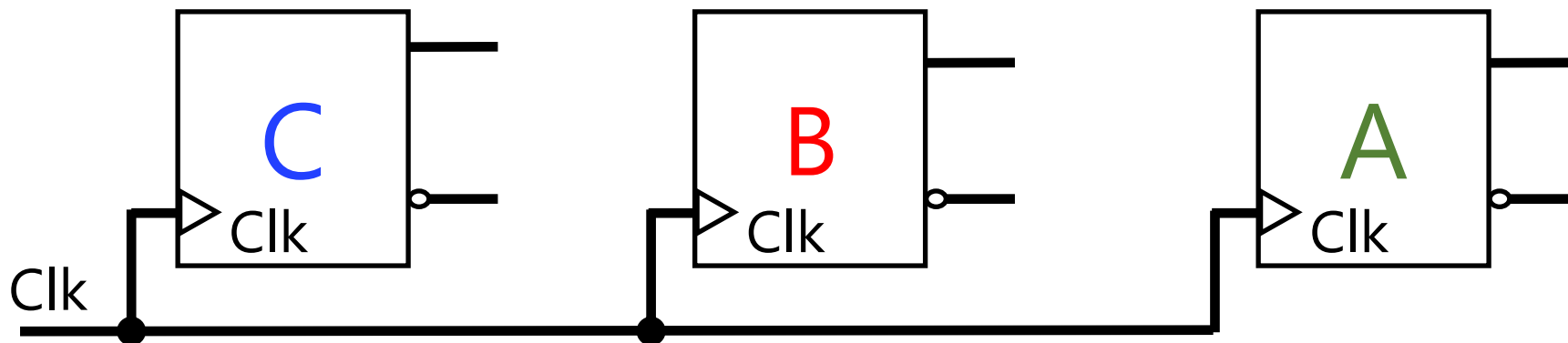
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			



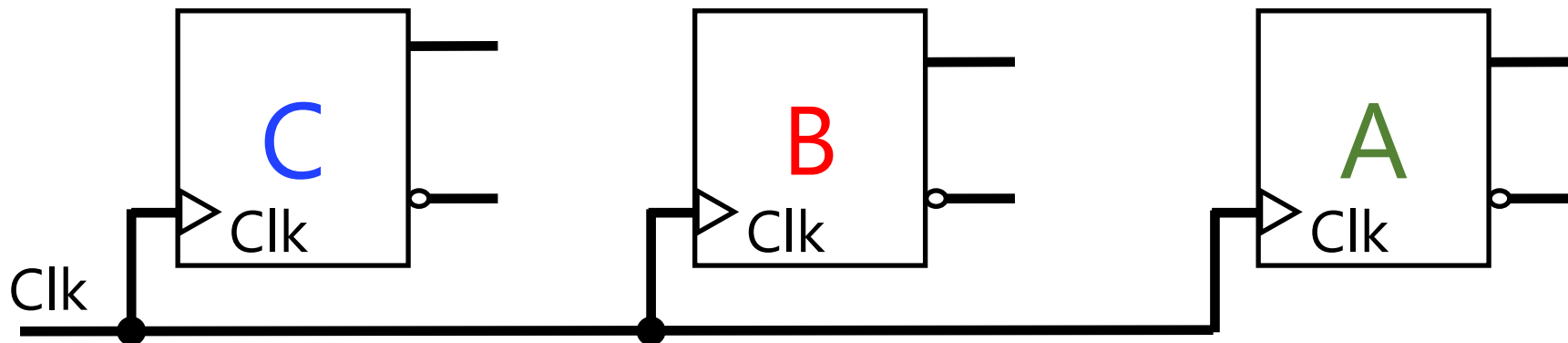
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0			
1	0	1			
1	1	0			
1	1	1			



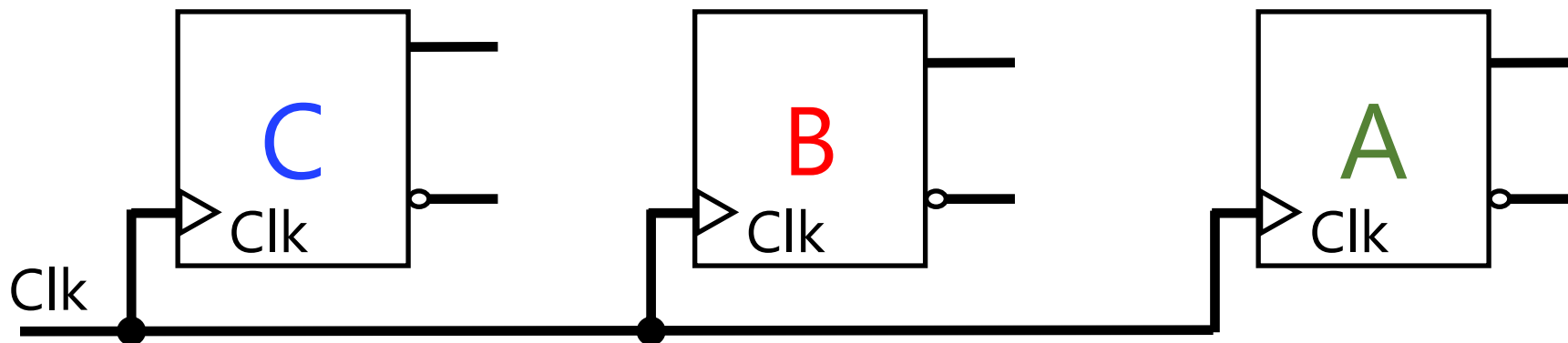
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1			
1	1	0			
1	1	1			



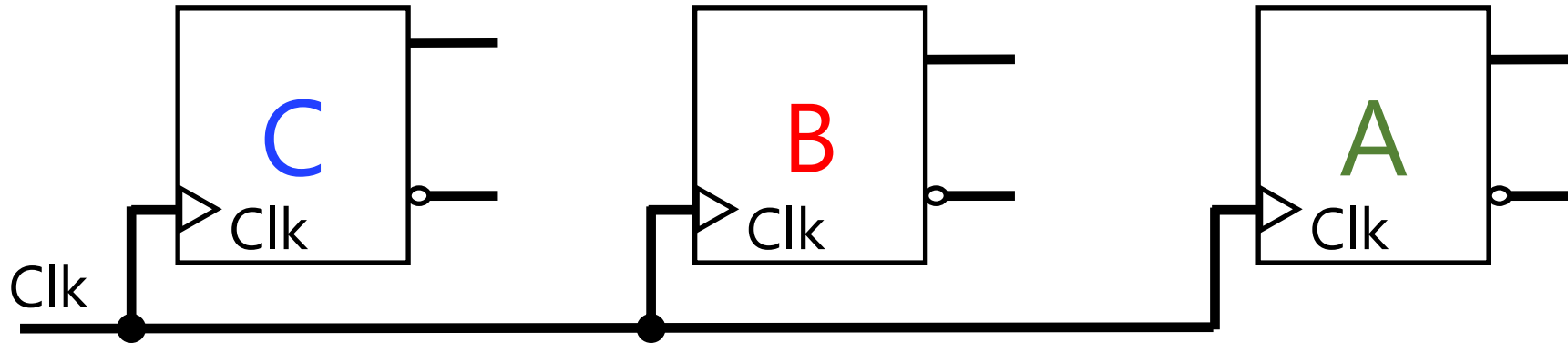
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0			
1	1	1			



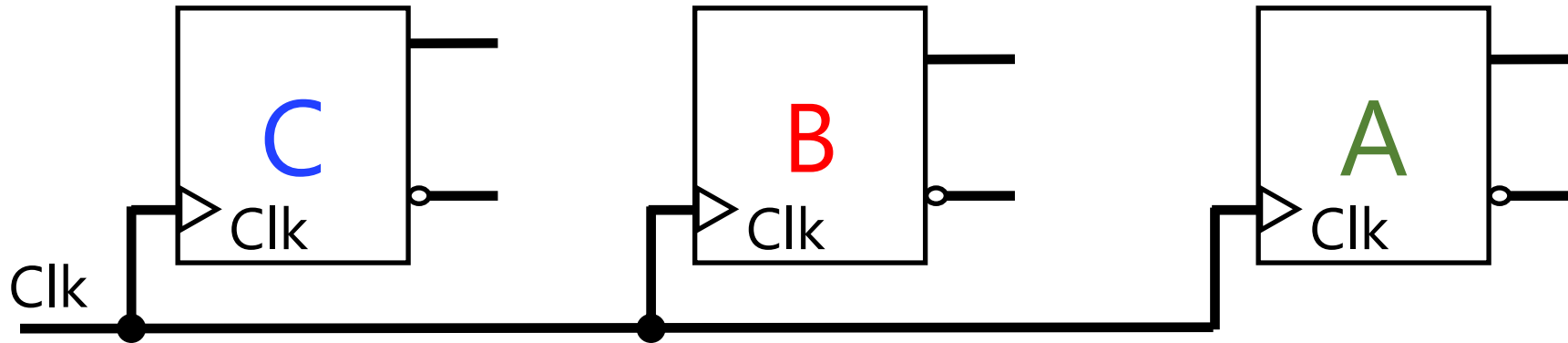
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1			



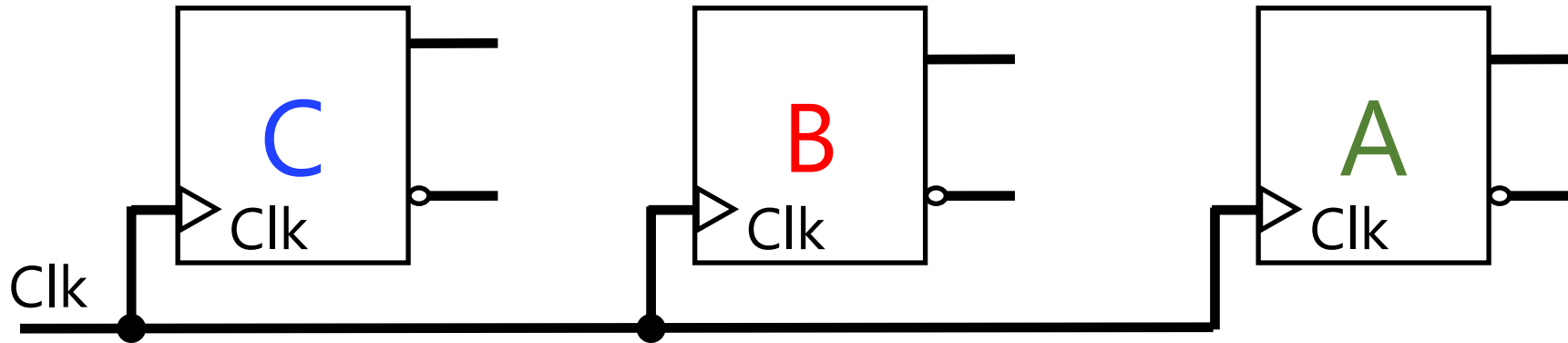
Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	?	?	?



Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	Loop to the beginning!	1
1	0	1	1		0
1	1	0	1		1
1	1	1	0	0	0



Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	Stuck in 7 Just one time counter!	1
1	0	1	1		0
1	1	0	1		1
1	1	1	1		1



Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	Our Design Choice!	1
1	0	1	1		0
1	1	0	1		1
1	1	1	0	0	0

Design

5. What type of storage (flip-flop)?

RS, D, T, JK, or Mixed

Design

5. What type of storage (flip-flop)?

RS, D, T, JK, or Mixed

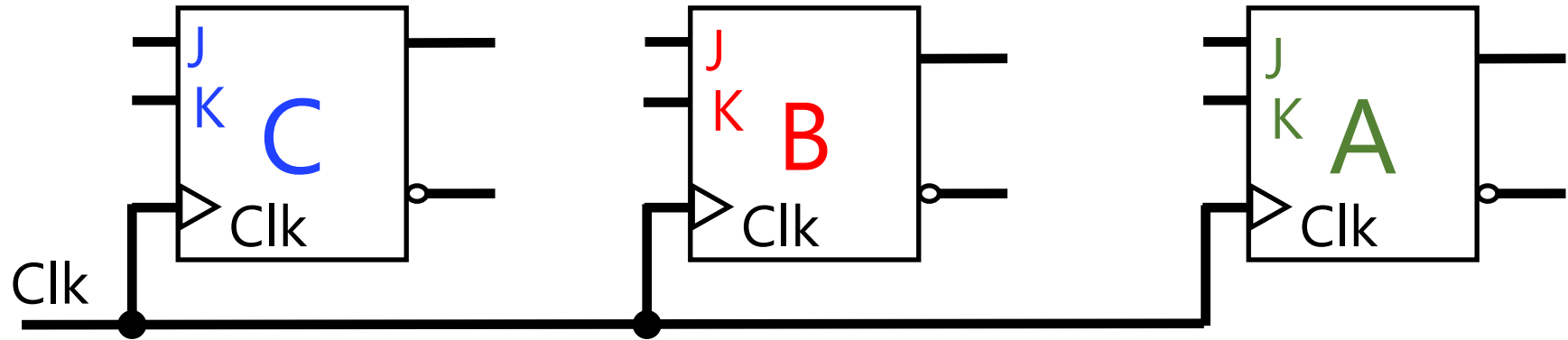
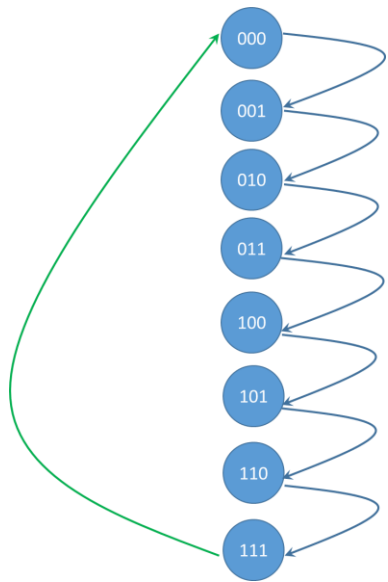
In terms of design, does not matter.

In terms of **efficiency**, matters!

Counter

Count from 0 to $N=7$

Let's select **JK**, the complete FF.



Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0

Design

6. Boolean expression for the flip-flops' input?
input equations, aka, *excitation* equations

Counter

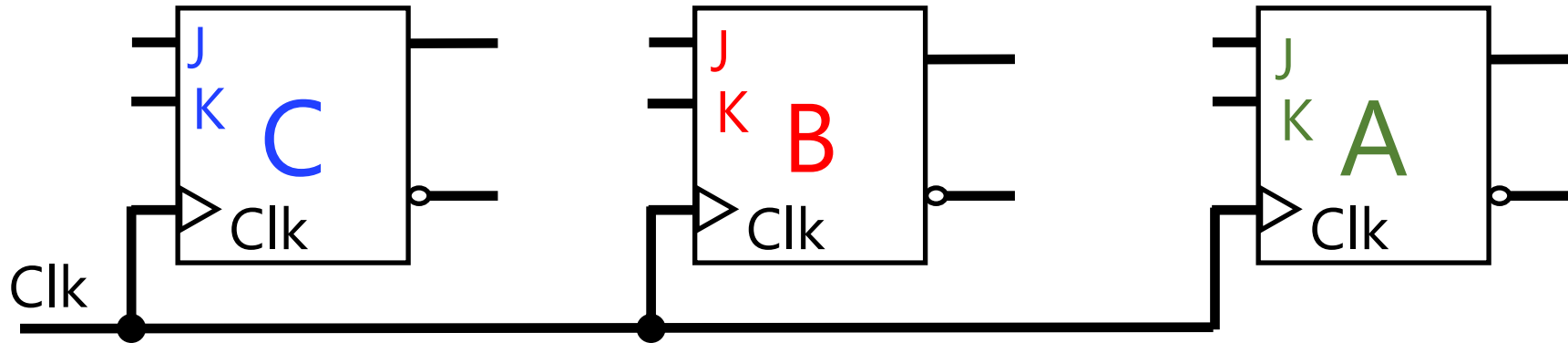
Count from 0 to $N=7$

A	$J_A =$	$K_A =$
B	$J_B =$	$K_B =$
C	$J_C =$	$K_C =$

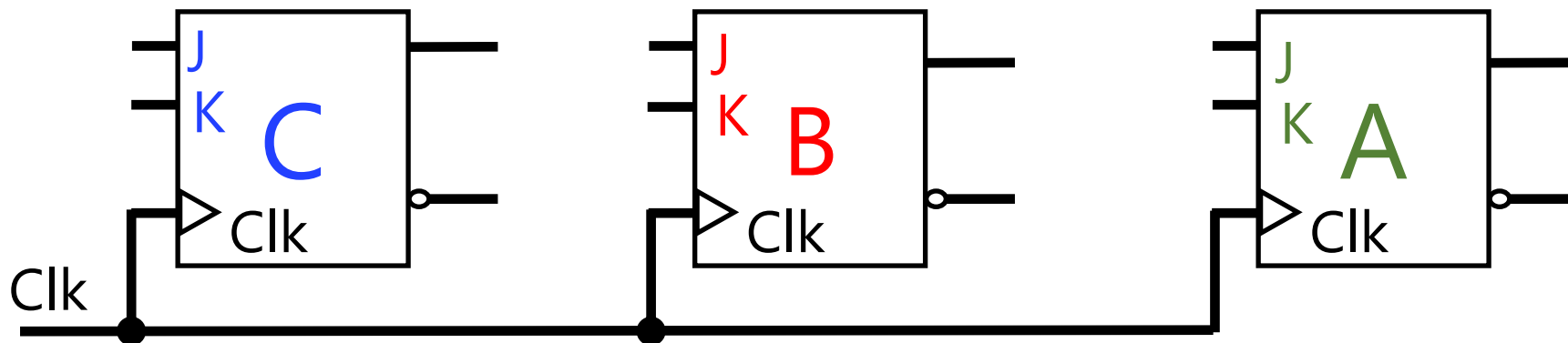
Counter

Count from 0 to $N=7$

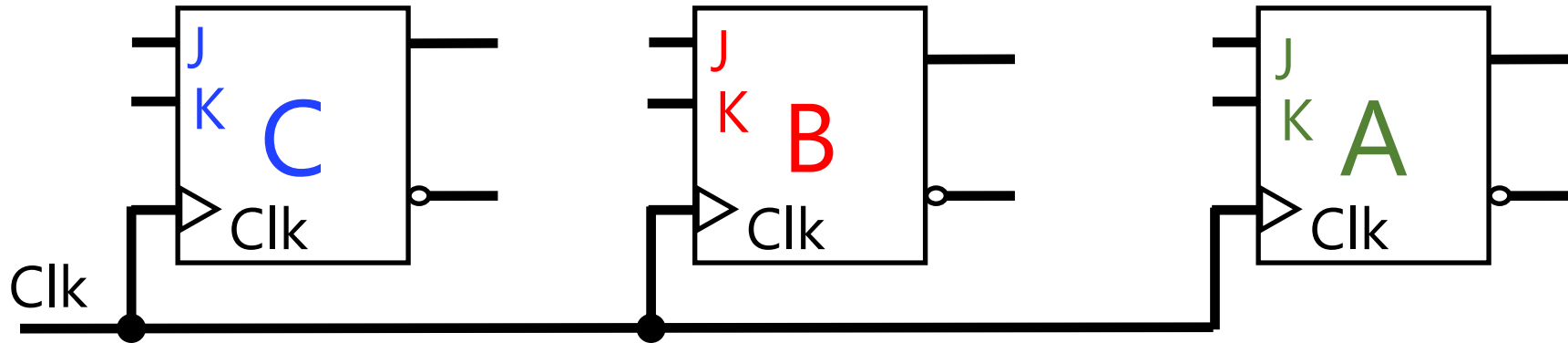
A	$J_A =$	$K_A =$
B	$J_B =$	$K_B =$
C	$J_C =$	$K_C =$



Q(T)			Q(T+1)		
C	B	A	C	B	A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0

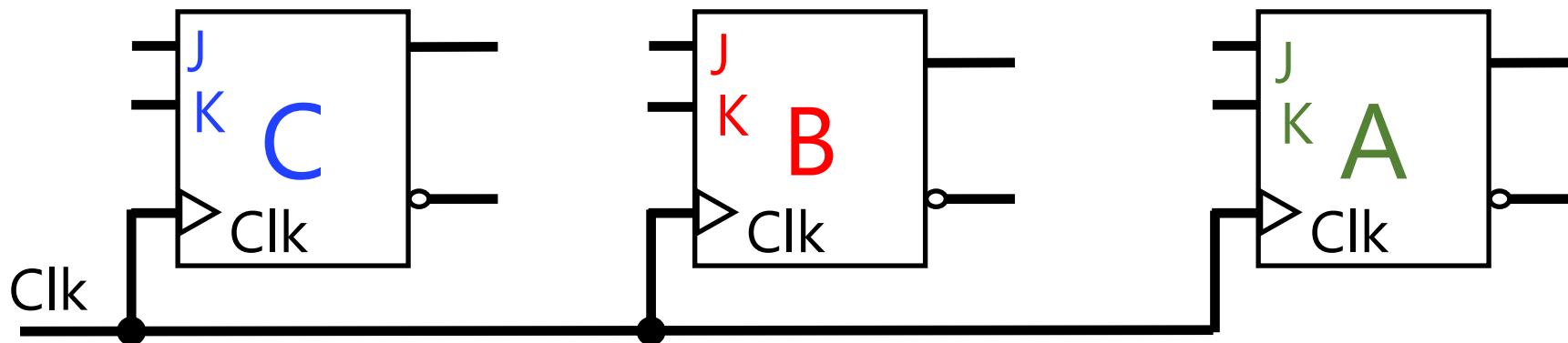


Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J_A	K_A
0	0	0	0	0	1			
0	0	1	0	1	0			
0	1	0	0	1	1			
0	1	1	1	0	0			
1	0	0	1	0	1			
1	0	1	1	1	0			
1	1	0	1	1	1			
1	1	1	0	0	0			

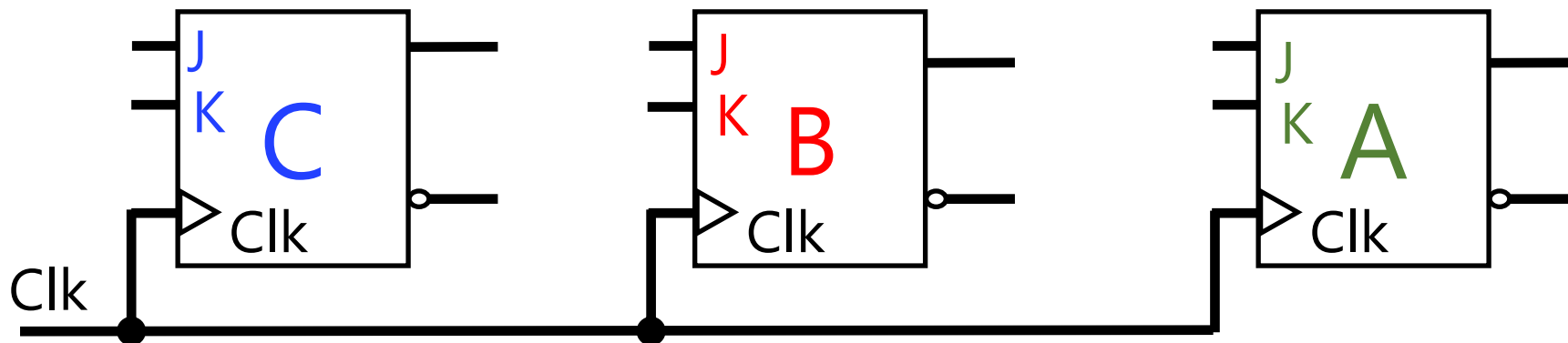


Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J_A	K_A
0	0	0	0	0	1	Set	1	0
0	0	1	0	1	0			
0	1	0	0	1	1			
0	1	1	1	0	0			
1	0	0	1	0	1			
1	0	1	1	1	0			
1	1	0	1	1	1			
1	1	1	0	0	0			

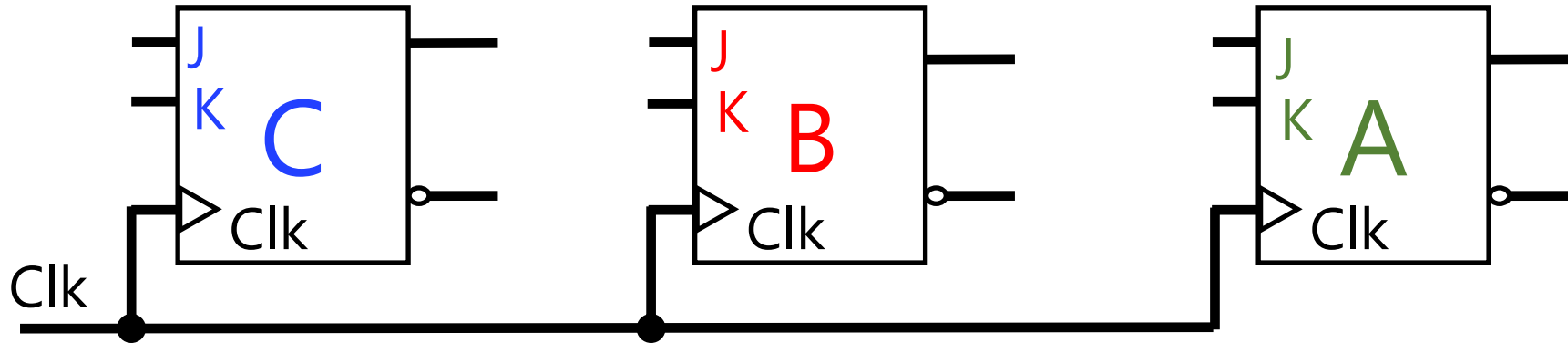
OR



Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J_A	K_A
0	0	0	0	0	1	Comp	1	1
0	0	1	0	1	0			
0	1	0	0	1	1			
0	1	1	1	0	0			
1	0	0	1	0	1			
1	0	1	1	1	0			
1	1	0	1	1	1			
1	1	1	0	0	0			

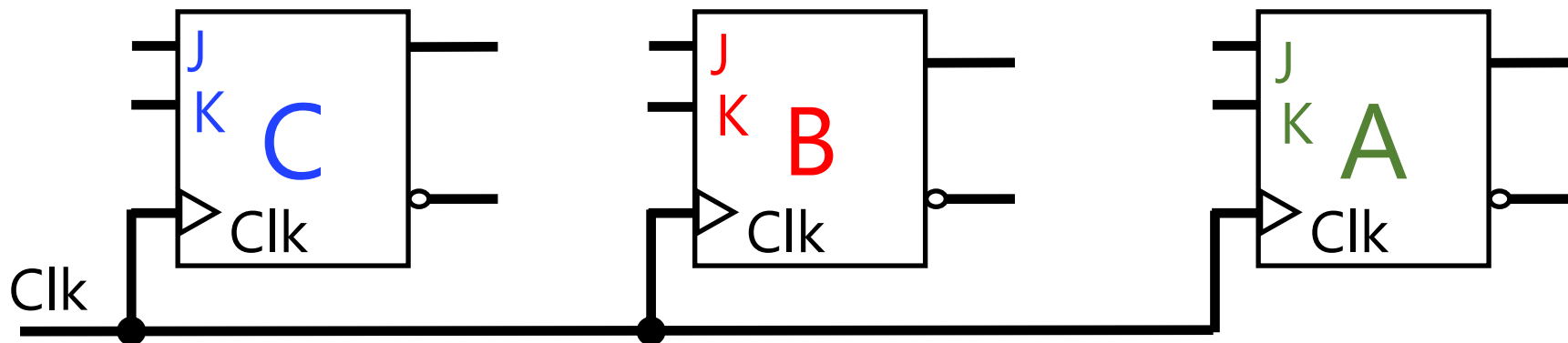


Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J_A	K_A
0	0	0	0	0	1	Set/Comp	1	$0/1 \rightarrow \times$
0	0	1	0	1	0			
0	1	0	0	1	1			
0	1	1	1	0	0			
1	0	0	1	0	1			
1	0	1	1	1	0			
1	1	0	1	1	1			
1	1	1	0	0	0			

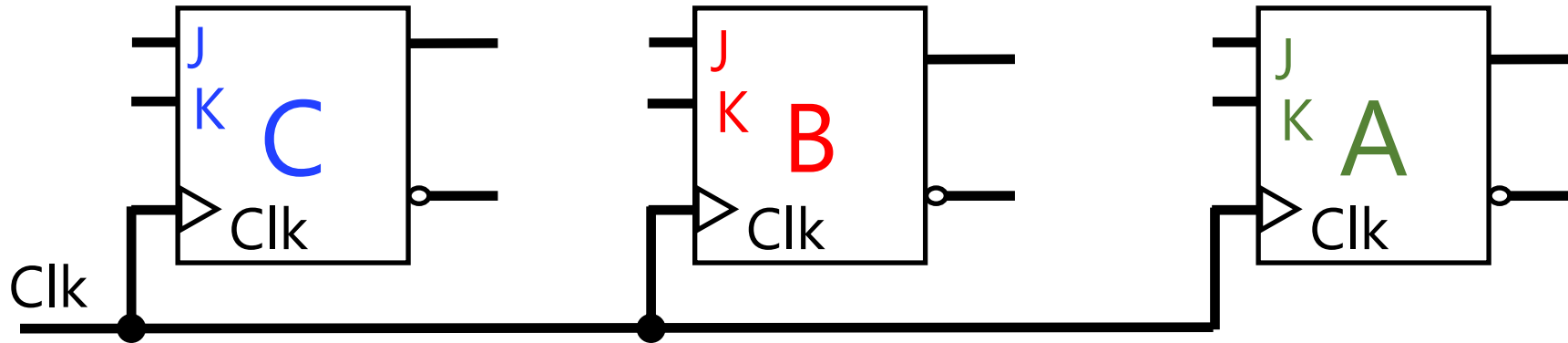


Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J_A	K_A
0	0	0	0	0	1	Set/Comp	1	$\times$
0	0	1	0	1	0	Reset	0	1
0	1	0	0	1	1			
0	1	1	1	0	0			
1	0	0	1	0	1			
1	0	1	1	1	0			
1	1	0	1	1	1			
1	1	1	0	0	0			

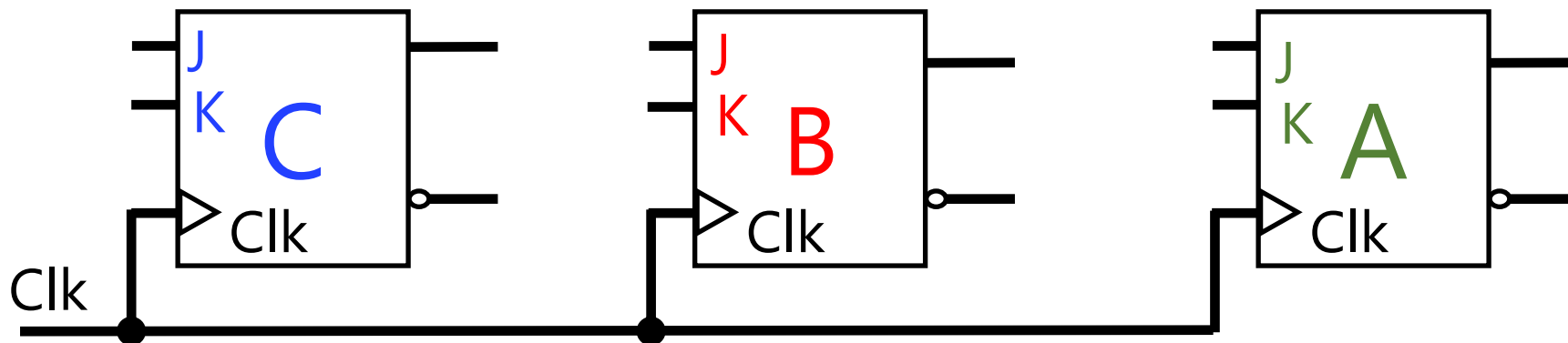
OR



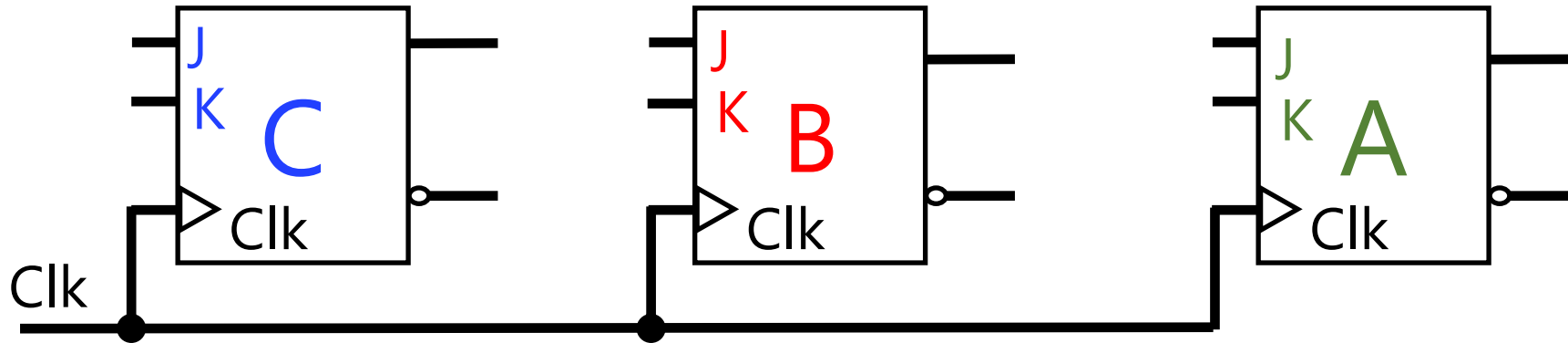
Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J_A	K_A
0	0	0	0	0	1	Set/Comp	1	$\times$
0	0	1	0	1	0	Comp.	1	1
0	1	0	0	1	1			
0	1	1	1	0	0			
1	0	0	1	0	1			
1	0	1	1	1	0			
1	1	0	1	1	1			
1	1	1	0	0	0			



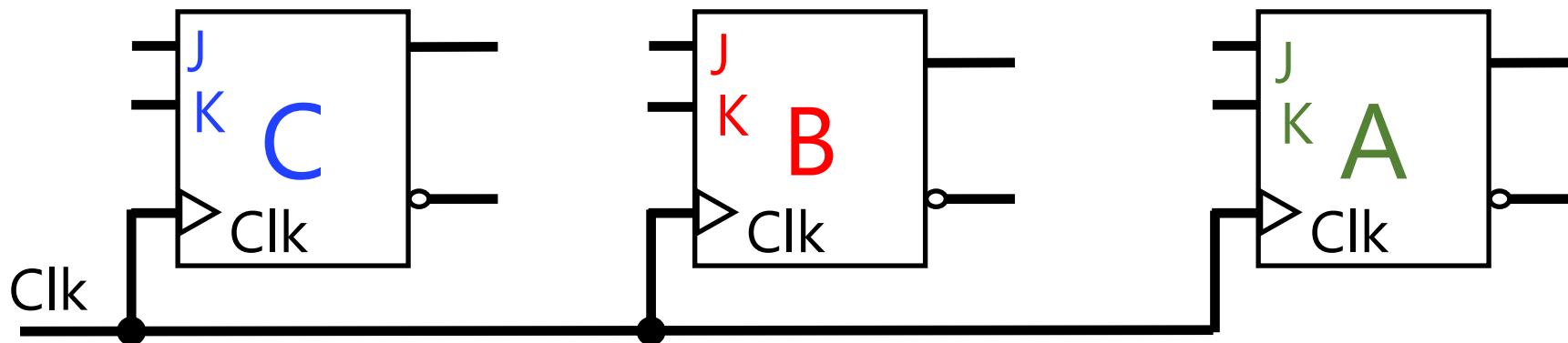
Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J_A	K_A
0	0	0	0	0	1	Set/Comp	1	$\times$
0	0	1	0	1	0	Reset/Comp	$\times$	1
0	1	0	0	1	1			
0	1	1	1	0	0			
1	0	0	1	0	1			
1	0	1	1	1	0			
1	1	0	1	1	1			
1	1	1	0	0	0			



Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J_A	K_A
0	0	0	0	0	1	Set/Comp	1	$\times$
0	0	1	0	1	0	Reset/Comp	$\times$	1
0	1	0	0	1	1	Set/Comp	1	$\times$
0	1	1	1	0	0			
1	0	0	1	0	1			
1	0	1	1	1	0			
1	1	0	1	1	1			
1	1	1	0	0	0			



Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J_A	K_A
0	0	0	0	0	1	Set/Comp	1	$\times$
0	0	1	0	1	0	Reset/Comp	$\times$	1
0	1	0	0	1	1	Set/Comp	1	$\times$
0	1	1	1	0	0	Reset/Comp	$\times$	1
1	0	0	1	0	1			
1	0	1	1	1	0			
1	1	0	1	1	1			
1	1	1	0	0	0			

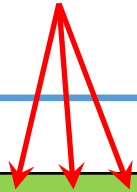


Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J _A	K _A
0	0	0	0	0	1	Set/Comp	1	×
0	0	1	0	1	0	Reset/Comp	×	1
0	1	0	0	1	1	Set/Comp	1	×
0	1	1	1	0	0	Reset/Comp	×	1
1	0	0	1	0	1	Set/Comp	1	×
1	0	1	1	1	0	Reset/Comp	×	1
1	1	0	1	1	1	Set/Comp	1	×
1	1	1	0	0	0	Reset/Comp	×	1

Counter

Count from 0 to N=7

Flip-Flops

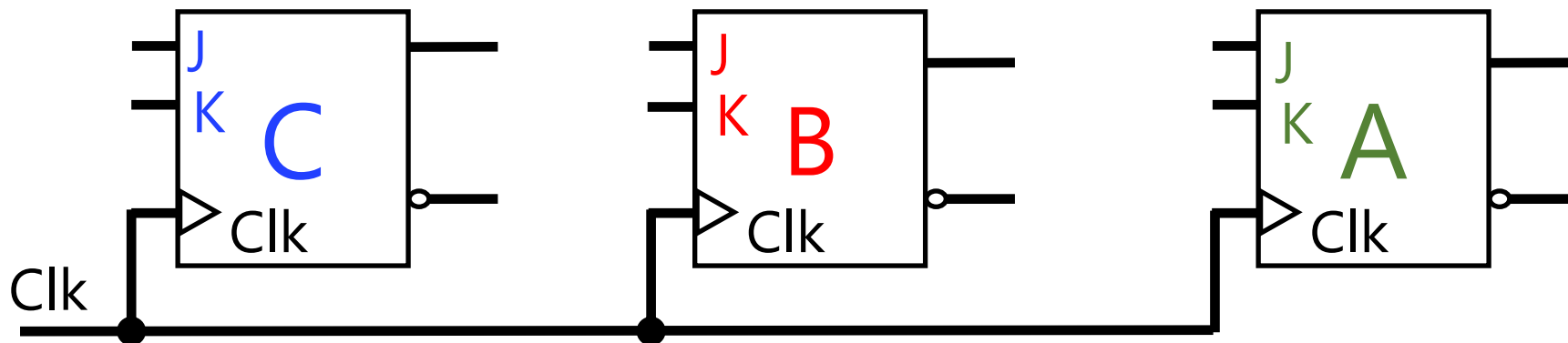


A	$J_A = F(C, B, A) = \sum(0, 2, 4, 6) + d(1, 3, 5, 7)$	$K_A = F(C, B, A) = \sum(1, 3, 5, 7) + d(0, 2, 4, 6)$
B	$J_B =$	$K_B =$
C	$J_C =$	$K_C =$

Counter

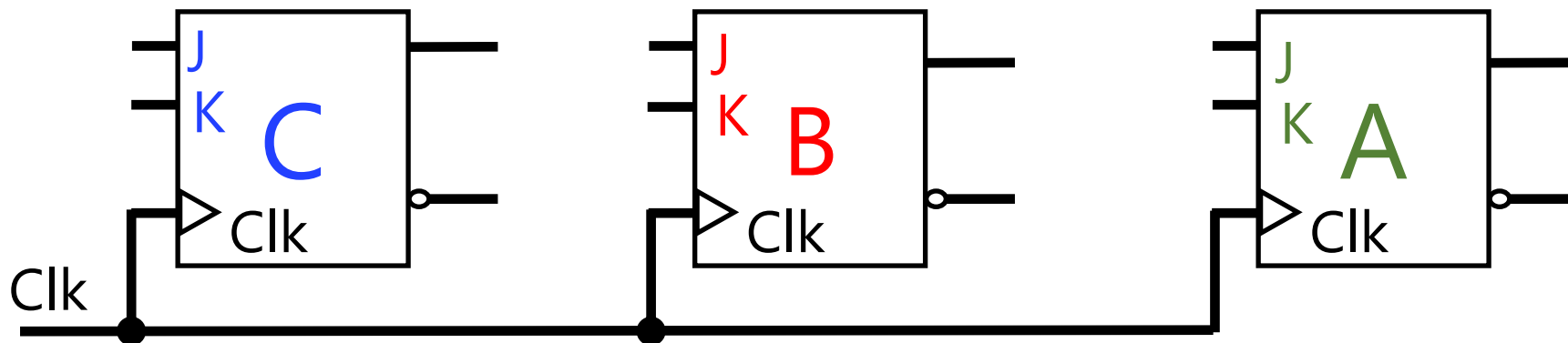
Count from 0 to N=7

A	$J_A = F(C,B,A) = \sum(0,2,4,6) + d(1,3,5,7)$	$K_A = F(C,B,A) = \sum(1,3,5,7) + d(0,2,4,6)$
B	$J_B =$	$K_B =$
C	$J_C =$	$K_C =$

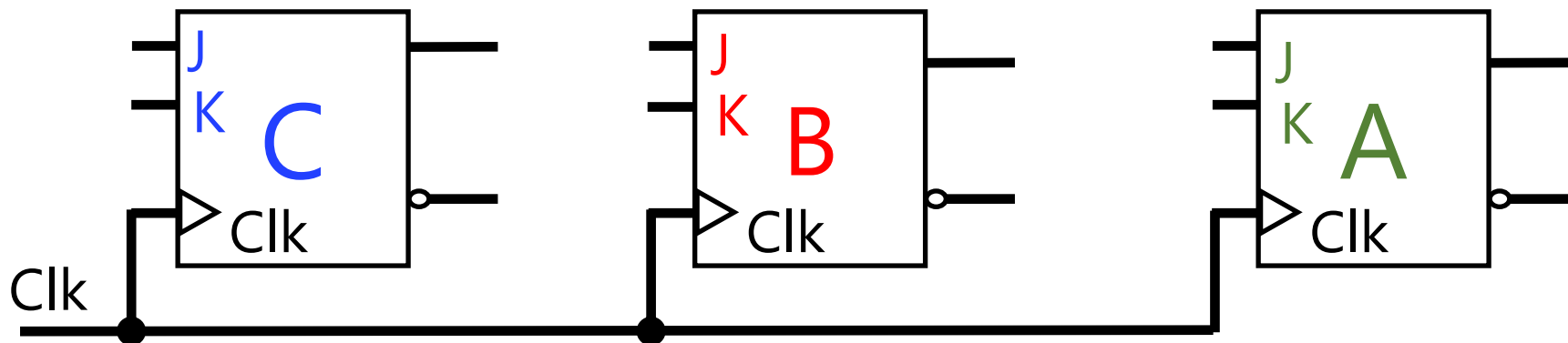


Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J_B	K_B
0	0	0	0	0	1	Store	0	0
0	0	1	0	1	0			
0	1	0	0	1	1			
0	1	1	1	0	0			
1	0	0	1	0	1			
1	0	1	1	1	0			
1	1	0	1	1	1			
1	1	1	0	0	0			

OR



Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J _B	K _B
0	0	0	0	0	1	Reset	0	1
0	0	1	0	1	0			
0	1	0	0	1	1			
0	1	1	1	0	0			
1	0	0	1	0	1			
1	0	1	1	1	0			
1	1	0	1	1	1			
1	1	1	0	0	0			



Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J _B	K _B
0	0	0	0	0	1	Store/Reset	0	×
0	0	1	0	1	0	Set/Comp	1	×
0	1	0	0	1	1	Store/Set	×	0
0	1	1	1	0	0	Reset/Comp	×	1
1	0	0	1	0	1	Store/Reset	0	×
1	0	1	1	1	0	Set/Comp	1	×
1	1	0	1	1	1	Store/Set	×	0
1	1	1	0	0	0	Reset/Comp	×	1

Counter

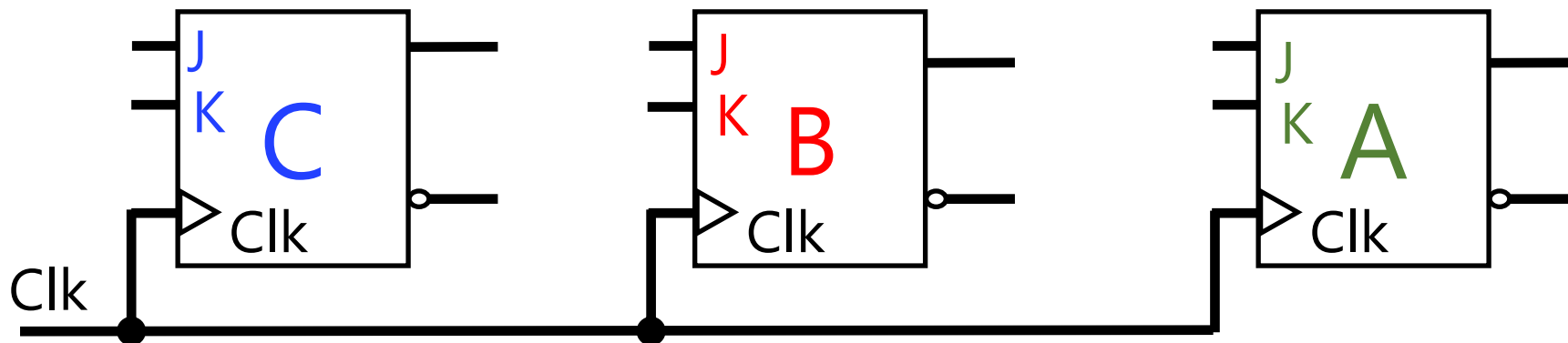
Count from 0 to N=7

A	$J_A = F(C,B,A) = \sum(0,2,4,6) + d(1,3,5,7)$	$K_A = F(C,B,A) = \sum(1,3,5,7) + d(0,2,4,6)$
B	$J_B = F(C,B,A) = \sum(1,5) + d(2,3,6,7)$	$K_B = F(C,B,A) = \sum(3,7) + d(0,1,4,5)$
C	$J_C =$	$K_C =$

Counter

Count from 0 to N=7

A	$J_A = F(C,B,A) = \sum(0,2,4,6) + d(1,3,5,7)$	$K_A = F(C,B,A) = \sum(1,3,5,7) + d(0,2,4,6)$
B	$J_B = F(C,B,A) = \sum(1,5) + d(2,3,6,7)$	$K_B = F(C,B,A) = \sum(3,7) + d(0,1,4,5)$
C	$J_C =$	$K_C =$



Q(T)			Q(T+1)			Not part of state table!		
C	B	A	C	B	A	Action	J_C	K_C
0	0	0	0	0	1	Store/Reset	0	×
0	0	1	0	1	0	Store/Reset	0	×
0	1	0	0	1	1	Store/Reset	0	×
0	1	1	1	0	0	Comp/Set	1	×
1	0	0	1	0	1	Store/Set	×	0
1	0	1	1	1	0	Store/Set	×	0
1	1	0	1	1	1	Store/Set	×	0
1	1	1	0	0	0	Comp/Reset	×	1

Counter

Count from 0 to N=7

A	$J_A = F(C,B,A) = \sum(0,2,4,6) + d(1,3,5,7)$	$K_A = F(C,B,A) = \sum(1,3,5,7) + d(0,2,4,6)$
B	$J_B = F(C,B,A) = \sum(1,5) + d(2,3,6,7)$	$K_B = F(C,B,A) = \sum(3,7) + d(0,1,4,5)$
C	$J_C = F(C,B,A) = \sum(3) + d(4,5,6,7)$	$K_C = F(C,B,A) = \sum(7) + d(0,1,2,3)$

Design

7. Minimization of input (*excitation*) equations

Counter

Count from 0 to N=7

3-Variable K-Map

A	$J_A = F(C,B,A) = \sum(0,2,4,6) + d(1,3,5,7)$	$K_A = F(C,B,A) = \sum(1,3,5,7) + d(0,2,4,6)$
B	$J_B = F(C,B,A) = \sum(1,5) + d(2,3,6,7)$	$K_B = F(C,B,A) = \sum(3,7) + d(0,1,4,5)$
C	$J_C = F(C,B,A) = \sum(3) + d(4,5,6,7)$	$K_C = F(C,B,A) = \sum(7) + d(0,1,2,3)$

		BA			
		00	01	11	10
C	0	0 m_0	0 m_1	1 m_3	0 m_2
	1	X m_4	X m_5	X m_7	X m_6

$$J_C = F(C, B, A) = \sum(3) + d(4, 5, 6, 7)$$

$$J_C = BA$$

		BA			
		00	01	11	10
C	0	0 m_0	1 m_1	X m_3	X m_2
	1	0 m_4	1 m_5	X m_7	X m_6

$$J_B = F(C, B, A) = \sum(1, 5) + d(2, 3, 6, 7)$$

$$J_B = A$$

		BA			
		00	01	11	10
C	0	1 m_0	X m_1	X m_3	1 m_2
	1	1 m_4	X m_5	X m_7	1 m_6

$$J_A = F(C, B, A) = \sum(0, 2, 4, 6) + d(1, 3, 5, 7)$$

$$J_A = 1$$

		BA			
		00	01	11	10
C	0	X m_0	X m_1	X m_3	X m_2
	1	0 m_4	0 m_5	1 m_7	0 m_6

$$K_C = F(C, B, A) = \sum(7) + d(0, 1, 2, 3)$$

$$K_C = BA$$

		BA			
		00	01	11	10
C	0	X m_0	X m_1	1 m_3	0 m_2
	1	X m_4	X m_5	1 m_7	0 m_6

$$K_B = F(C, B, A) = \sum(3, 7) + d(0, 1, 4, 5)$$

$$K_B = A$$

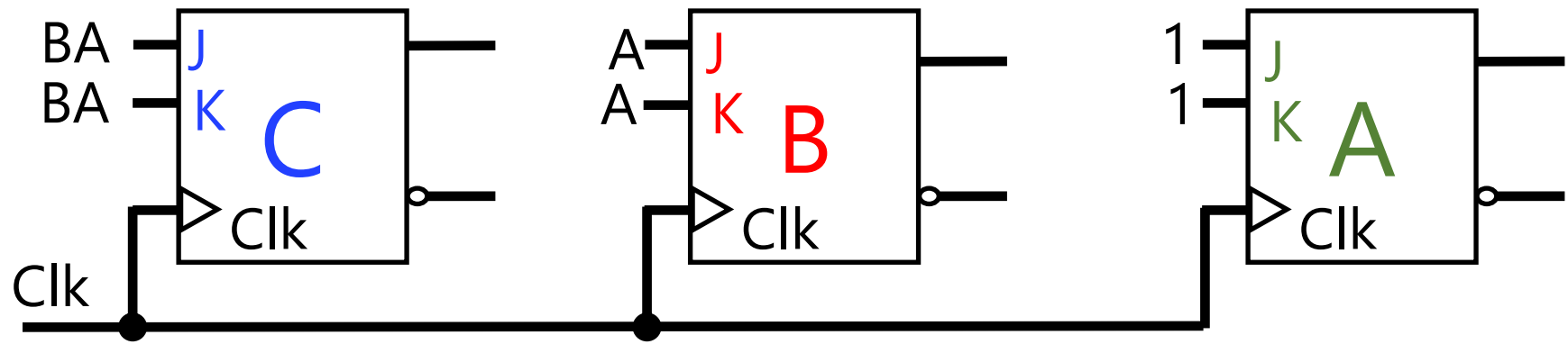
		BA			
		00	01	11	10
C	0	X m_0	1 m_1	1 m_3	X m_2
	1	X m_4	1 m_5	1 m_7	X m_6

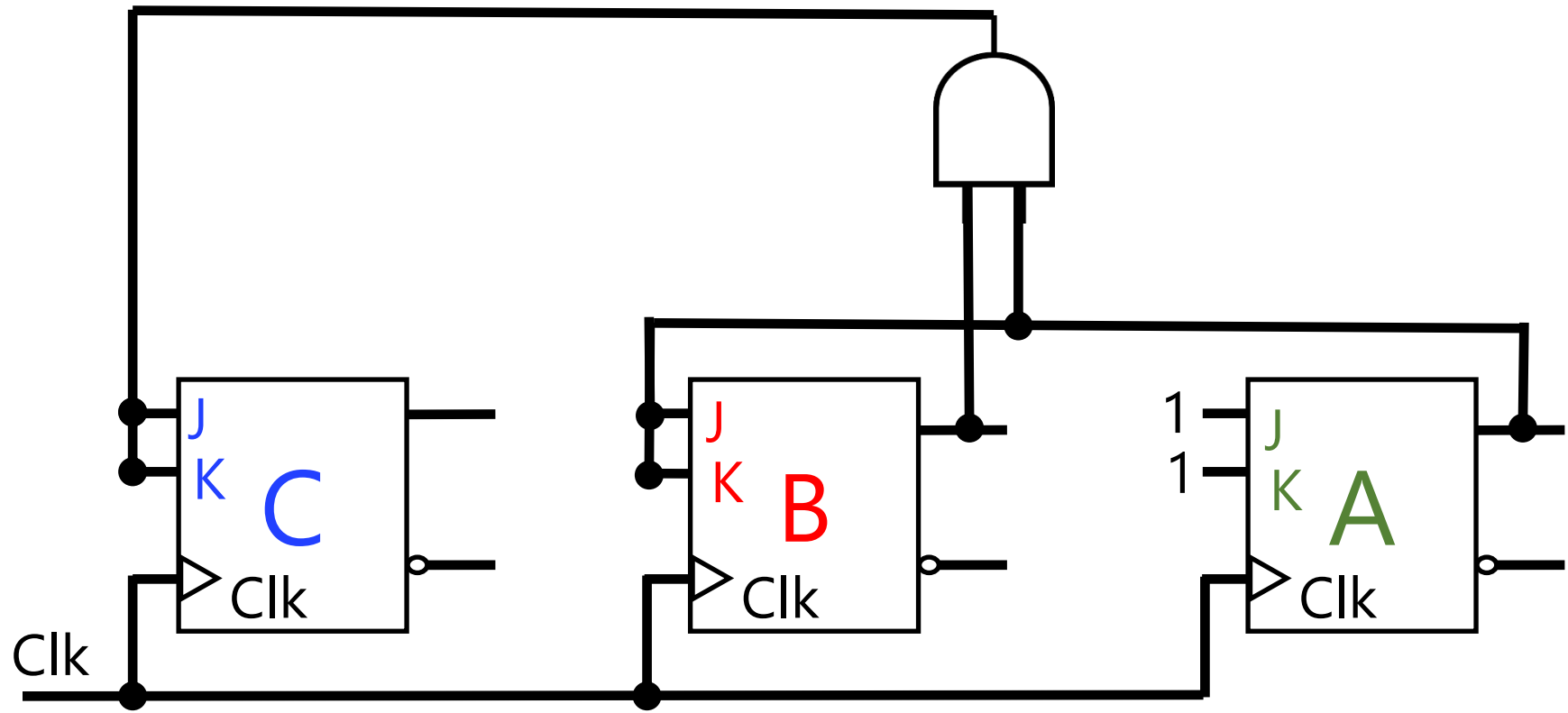
$$K_A = F(C, B, A) = \sum(1, 3, 5, 7) + d(0, 2, 4, 6)$$

$$K_A = 1$$

Design

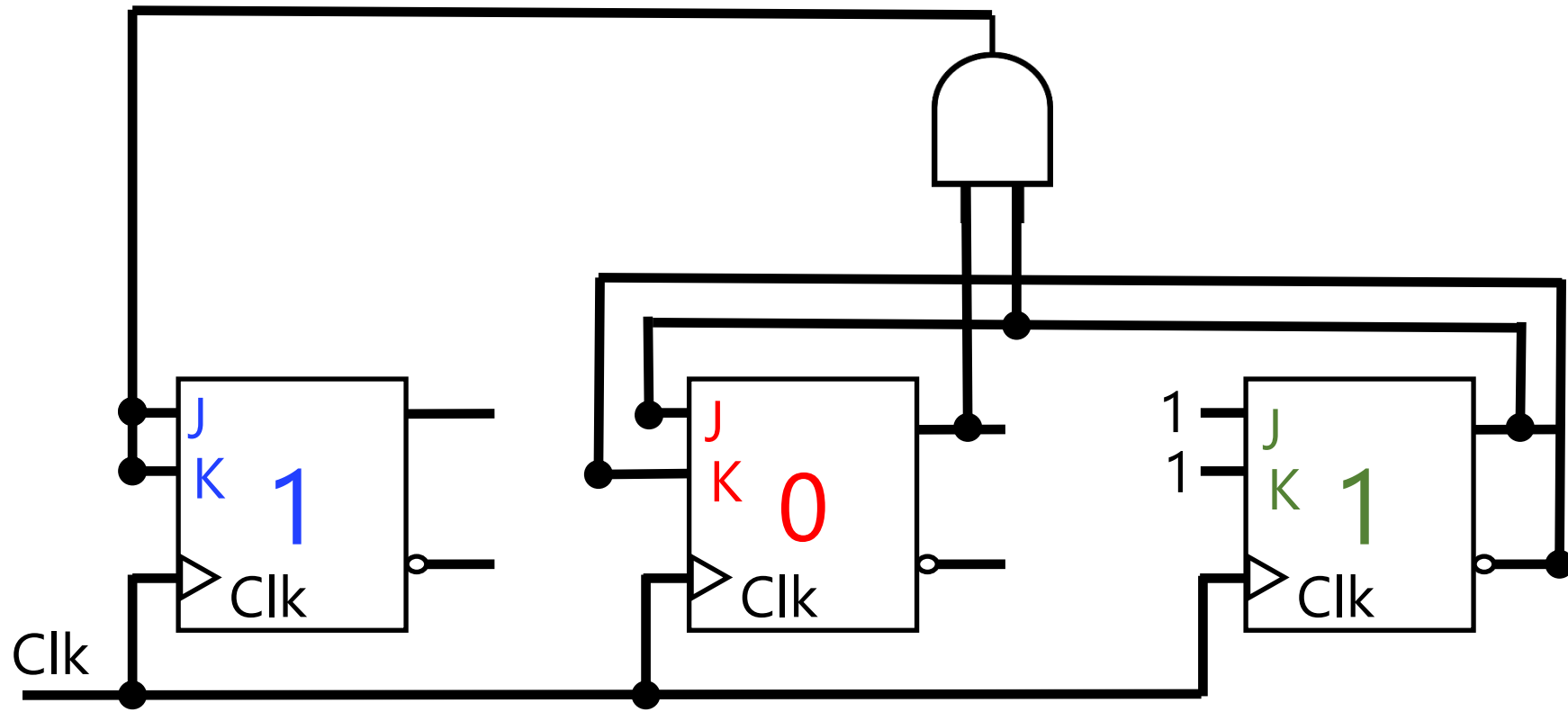
8. Draw/Sketch Logic Circuit



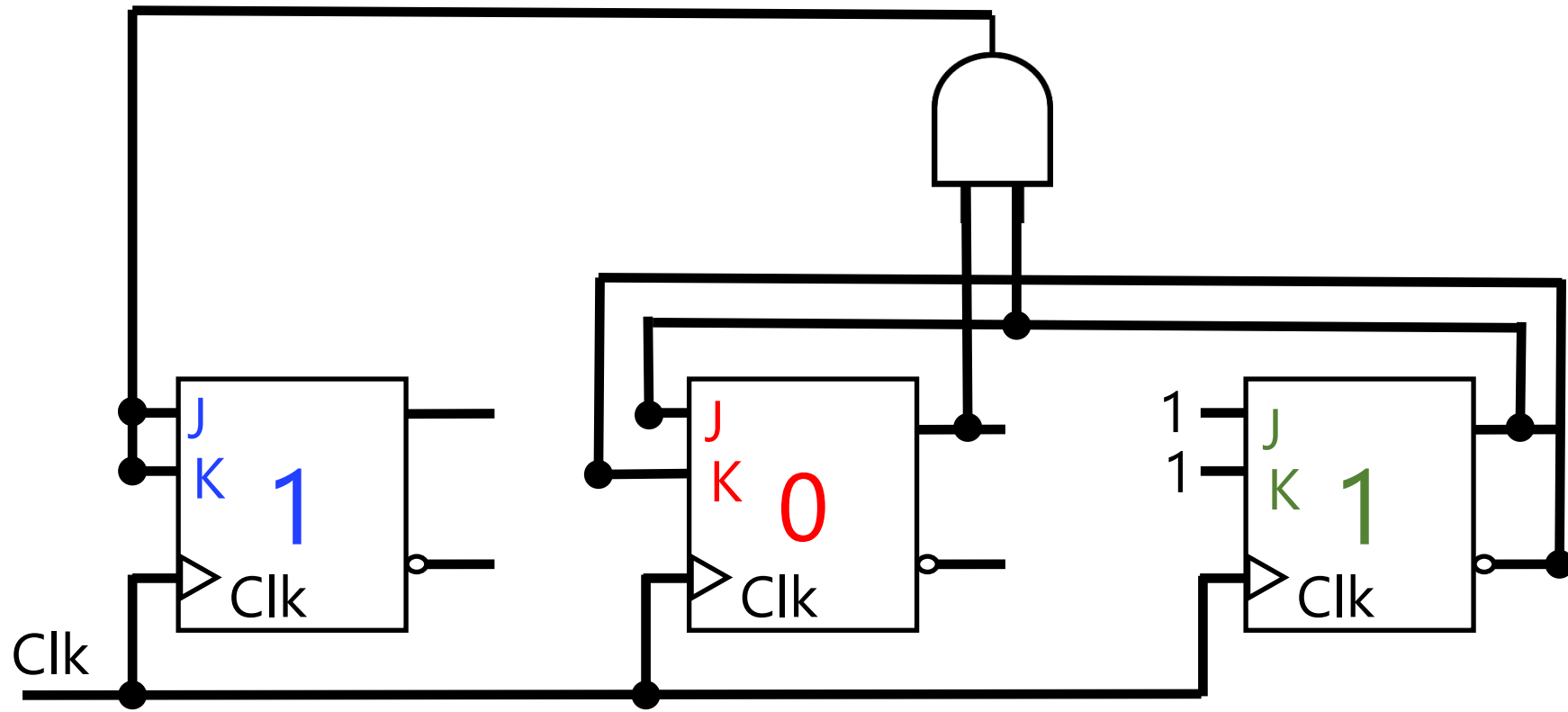


Design

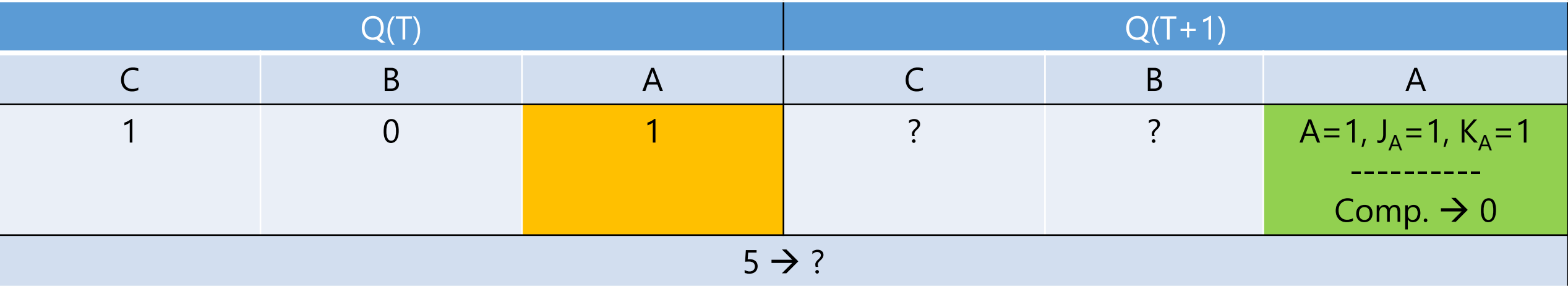
9. (Optional) Test

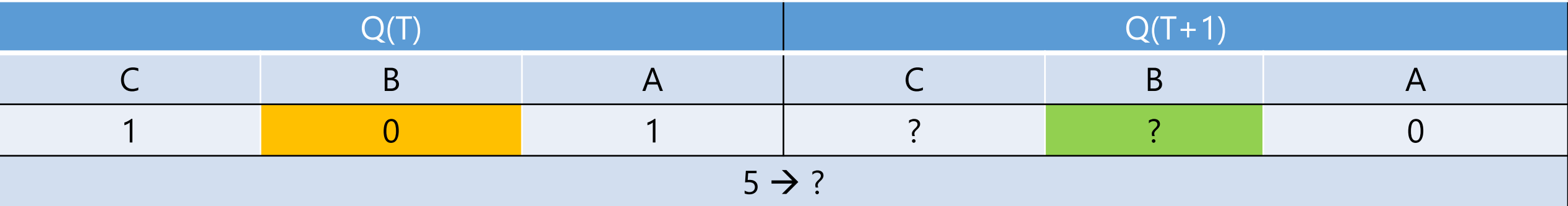


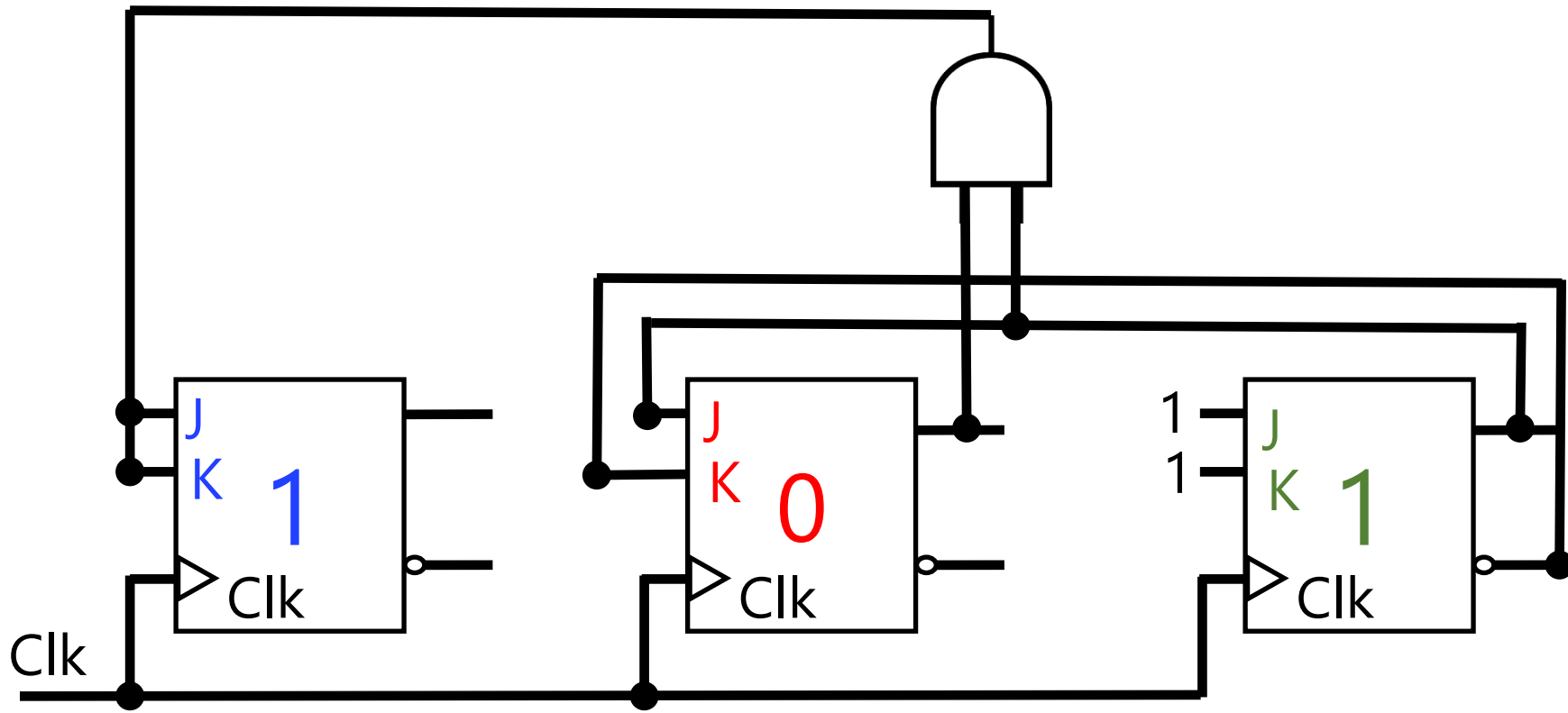
Q(T)			Q(T+1)		
C	B	A	C	B	A
1	0	1	?	?	?
5 → ?					



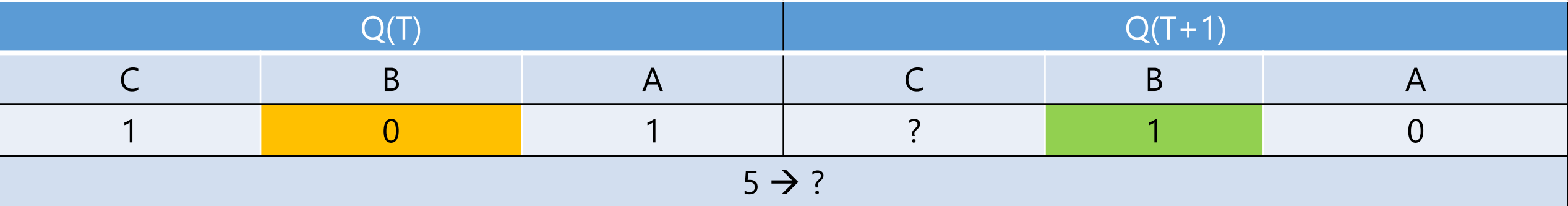
Q(T)			Q(T+1)		
C	B	A	C	B	A
1	0	1	?	?	?
5 → ?					

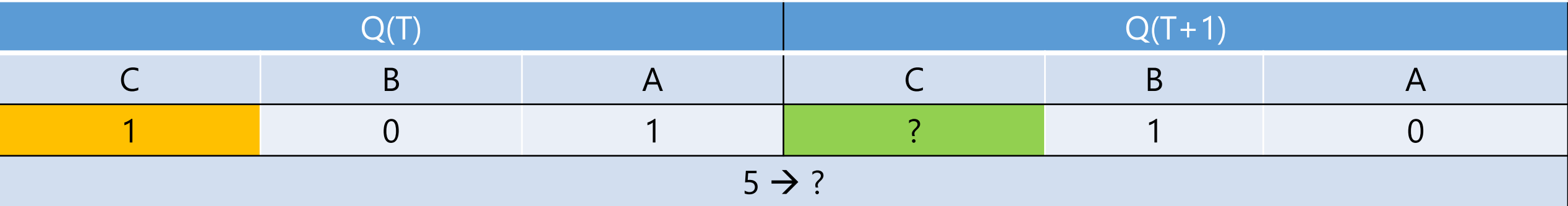


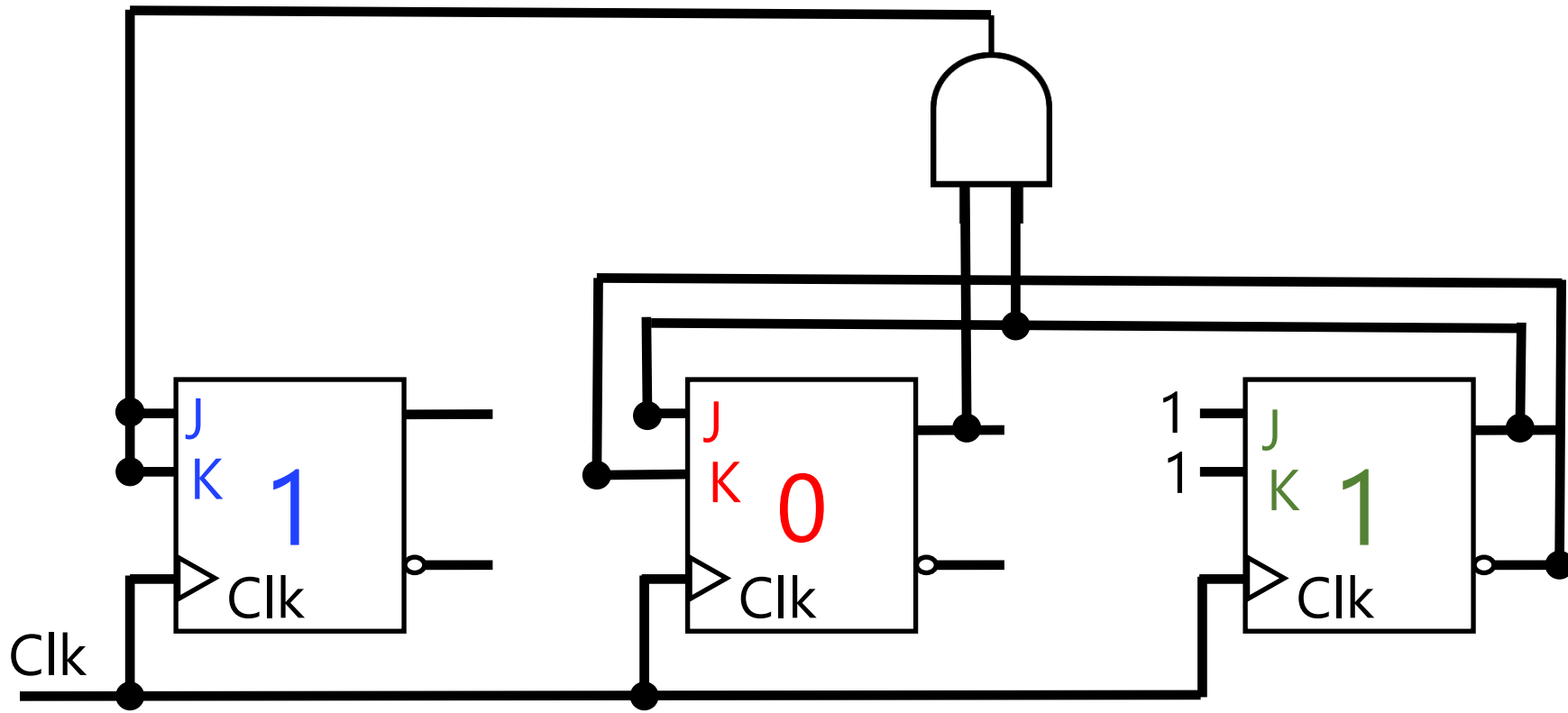




Q(T)			Q(T+1)		
C	B	A	C	B	A
1	0	1	?	B=0, $J_B=A=1$, $K_B=A'=0$ ----- Set $\rightarrow 1$	0
5 $\rightarrow$?					

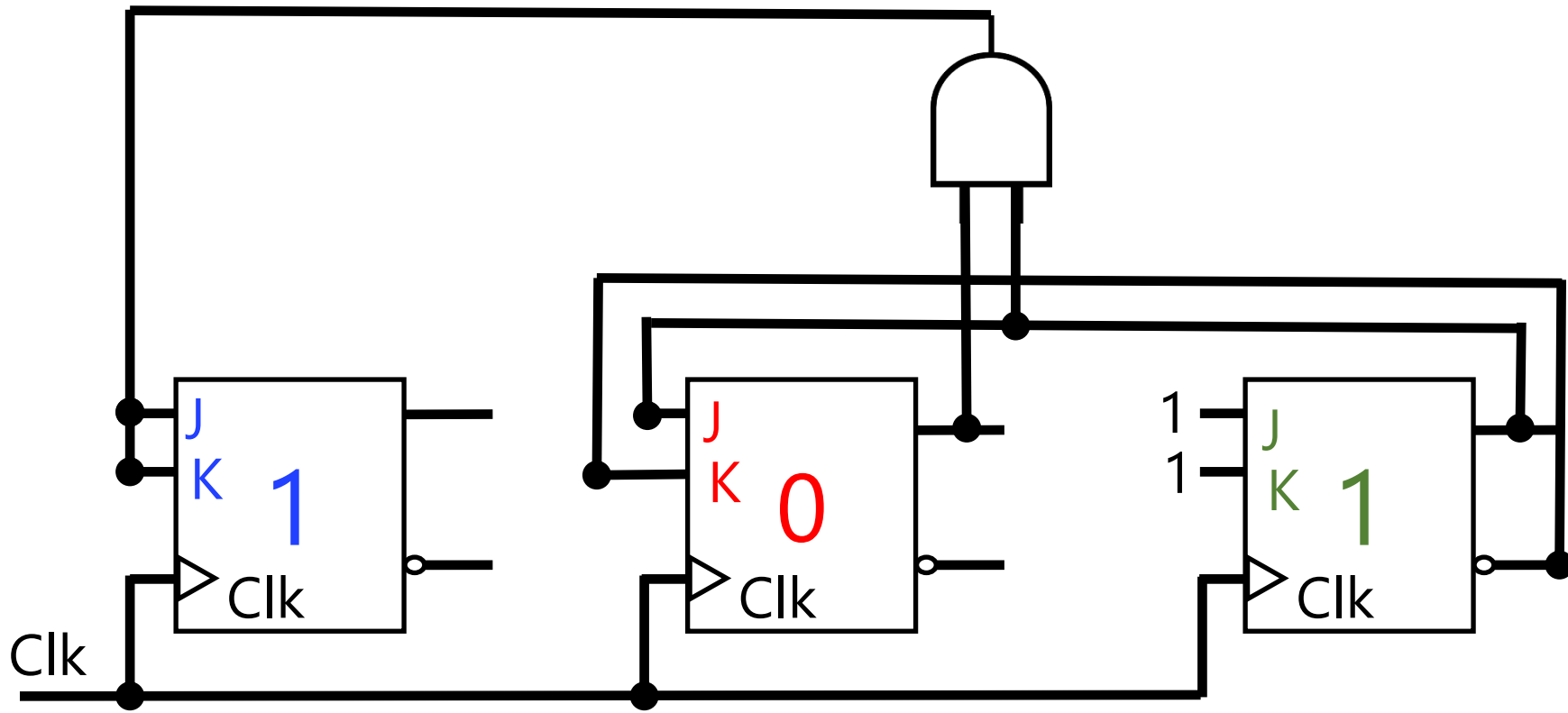






Q(T)			Q(T+1)		
C	B	A	C	B	A
1	0	1	$C=1, J_C=BA=01=0$ $K_C=BA=01=0$ ----- Store $\rightarrow 1$	1	0

5 $\rightarrow$?



Q(T)			Q(T+1)		
C	B	A	C	B	A
1	0	1	1	1	0
5 → 6					



Design (Recap)

0. Do we need combinational logic or sequential logic? Do we need memory?
 1. How many storage (flip-flops)? #FF
 2. Form the state (transition) diagram
 3. Form the state table
 4. Fill the state table
 5. What type of storage (flip-flop)? RS, D, T, JK, or Mixed
 6. Input (*excitation*) equations for each FF
 7. Minimization of input (*excitation*) equations
 8. Draw/Sketch Logic Circuit
 9. (Optional) Test
-

Design (Recap)

0. Do we need combinational logic or sequential logic? Do we need memory?
 1. How many storage (flip-flops)? #FF
 2. Form the state (transition) diagram
 3. Form the state table
 4. Fill the state table
 5. What type of storage (flip-flop)? RS, D, T, JK, or Mixed
 6. Input (*excitation*) equations for each FF
 7. Minimization of input (*excitation*) equations
 8. Draw/Sketch Logic Circuit
 9. (Optional) Test
-

Design (Advanced)

0. Do we need combinational logic or sequential logic? Do we need memory?

1. How many storage (flip-flops)? #FF

2. Form the state (transition) diagram

2.1. State Reduction

3. Form the state table

4. Fill the state table

5. What type of storage (flip-flop)? RS, D, T, JK, or Mixed

6. Input (*excitation*) equations for each FF

7. Minimization of input (*excitation*) equations

8. Draw/Sketch Logic Circuit

9. (Optional) Test

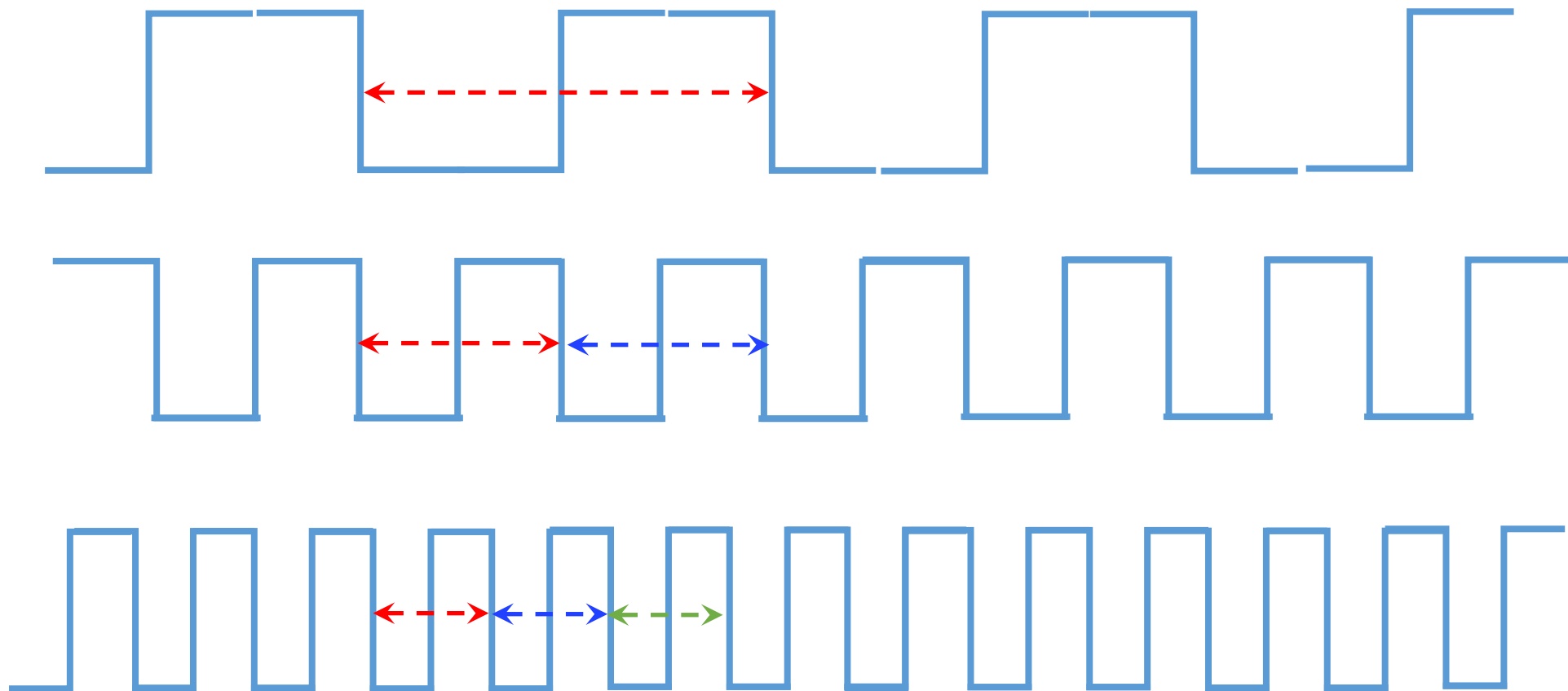
Theory of Automata

COMP-2140: Computer Languages, Grammars, and Translators

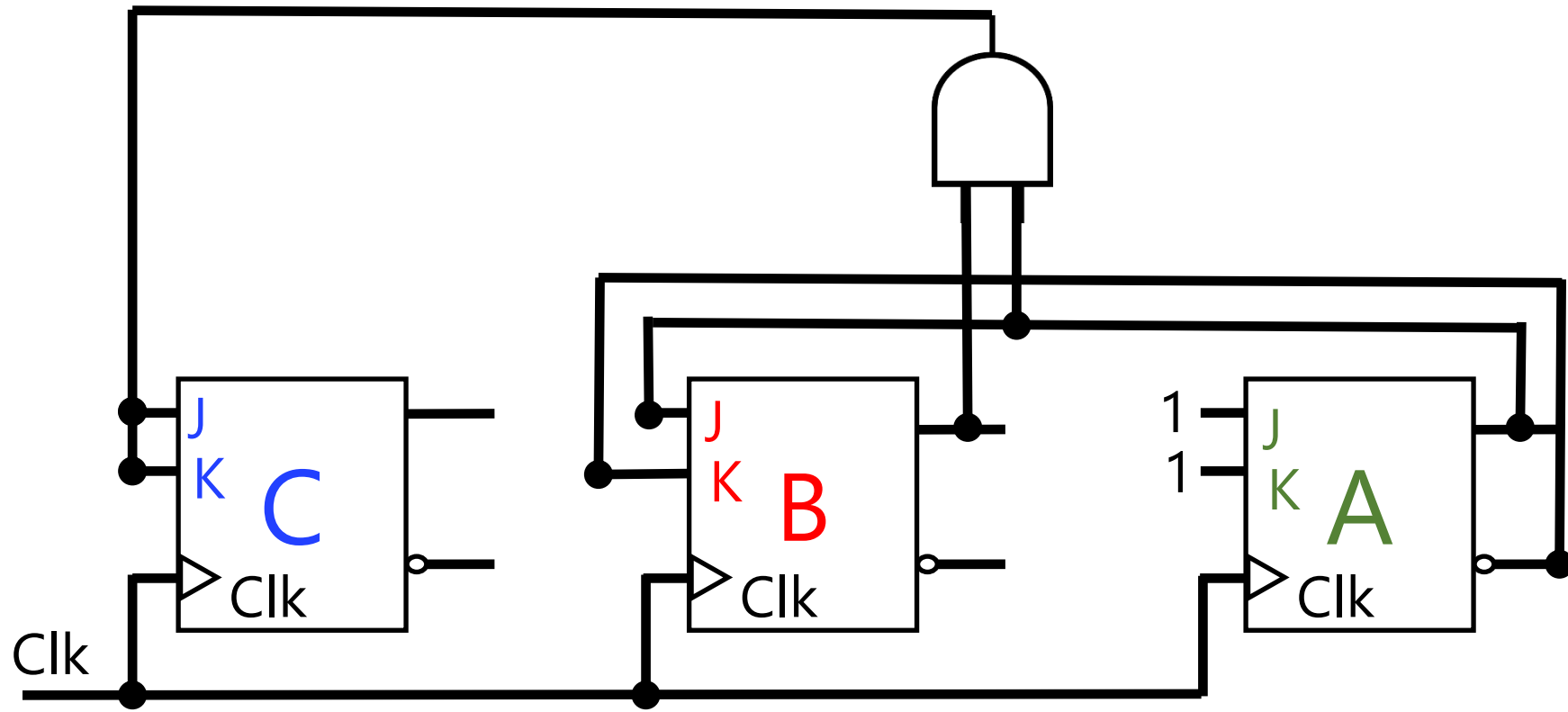
Clock (revisited)

shortened as *clk*

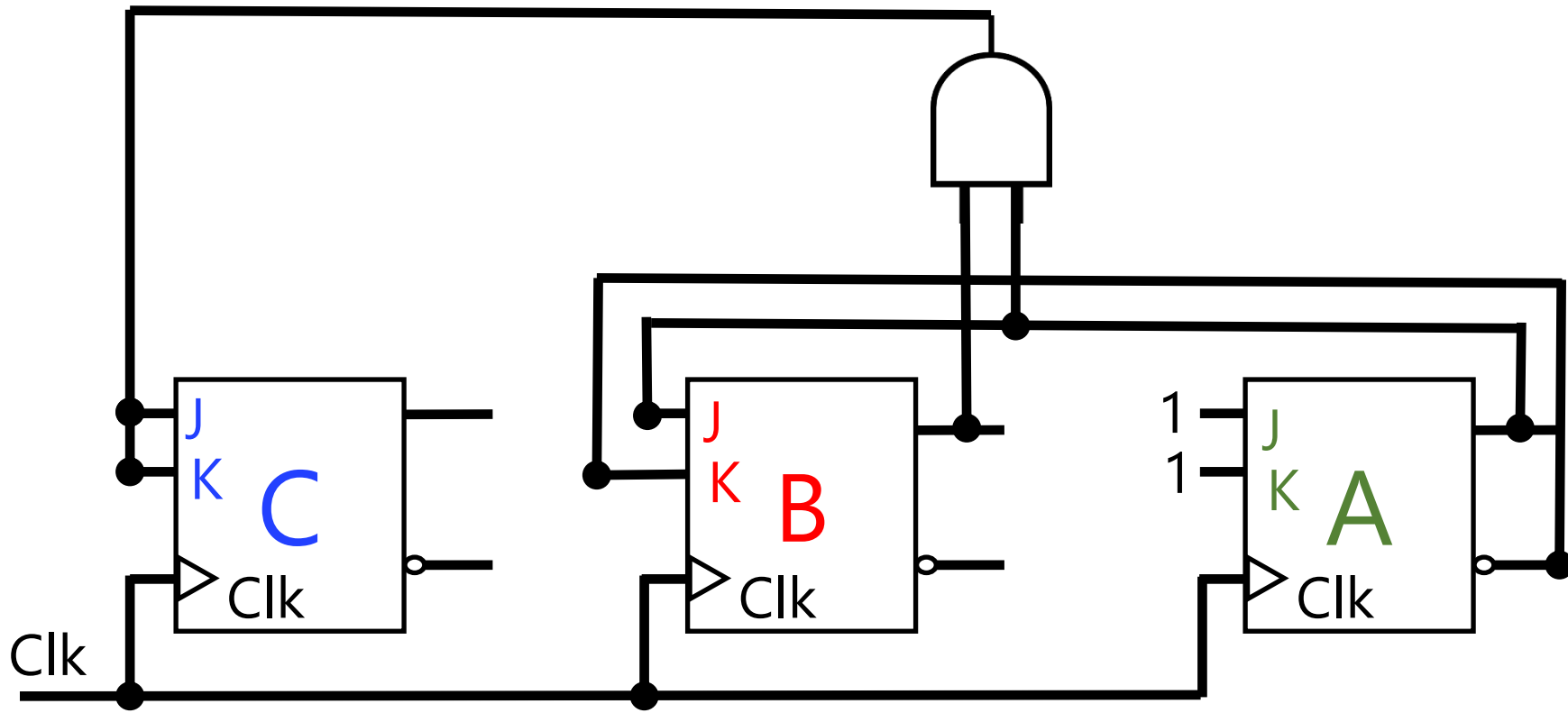
timing device that generates a train of pulses



How many pulse in **1** sec?

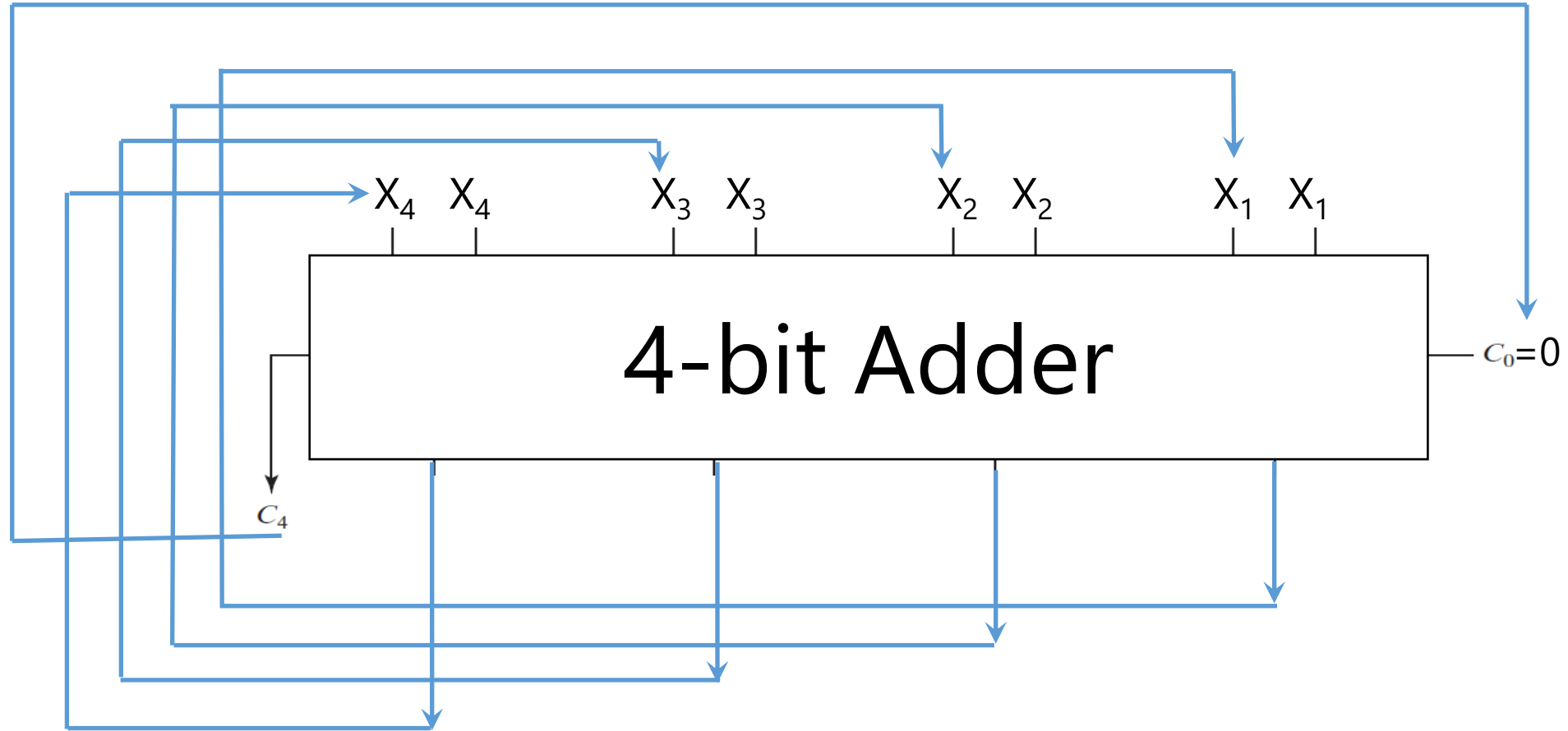


Counter 0 to 7

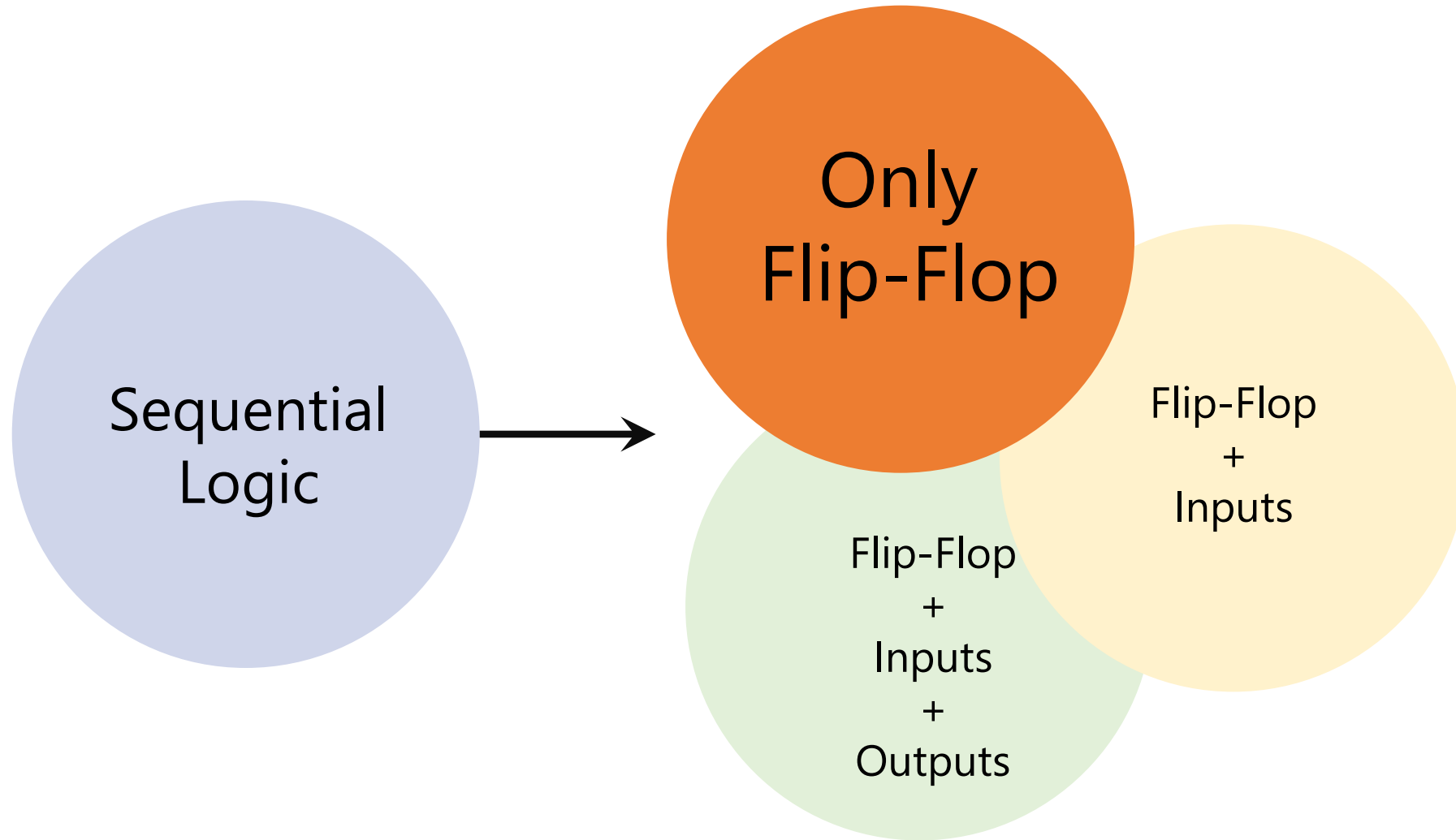


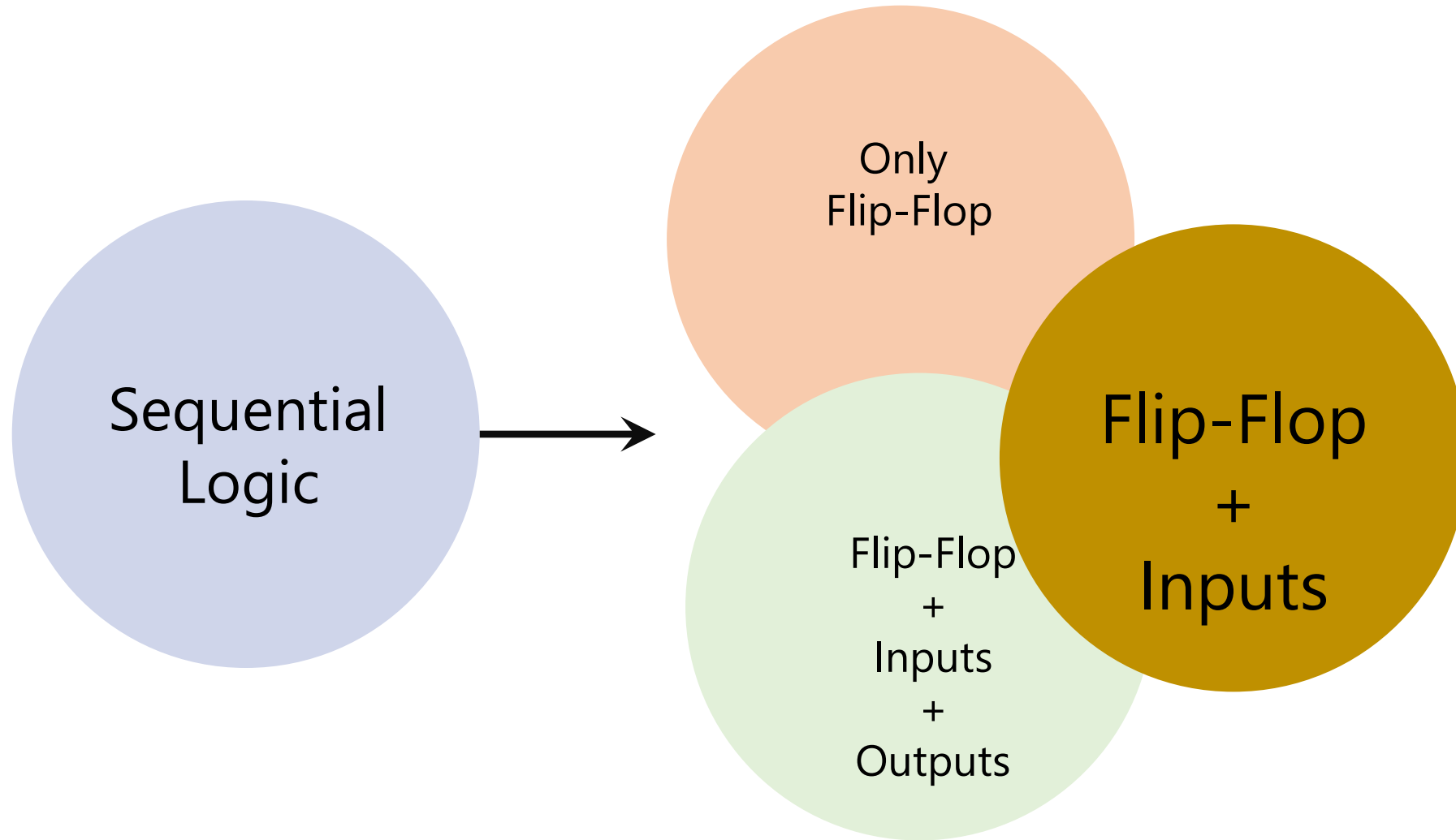
Counter 0 to 7

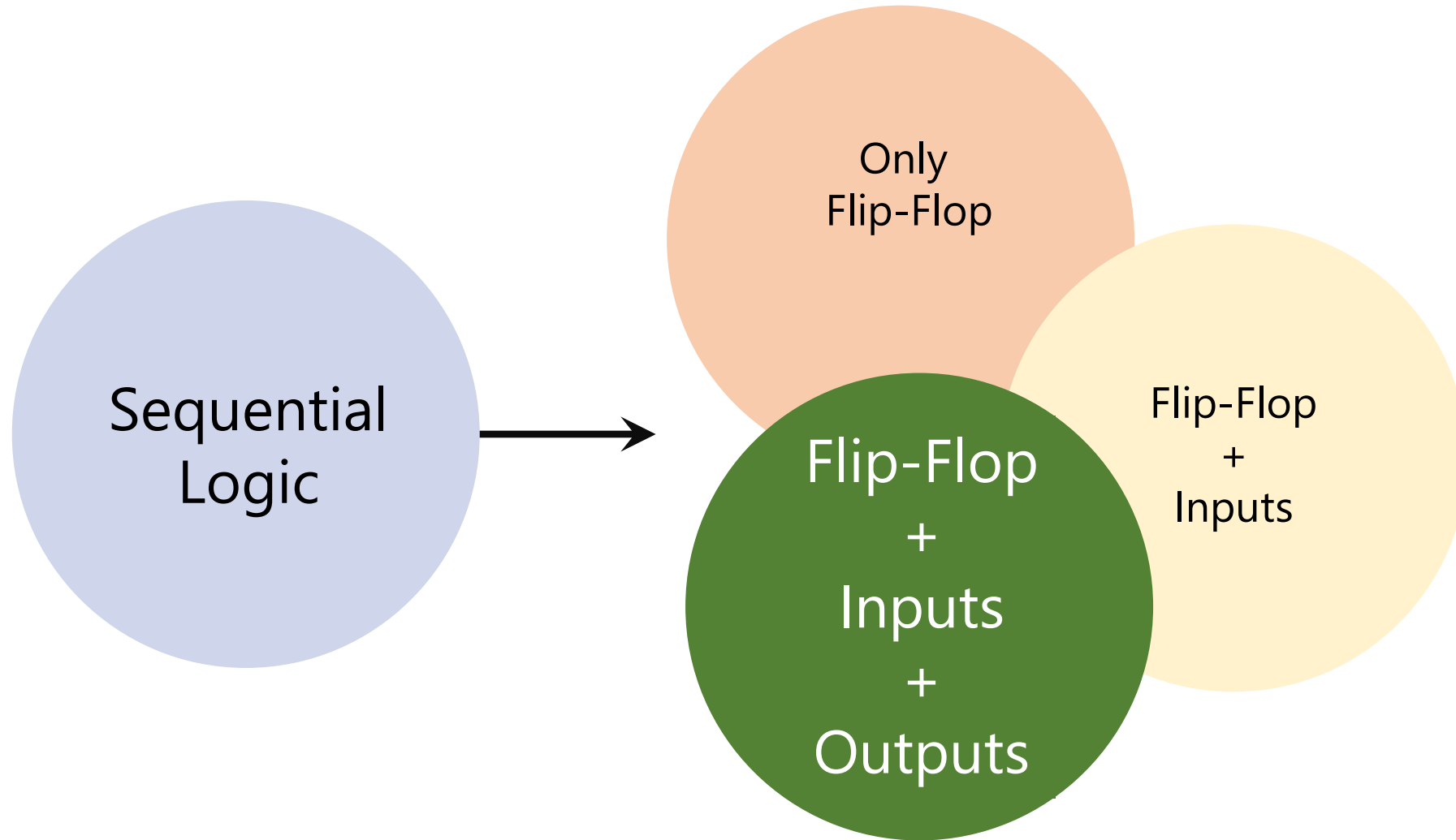
How many pulses? Elapsed time?



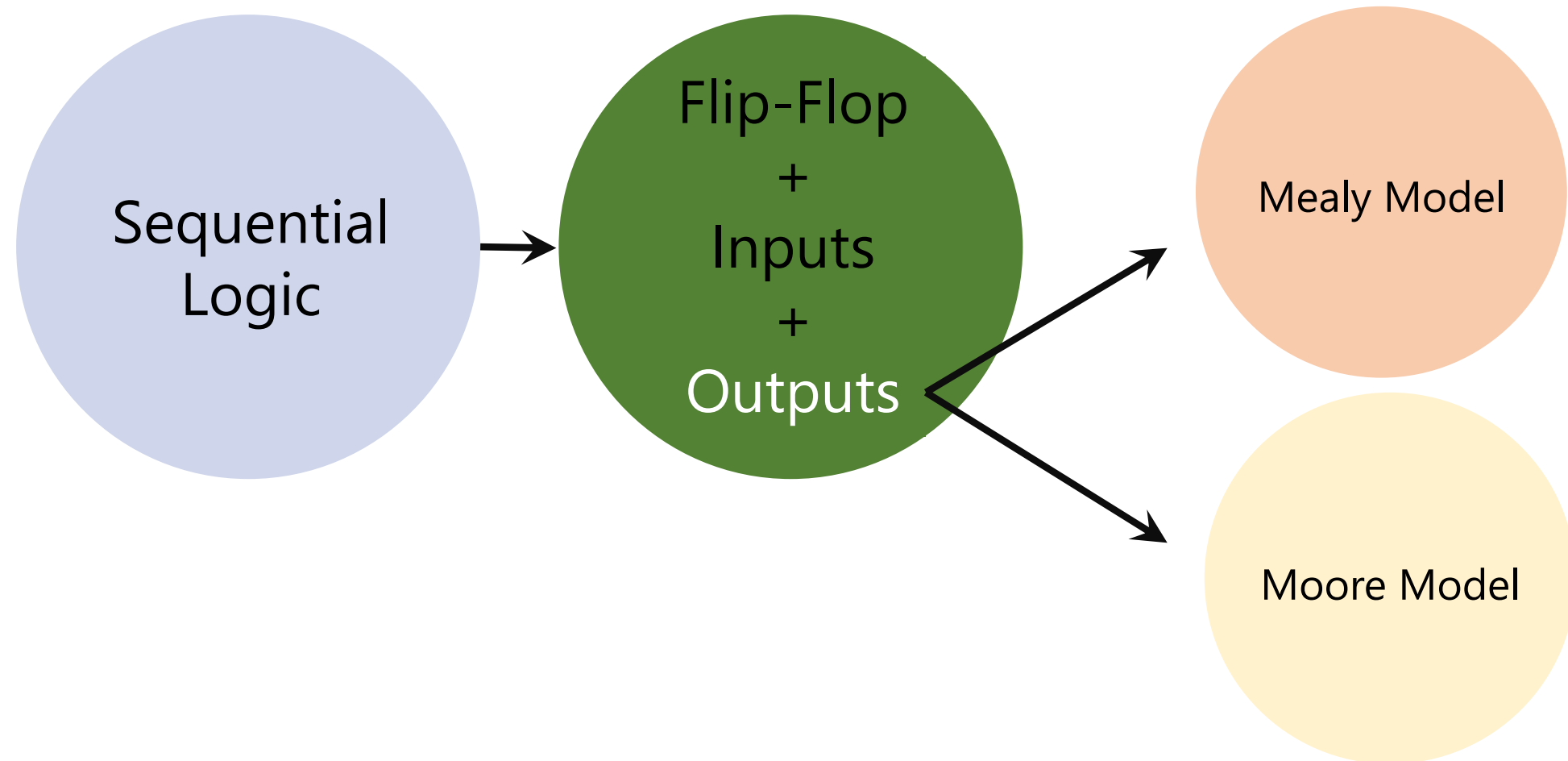
$X \times Y = X + \dots + X \rightarrow$ When to stop?
Feedback $\rightarrow$ Sequential Logic







Analysis
vs.
Design





George H. Mealy

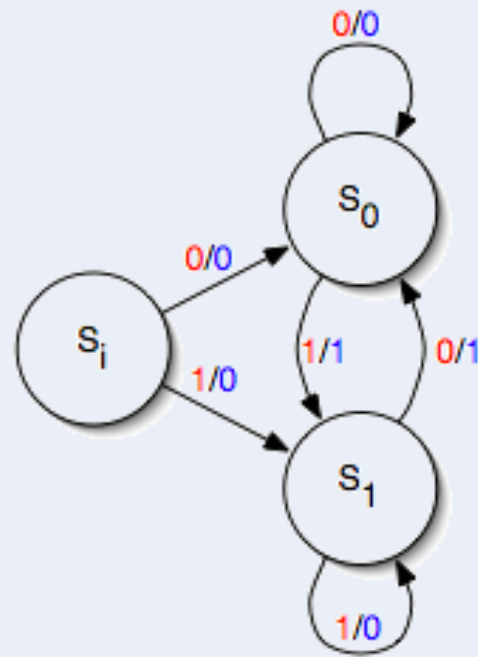
(1927 – 2010)

Mathematician and Computer Scientist

Invented Mealy Machine

Also a pioneer of modular programming

Outputs = Function(Current State, Inputs)



Edward Forrest Moore

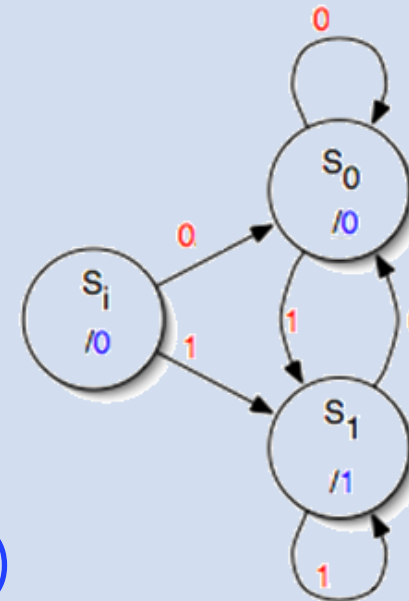
(1925 – 2003)

Mathematician and Computer Scientist

Inventor of the Moore Machine

Also an early pioneer of artificial life

Outputs = Function(Current State, Inputs)



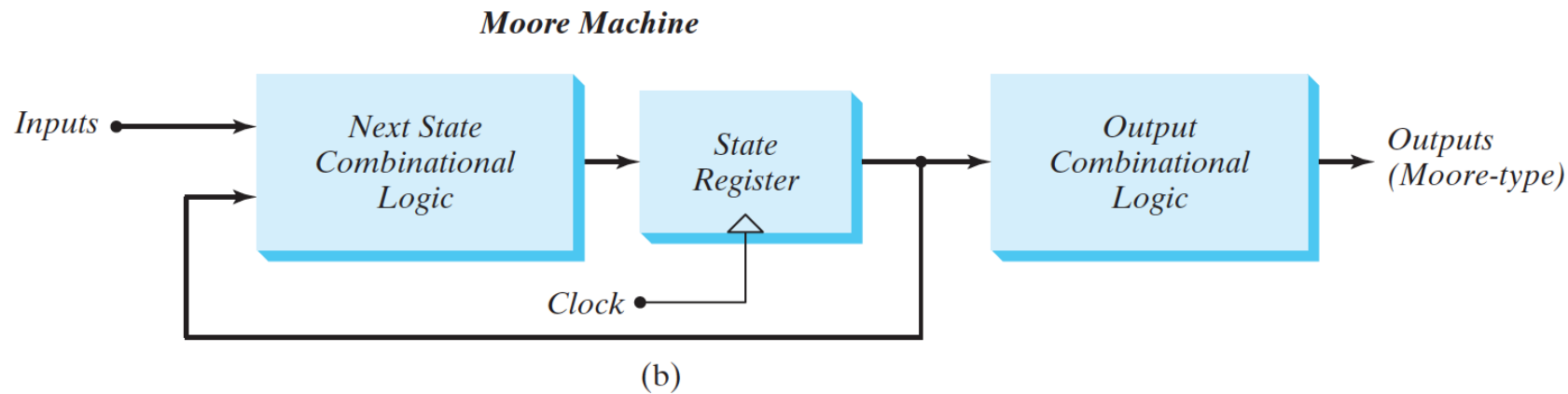
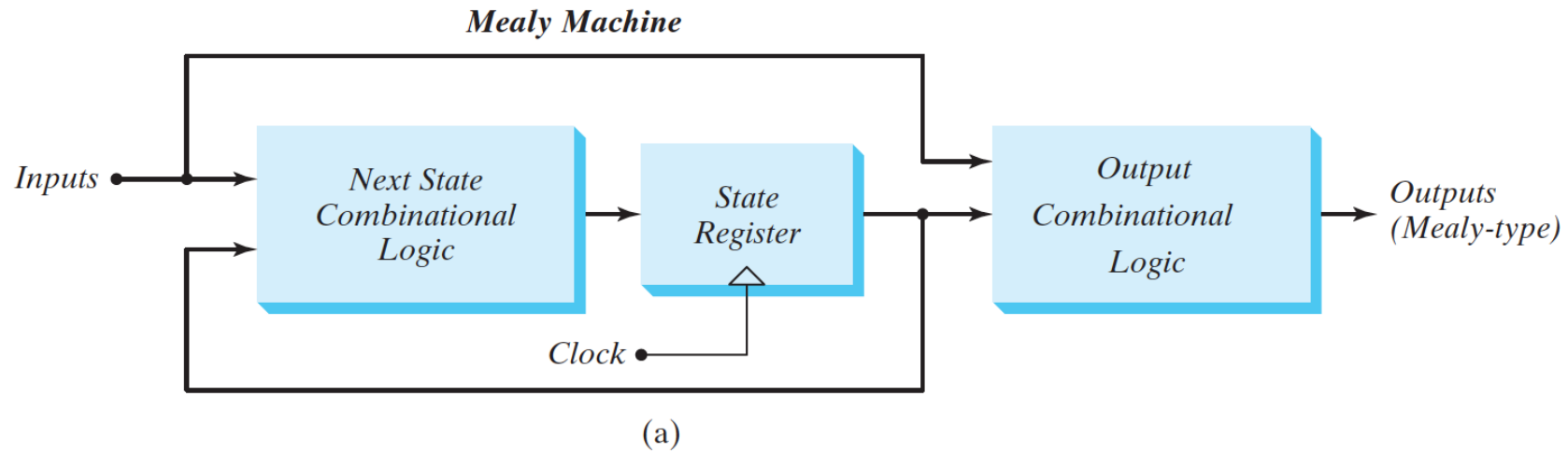


FIGURE 5.21
Block diagrams of Mealy and Moore state machines



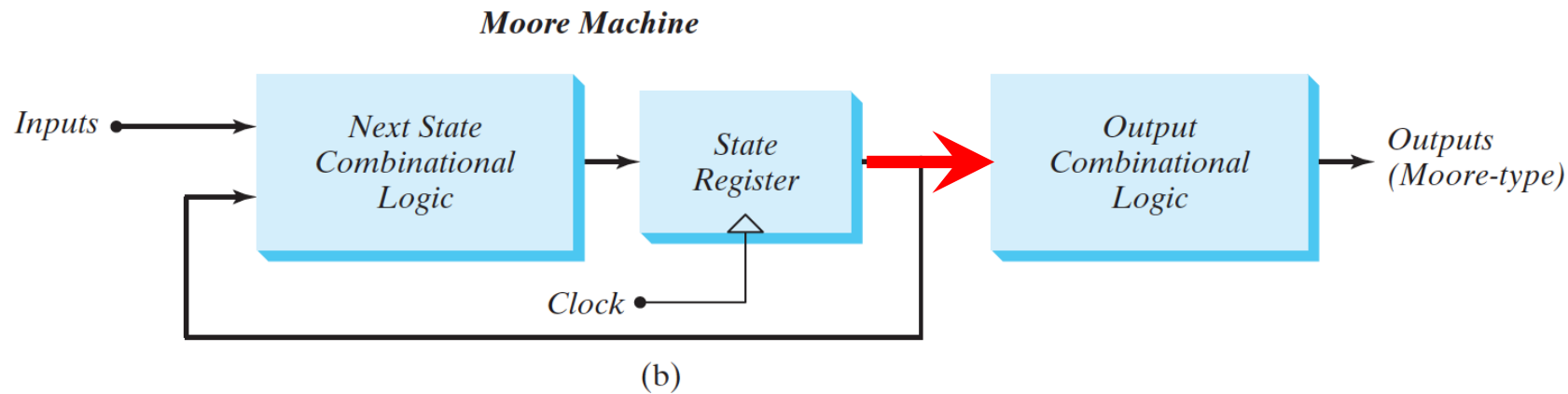
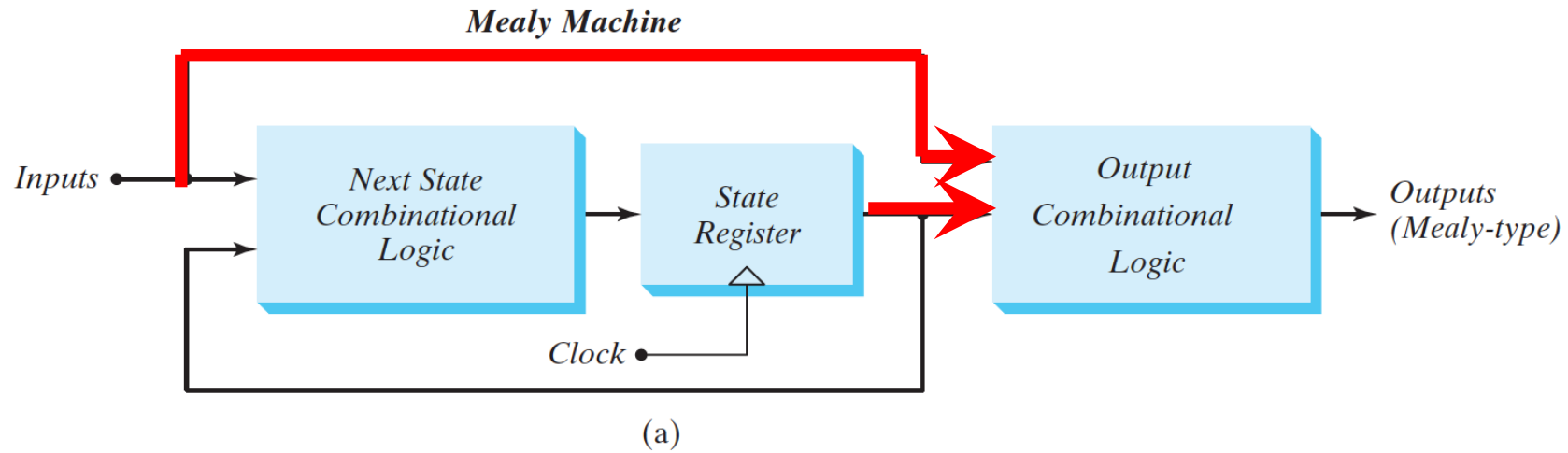


FIGURE 5.21
Block diagrams of Mealy and Moore state machines



Sequential
Logic

Flip-Flop
+
Inputs
+
Outputs

Mealy Model



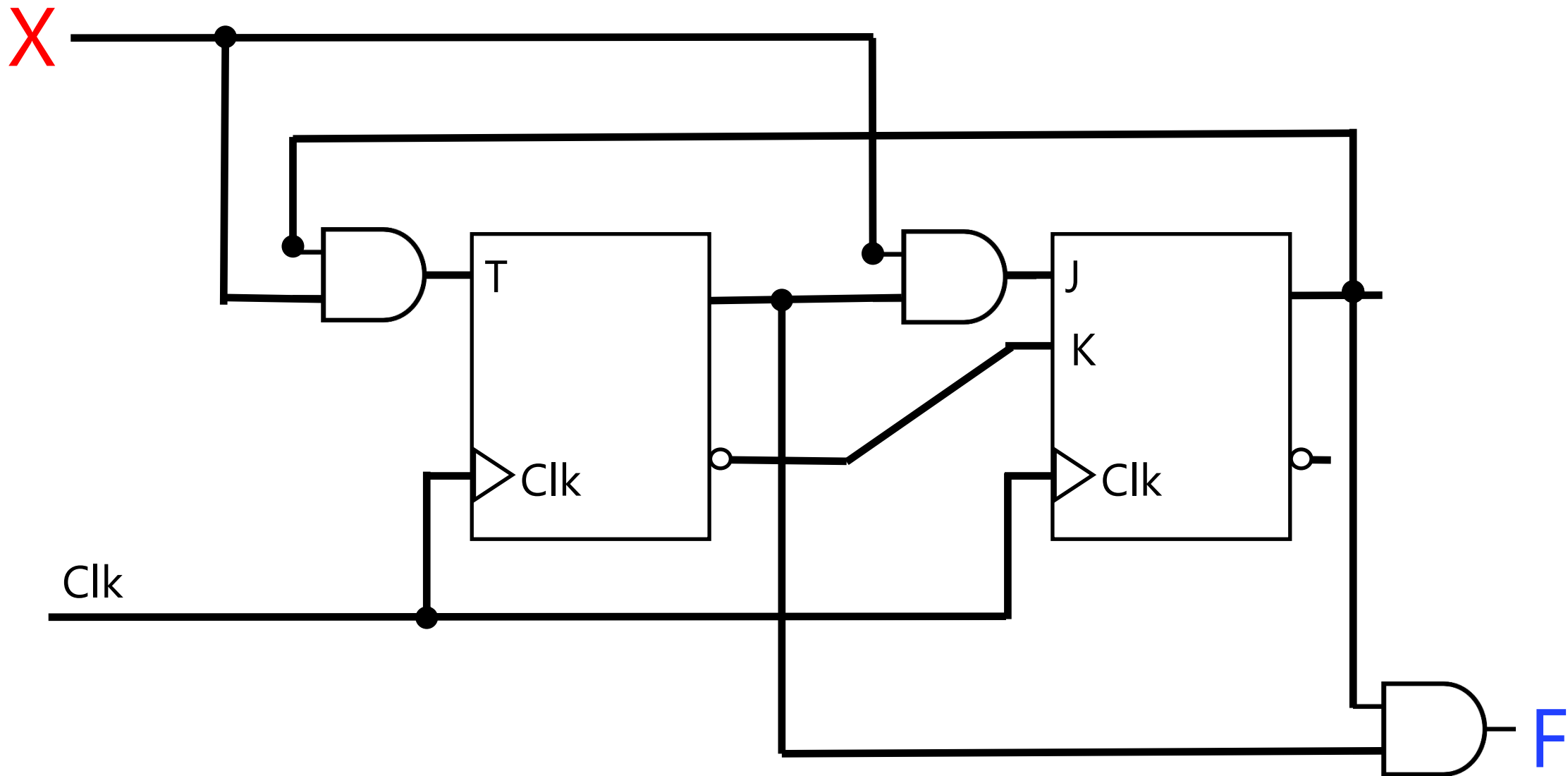
Analysis (Moore model in output) by an example

Analysis (Recap)

0. Is the circuit sequential or combinational? Any FF or feedback → Sequential
 1. What are the flip-flops? RS, D, T, JK, or mixed (e.g., 2 JK, 1 RS, ...)
 2. What are the state combinations? $2^{\#FF}$
 3. Form "State" table:
 - a) Columns: for each FF, two columns:
 - one for current state,
 - one for next state
 - b) Rows: for each state combination
 - In total: $2^{\#FF}$
 4. Fill the state table for next state columns based on:
 - a) the current state
 - b) the inputs to the FFs
 5. Form State Transition Diagram
 6. (Optional) Analyze paths and states in state transition diagram
-

Analysis (+ Input + Moore Model Output)

0. Is the circuit sequential or combinational? Any FF or feedback → Sequential
 1. What are the flip-flops? RS, D, T, JK, or mixed (e.g., 2 JK, 1 RS, ...)
 2. What are the state combinations? ~~$2^{\#FF}$~~
 3. Form "State" table:
 - a) Columns: for each FF, two columns:
 - one for current state,
 - one for next state
 - b) Rows: for each state combination
 - In total: ~~$2^{\#FF}$~~
 4. Fill the state table for next state columns based on:
 - a) the current state
 - b) the inputs to the FFs
 5. Form State Transition Diagram
 6. (Optional) Analyze paths and states in state transition diagram
-

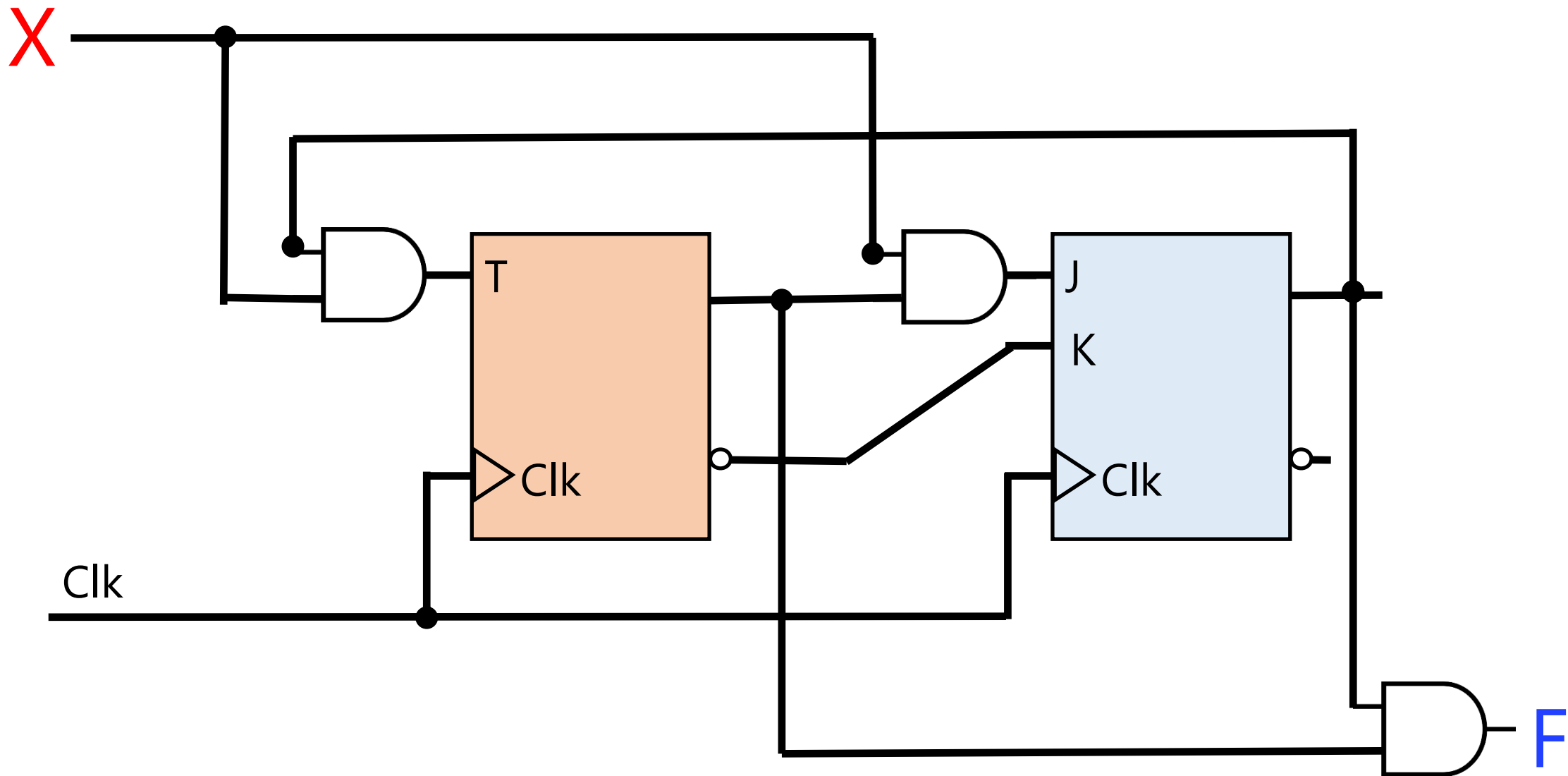


Analysis (Recap)

0. Is the circuit sequential or combinational? Any FF or feedback → Sequential

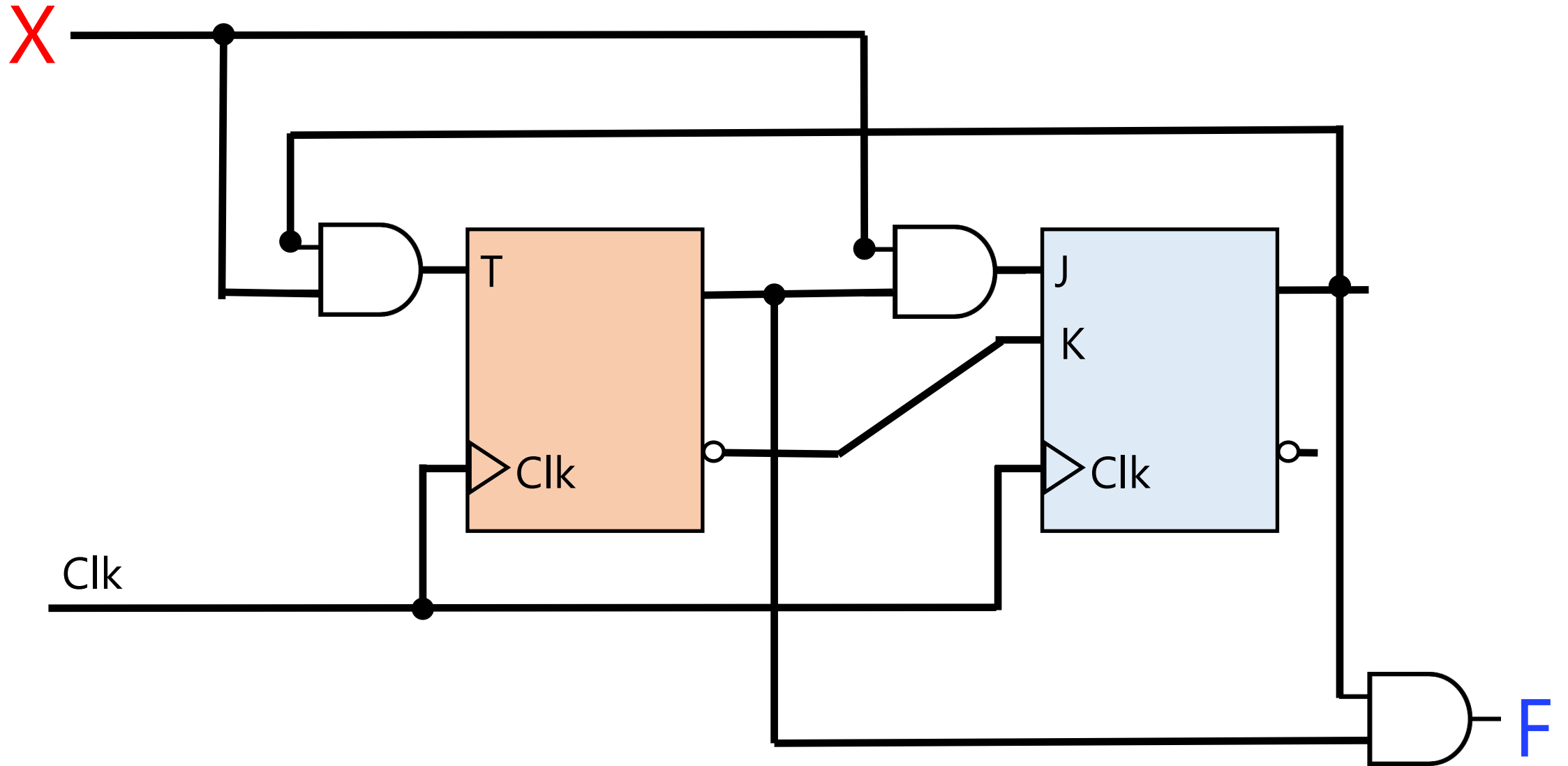
Analysis (Recap)

- 0. Is the circuit sequential or combinational? **Sequential**
- 1. What are the flip-flops?



Analysis (Recap)

0. Is the circuit sequential or combinational? Sequential
1. What are the flip-flops? T, JK
2. What are the state combinations?

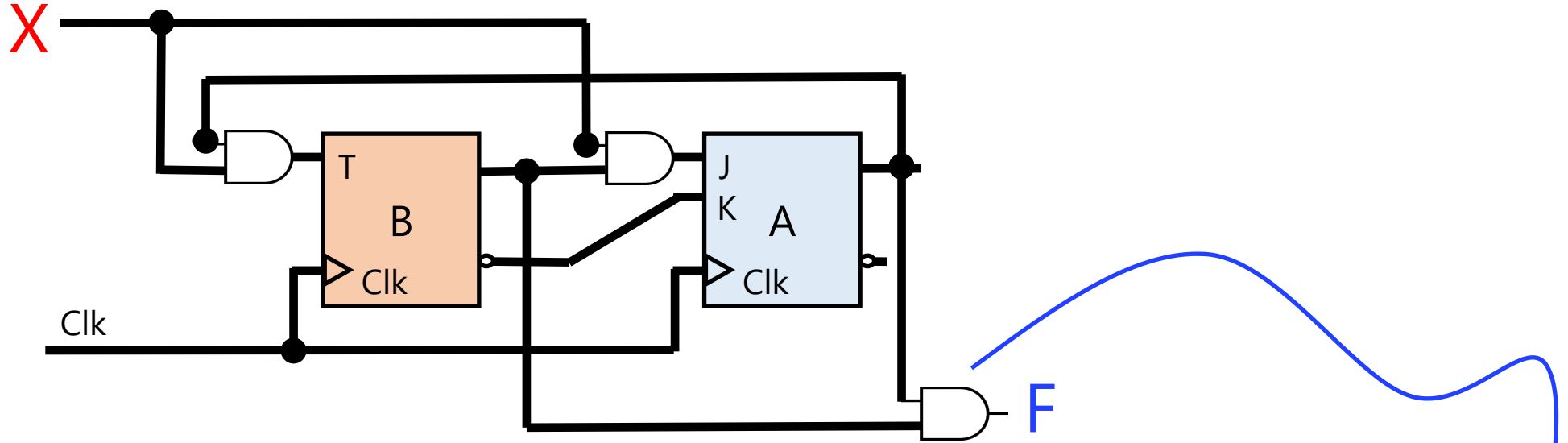


#FFs + #Inputs = 2+1 $\rightarrow 2^3 = 8$ combinations

Analysis (Recap)

0. Is the circuit sequential or combinational? Sequential
 1. What are the flip-flops? T, JK
 2. What are the state combinations? $2^{\#FF} \times 2^{\#inputs} = 2^{\#FF+\#inputs} = 2^3 = 8$
 3. Form "State" table:
 - a) Columns:
 - For each FF, two columns: one for current state, one for next state
 - For each input, one column
 - For each output, one column
 - b) Rows: See item 2
-

F

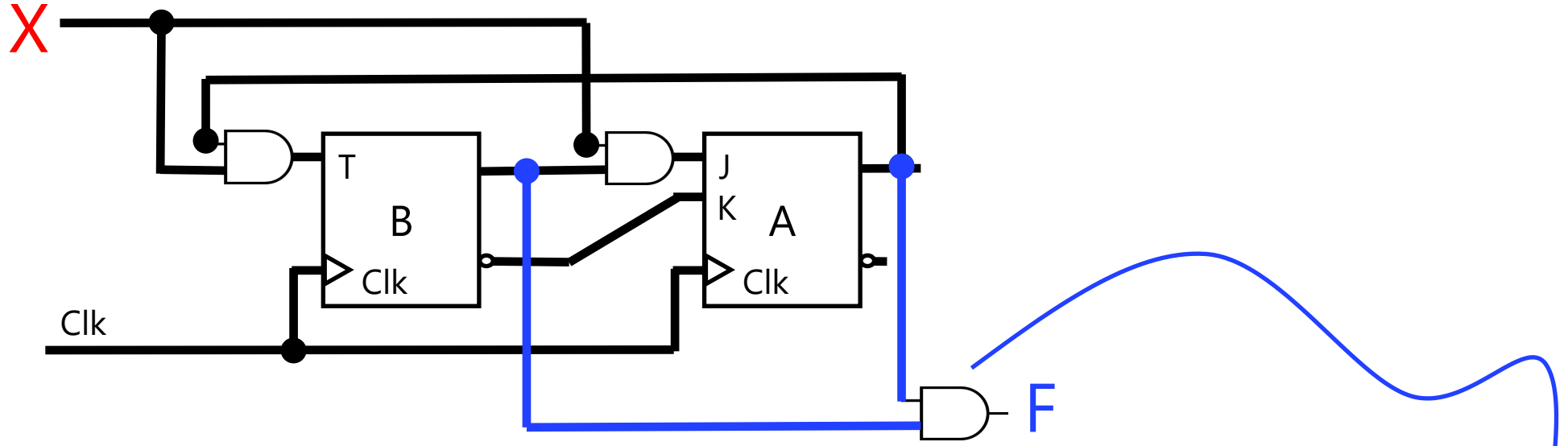


Q(T)		Q(T+1) when X=0		Q(T+1) when X=1		Outputs
B	A	B	A	B	A	F
						Moore model: does not depend on X

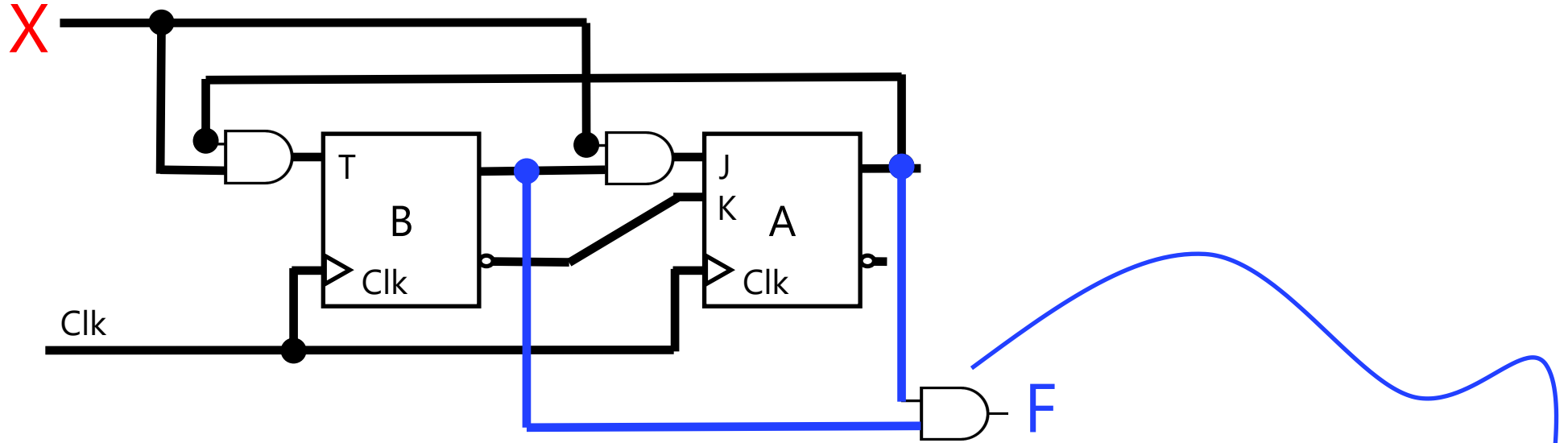
Alternative State Table

Analysis (Recap)

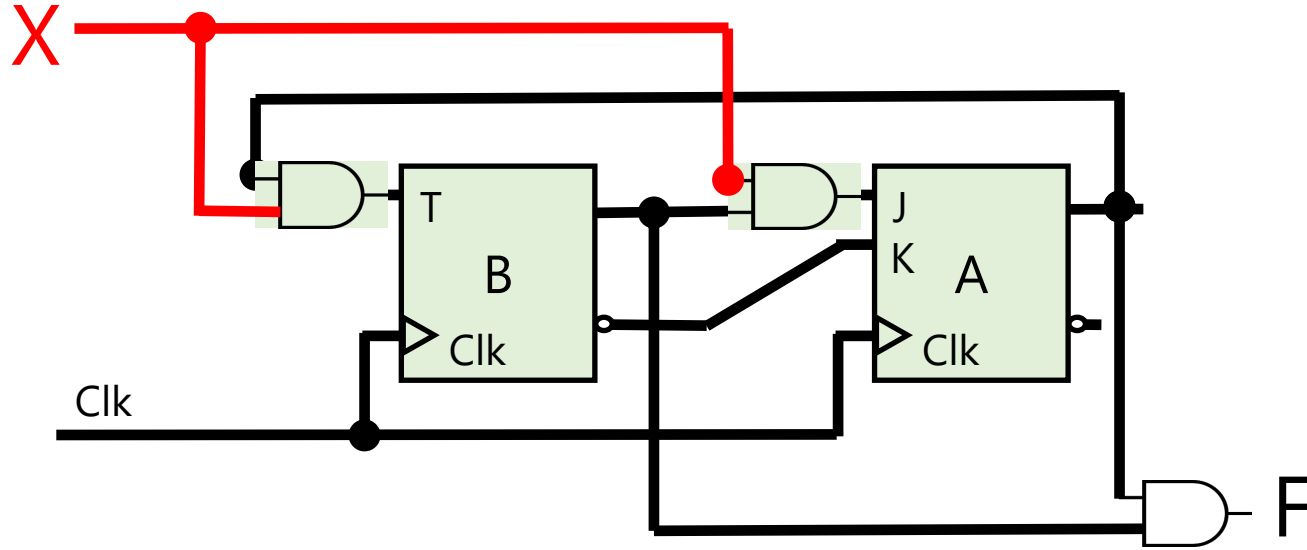
0. Is the circuit sequential or combinational? Sequential
 1. What are the flip-flops? T, JK
 2. What are the state combinations? $2^{\#FF} \times 2^{\#inputs} = 2^{\#FF+\#inputs} = 2^3 = 8$
 3. Form "State" table:
 - a) Columns:
 - For each FF, two columns: one for current state, one for next state
 - For each input, one column
 - For each output, one column
 - b) Rows: See item 2
 4. Fill the state table for
 - a) next state columns
 - b) the output value
-



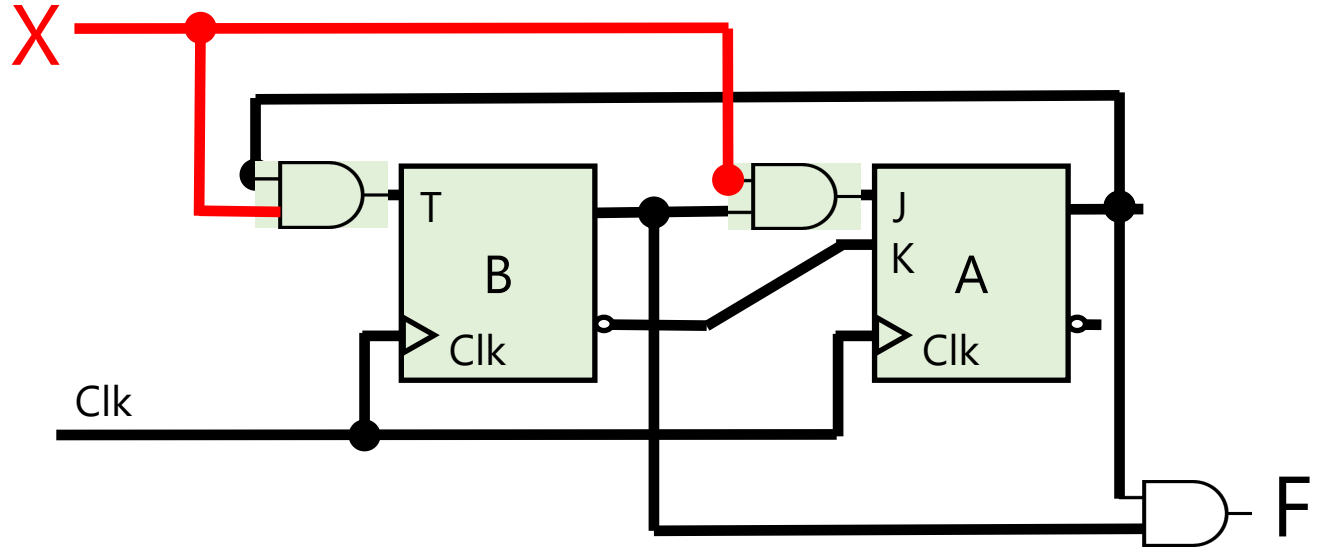
Inputs	Q(T)		Q(T+1)		Outputs
X	B	A	B	A	F=BA
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			



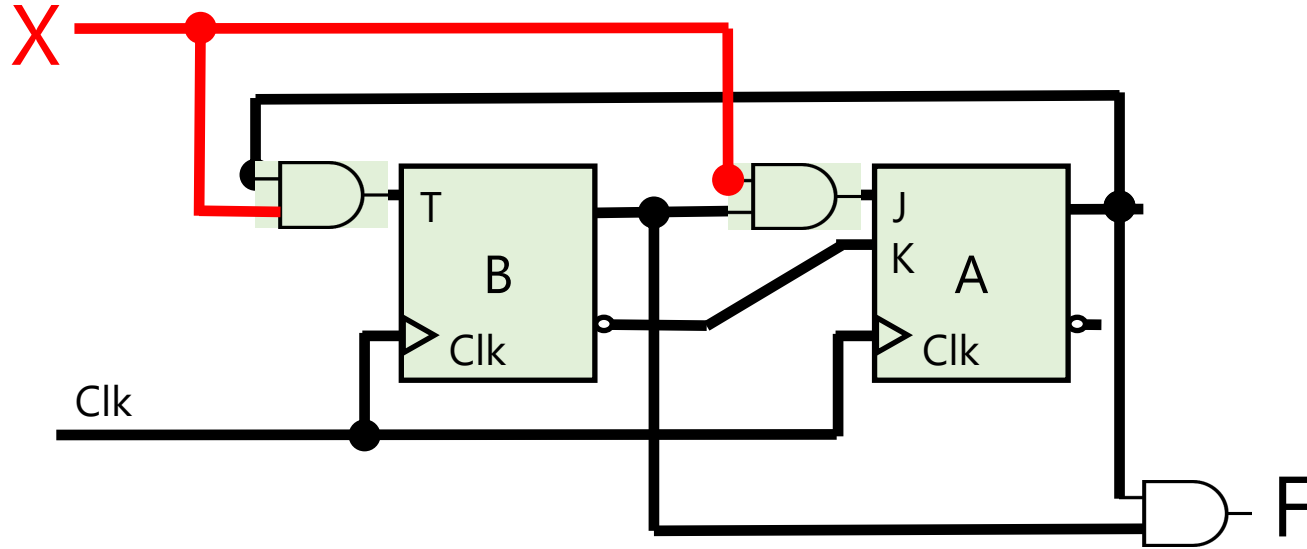
Inputs	Q(T)		Q(T+1)		Outputs
X	B	A	B	A	F=BA
0	0	0			0
0	0	1			0
0	1	0	Moore Model Only depends on current state X is not involved!		0
0	1	1			1
1	0	0			0
1	0	1			0
1	1	0			0
1	1	1			1



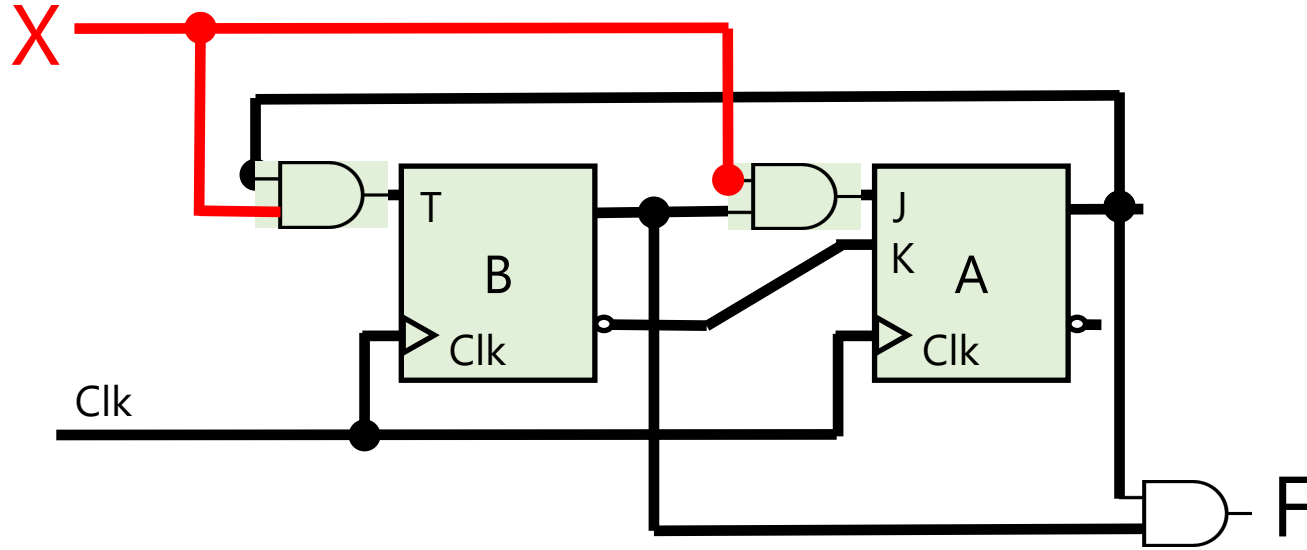
Inputs	Q(T)		Q(T+1)		Outputs
X	B	A	B	A	F=BA
0	0	0			0
0	0	1			0
0	1	0			0
0	1	1			1
1	0	0			0
1	0	1			0
1	1	0			0
1	1	1			1



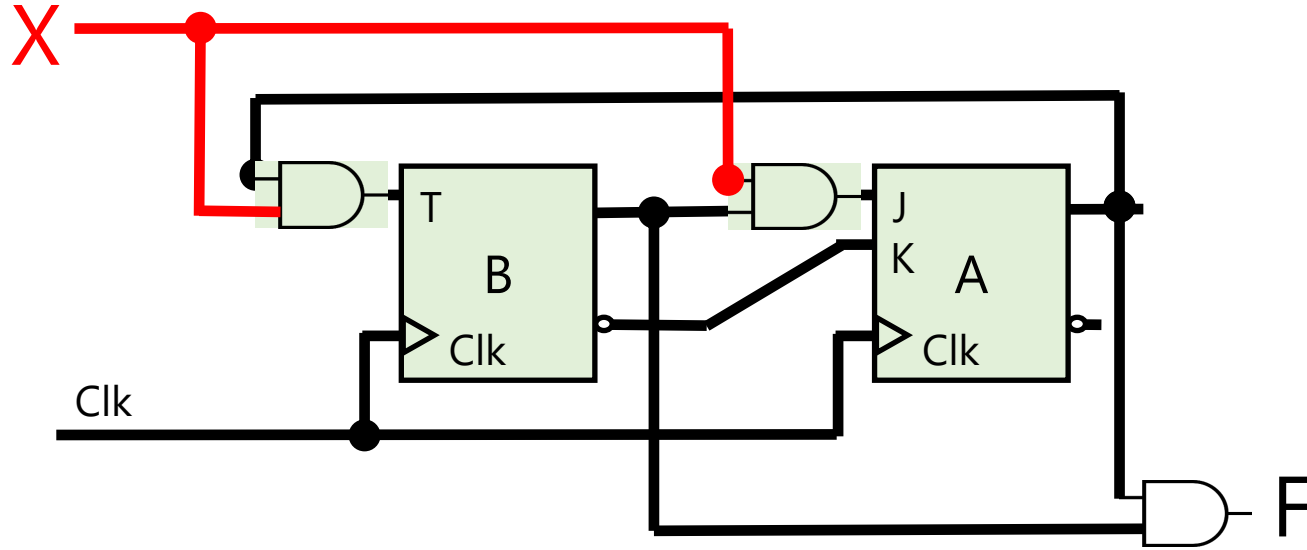
Inputs	Q(T)		Q(T+1)		Outputs
X	B	A	B	A	F=BA
0	0	0		A=0 $J_A = XB = 00 = 0$ $K_A = B' = 0' = 1$ ----- Reset $\rightarrow 0$	0
0	0	1			0
0	1	0			0
0	1	1			1
1	0	0			0



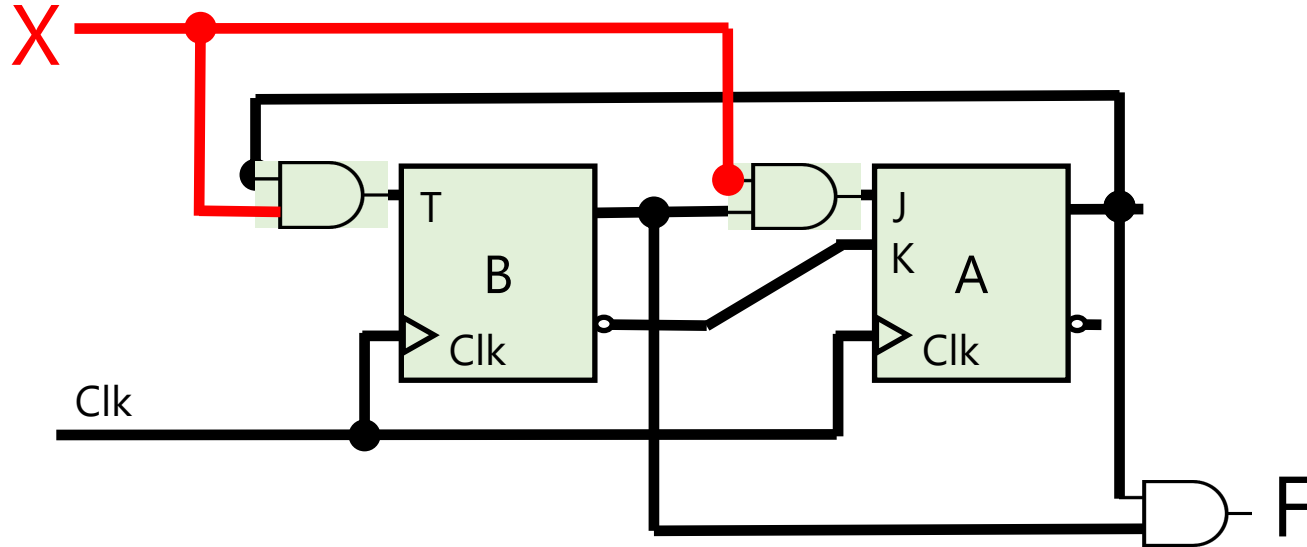
Inputs	Q(T)		Q(T+1)		Outputs
X	B	A	B	A	$F=BA$
0	0	0	B=0 $T_B = XA = 00 = 0$ ----- Store $\rightarrow 0$	0	0
0	0	1			0
0	1	0			0
0	1	1			1
1	0	0			0
1	0	1			0



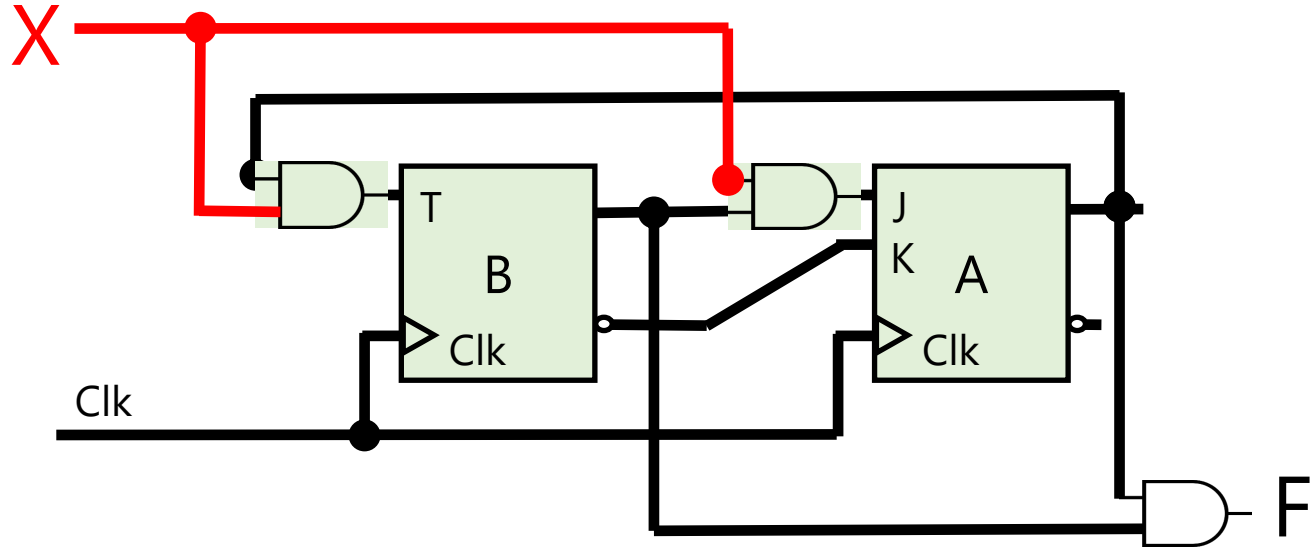
Inputs	Q(T)		Q(T+1)		Outputs
X	B	A	B	A	F=BA
0	0	0	0	0	0
0	0	1			0
0	1	0			0
0	1	1			1
1	0	0			0
1	0	1			0
1	1	0			0
1	1	1			1



Inputs	Q(T)		Q(T+1)		Outputs
X	B	A	B	A	F=BA
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	1	0	0
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	?	0	0
1	1	0	1	1	0
1	1	1	0	1	1



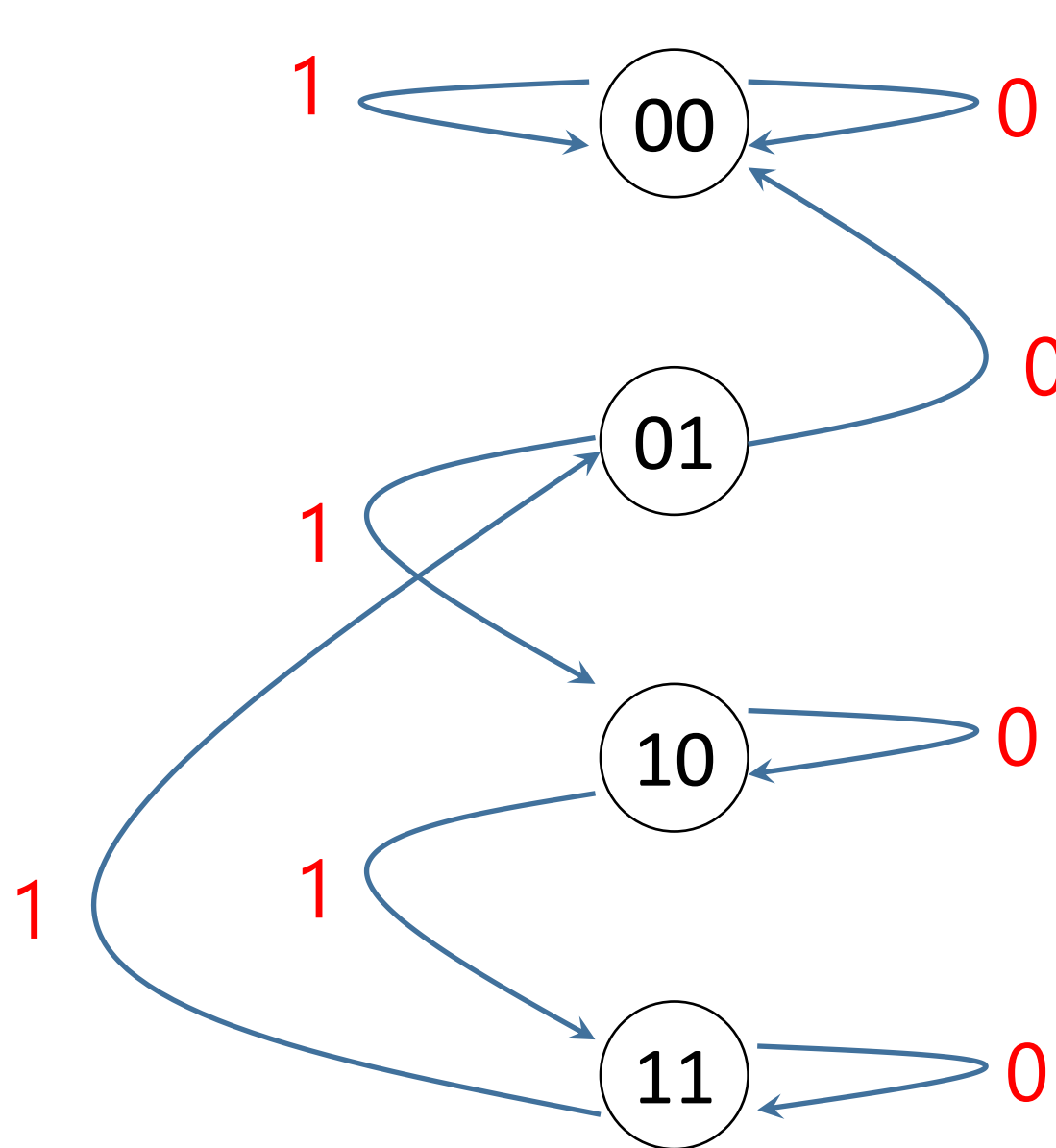
Inputs	Q(T)		Q(T+1)		Outputs
X	B	A	B	A	F=BA
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	1	0	0
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	B=0 TB=XA=11=1 ----- Comp. → 1		0



Inputs	Q(T)		Q(T+1)		Outputs
X	B	A	B	A	F=BA
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	1	0	0
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	1	0	0
1	1	0	1	1	0
1	1	1	0	1	1

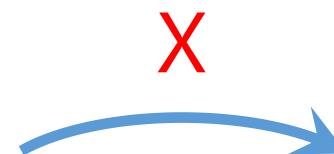
Analysis (Recap)

0. Is the circuit sequential or combinational? Sequential
 1. What are the flip-flops? T, JK
 2. What are the state combinations? $2^{\#FF} \times 2^{\#inputs} = 2^{\#FF+\#inputs} = 2^3 = 8$
 3. Form "State" table:
 - a) Columns:
 - For each FF, two columns: one for current state, one for next state
 - For each input, one column
 - For each output, one column
 - b) Rows: See item 2
 4. Fill the state table for
 - a) next state columns
 - b) the output value
 5. Form state (transition) diagram
 - a) nodes for states, directed edges for transitions between states
 - b) labels for edges by the value of input
 - c) labels for nodes by the value of output
-

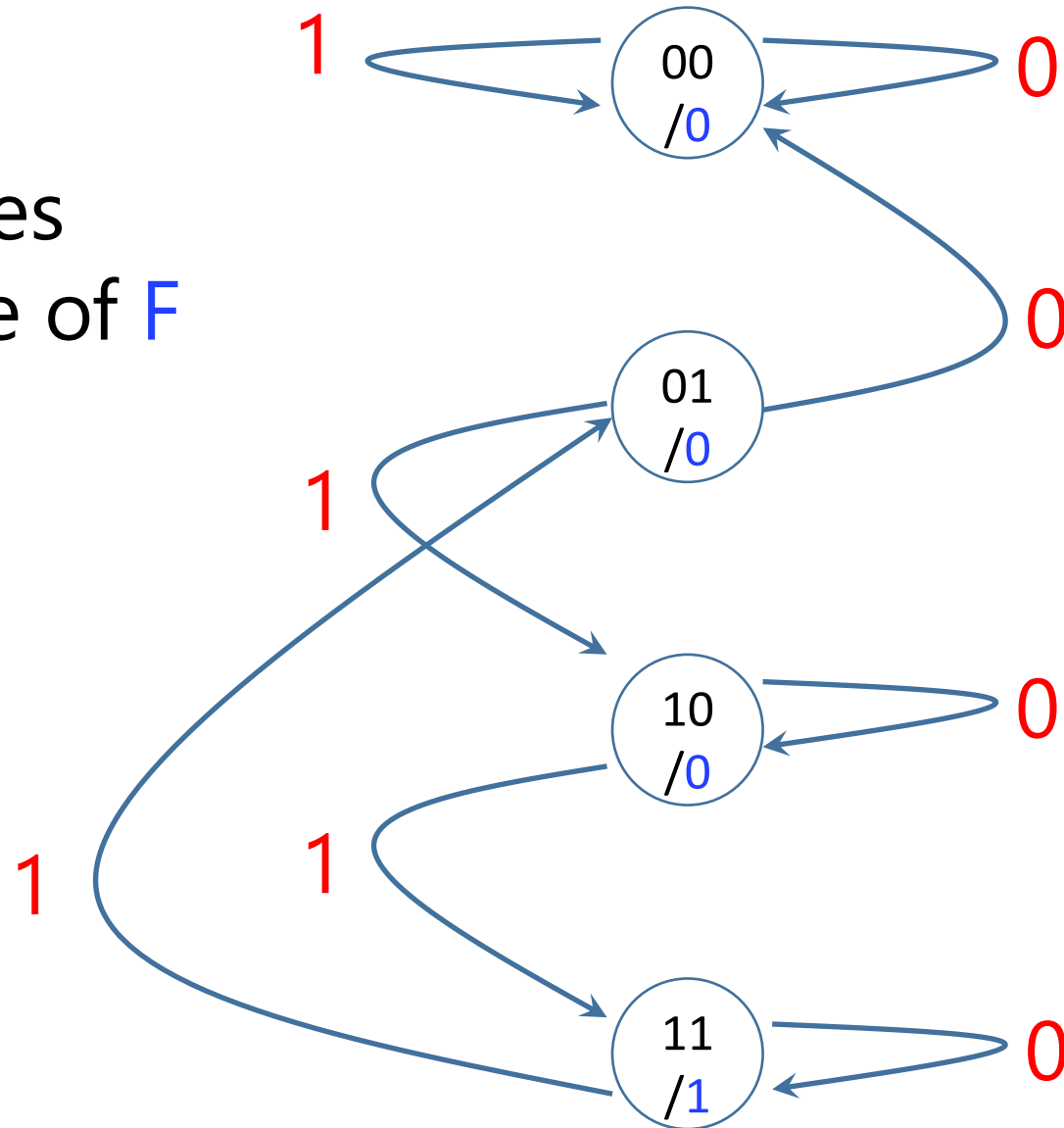
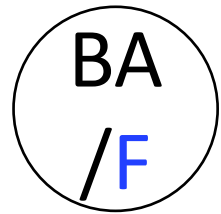


BA

Labels on edges
based on value of X



Labels on nodes
based on value of **F**



Analysis

6) (*Optional*) Path on State Transitions

Life lock!

